

Bachelor Thesis

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Faculty Computer Science

Limitations and Problems of Low-/No-Code Tools in the Context of E-commerce - A Practical Comparison of Webflow and Shopify

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Affidavit

I hereby declare that I have written this thesis independently, have not submitted it elsewhere for examination purposes, have not used any sources or aids other than those indicated, and have marked verbatim and analogous quotations as such.

Ingolstadt, 30.08.2023

A handwritten signature in dark ink, appearing to read 'Bergheimer', written in a cursive style.

Erik Bergheimer

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Abstract

In an increasingly digital world, it is becoming more and more essential to develop platforms for the web, whether it is apps, websites, or online stores. As a cheaper and more efficient alternative to full code development, Low-Code and No-Code tools are becoming increasingly popular. Webflow and Shopify are such Low-/No-Code tools which promise their users to develop online stores with little or no previous programming knowledge. This bachelor thesis aims to compare these two tools to show their limitations and problems. Therefore, this thesis's primary research question is, "What limitations and problems can be found while developing an online store utilizing the Low-/No-Code tools Webflow and Shopify?". For this purpose, a sample shop will be designed, which will be developed in both tools. A comparison table showing the limitations of both tools can then be created. This comparison reveals that Webflow has clear advantages in terms of online shop customization, while Shopify excels in better shop management tools and superior social media integrations. This also indicates that no clear recommendation for practical use can be given. Both tools have advantages and disadvantages, and the choice of which tool to use must be based on the specific use case and requirements.

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1 Introduction and Problem Definition

In an increasingly digital world, online shops are playing an ever-growing role. According to predictions from the platform eMarketer, nearly a quarter of total global retail sales are projected to take place on the internet by the year 2026 (Figure 1) [1].

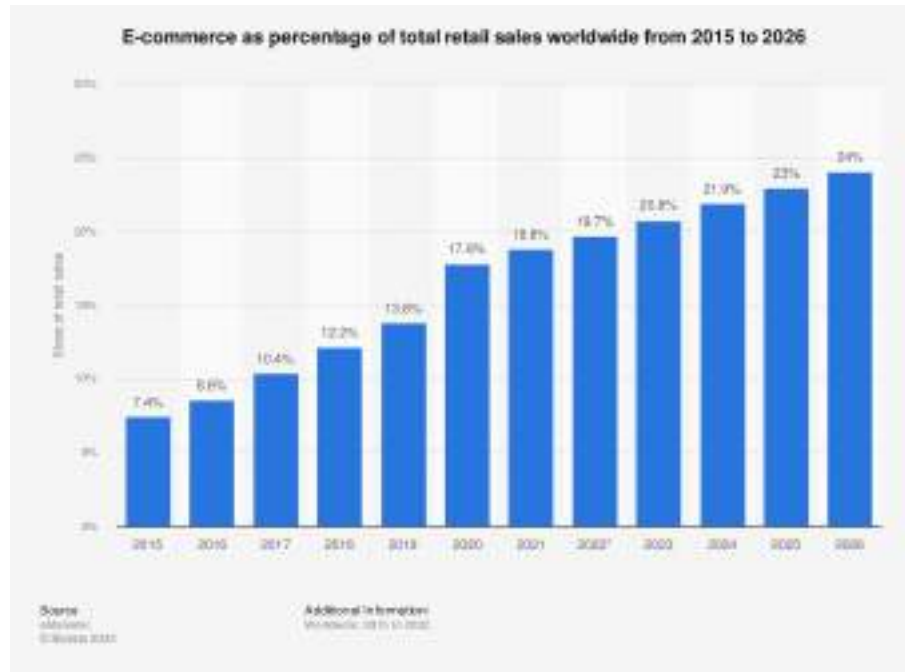


Figure 1: E-commerce as percentage of total retail sales worldwide from 2015 to 2026 [1].

These sales are facilitated by platforms ranging from large retailers, such as Amazon¹, Apple² and Walmart³ [2] to smaller online shops run by smaller brands. Either way, it is essential to develop these online shops. One option is to develop them entirely from scratch using a large number of different programming languages or tools like Perl, Java, ASP.NET, Javascript, PHP, and HTML [3, p.11]. Such a development approach, though, necessitates the expertise of software developers familiar with the chosen language or tool, which can significantly increase the costs, efforts and complexity associated with building an e-commerce platform [4, p.512].

In this context, the Low-/No-Code software development approach can provide a solution. "Low-code/no-code software development is an emerging approach delivering the opportunity to build software with a minimal need for manual coding and enhancing the involvement of non-programmers in software development [5]". Many different companies are developing tools for precisely this development approach like Bubble or Adalo, which are tools for traditional application development [6, p.29]. Indeed, tools like Wix and Webflow are becoming significant players in the Low-/No-Code market, especially in the field of web development [6, p.29], similar to Shopify, which enables users to create their own online stores without writing a single line of code [7].

¹<https://www.amazon.de/>

²<https://www.apple.com/>

³<https://www.walmart.com/>

Given that both Shopify and Webflow promise their users the ability to create an online store, using their chosen platform without requiring programming skills [8][9], the use of such tools is naturally quite appealing for businesses. They have the potential to save on developer resources, development time, and consequently, costs [5].

Besides the promises and development possibilities offered by Shopify and Webflow to their users, this thesis will focus on a practical comparison between Webflow and Shopify to explore the inherent limitations and problems of Low-/No-Code technologies in the context of e-commerce. Since these can have a negative impact on the performance and functionality of an online shop, it is important to identify and understand these limitations and problems in order to minimise potential risks and make the most of the benefits of Low-/No-Code tools. This study can also help companies and web developers make the best decision when choosing tools for their specific needs. Furthermore, this thesis shows the potential and development possibilities of Low-/No-Code tools, especially Webflow and Shopify.

1.1 Aim of this Thesis and Research Questions

This thesis focuses on introducing the topic of Low-/No-Code and the two tools, Webflow and Shopify. Subsequently, an example online shop will be designed in Figma and then developed in Webflow and Shopify. The objective of this study is to identify the differences that arise in the development process and to ascertain any limitations or issues that can be derived from it. The research questions are as follows:

1. **What limitations and problems can be found while developing an online shop utilizing the Low-/No-Code tools Webflow and Shopify?**
2. **Which recommendations can then be given for the application of both tools?**

In order to address these research questions, the first step is to create a sample brand with associated products and an online store, which is then developed using both tools. Subsequently, comparison criteria are established and the outcomes of both developments are compared.

1.2 Distinguishing Factors from Other Works

Numerous comparisons of Low/No-Code tools can be found on the internet, including evaluations of Webflow and Shopify's functionalities by online media. This work distinguishes itself from these existing studies by its substantial practical approach, a dimension often absent in prior Webflow and Shopify comparisons. Previous comparisons typically lacked the development of a sample online store and the subsequent evaluation of the development process in both tools against the defined comparison criteria. This distinctive methodology sets this study apart from others in the field, offering a unique perspective and depth of analysis.

1.3 Overview of the Thesis

The thesis is divided into six chapters. Chapter 2 explains the concepts of Low-Code and No-Code, and introduces the tools Webflow and Shopify. Additionally, an overview of the comparison criteria is provided in this chapter. Chapter 3 focuses on the creation of the example online shop, which was designed in Figma, and the Expert Review conducted with an e-commerce design expert. In Chapter 4, the implementation of the design from Chapter 3 in Webflow and Shopify is discussed, along with the considerations during development. Chapter 5 delves into the details of the table created for comparing the two tools. Furthermore, a comparison is drawn

based on the key comparison criteria, leading to Chapter 6, which addresses open research questions and the limitations of the study. Finally, a future outlook is provided, followed by a conclusion.

2 Introduction to Low-/No-Code

This chapter focuses on the fundamentals of the Low-/No-Code approach, explaining the What You See is What You Get principle, and exploring the concept of Citizen Development. Following that, Webflow and Shopify are introduced as tools, and the comparison criteria used in the research process for this comparison are discussed.

2.1 Fundamentals of the Low-/No-Code Approach

As briefly mentioned in the introduction, there are three different principles for software development: Full-Code, Low-Code, and No-Code. Each of these principles are defined differently and offer various advantages and disadvantages.

2.1.1 Full-Code

The Full-Code approach designates the conventional approach of developing software, in which programmers write lines of code to produce a software. The choice between using Full-Code or No-Code depends on a number of variables, including the project's complexity, the resources at hand, and the time frame. For large-scale projects that need unique features and functionalities, Full-Code is suitable. Developers may customize every component of the program since they have total control over it. Full-Code development, however, takes a lot of time, money, and technical know-how [10].

In the complete custom development of e-commerce platforms, a variety of different programming languages and tools are typically used. Some of the most well-known ones include Perl, Java, ASP.NET, JavaScript, PHP, and HTML. As e-commerce applications are inherently user-centered and data-driven platforms, Database Management Systems are used for (user) data storage. Some of the well-known Database Management Systems include IBM's DB2, Sybase, Microsoft Structured Query Language (SQL), or MySQL [3, p.11].

2.1.2 Low-Code

The Low-Code approach lets experienced software engineers achieve their task more quickly by using pre-built components. However, they do not eliminate all code. Low-Code includes a lot more developer-focused capabilities, such as several application contexts (development, staging, and production features). They are not a tool for everyone though because they do require some programming experience [11].

According to OutSystems, nearly all Low-Code platforms have the capabilities to create business process automation solutions, web apps, workflow management, and data visualization [12]. Typical Low-Code development platforms are for example Microsoft PowerApps, Mendix, Appian, and Wavemaker [13, p.60] [14] [15].

2.1.3 No-Code

Last but not least the No-Code approach enables non-programmers to create software. In other words, No-Code platforms enable people with limited technical knowledge to create apps that integrate with an organization's present infrastructure and operations. No-Code tools provide

features like drag-and-drop functionality, process flows, visual tools, and more, much like Low-Code tools do. Businesses find No-Code development appealing since it is flexible, simple to create, and people without any coding knowledge can use it [14].

Typical use cases for the utilization of No-Code development tools encompass the creation of internal tools and workflows, data collection and reporting, prototyping and Minimum Viable Product (MVP) development, business process automation, and website and landing page creation [16]. Common tools in this context include, Webflow and Wix for website creation [6, p.29], Shopify for building online stores [7], as well as Bubble for app development [6, p.29].

2.1.4 Differences between Low-Code and No-Code

No-Code and Low-Code platforms, as mentioned earlier, have several similarities. The differences between Low-Code and No-Code platforms will be addressed in the following.

Low-Code development necessitates some coding from users, although significantly less than what is typically required in traditional application development. Skilled developers and programmers utilize Low-Code platforms to rapidly create applications and redirect their focus from mundane programming tasks to more intricate and innovative work that holds greater significance and value for the organization. Additionally, non-IT professionals with basic programming knowledge also leverage Low-Code tools to build straightforward apps or extend the functionality of existing applications [17].

According to OutSystems, the main audience of Low-Code platforms are professional developers, so coding expertise is highly recommended [18]. Another distinguishing feature of Low-Code is its project suitability. These include business critical solutions and sophisticated apps, where "developer[s] can integrate with any enterprise system of record" [18]. Moreover, when necessary, developers should have the possibility in Low-Code tools to add custom code into their projects [18].

In contrast No-Code development is aimed at individuals in different business roles who lack technical expertise and programming skills but have a good understanding of business requirements and rules. These 'citizen developers' (Chapter 2.1.6) can utilize No-Code platforms to effortlessly and rapidly create, test, and deploy their business applications, as long as the chosen tools align with the necessary functions and capabilities [17]. The perfect type of project for the use of No-Code platforms are simple, departmental apps or software. Lastly, they do not have to offer any kind of platform extensibility or custom code integration [18].

2.1.5 WYSIWYG - What You See Is What You Get

In the context of the terms Low-/No-Code tools, the abbreviation WYSIWYG is often mentioned and stands for What You See Is What You Get. "In computing and website development, this refers to a software whose user interface (UI) allows the user to view their real-time edits on a display that resembles the final published outcome" [19].

The WYSIWYG interface of word processors like Microsoft Word or even Wordpress enables the display to mimic document configurations (such fonts, line breaks, and final pagination) as they would appear on a printed page (Figure 2). When building a website using a WYSIWYG

platform, the editor shows exactly how each page would look to a user. Some WYSIWYG applications employ a variety of settings to draw attention to the minute variations between what the users see and what they receive [19].



Figure 2: Example of the usage of the WYSIWYG approach by Wordpress [20].

A WYSIWYG system differs from conventional editors in that it does not demand the user to use markup languages like HTML or descriptive codes. This system is frequently seen as more accessible or user-friendly because users do not need to know the commands to generate or modify the layout and content of printed documents or digital products [19]. WYSIWYG software has a streamlined process that enables the user to see the finished product as they create and change it, which sets it apart from conventional markup techniques. Anyone can quickly, and without interruptions, make changes in real time, notably developers and UX designers. With typical markup languages, users cannot examine the changes they make. Instead, they can only see the lines of code they enter [19].

Both Webflow and Shopify utilize this approach in the development process within their Website Editors. When an element is added to the canvas, it immediately appears on the canvas and can be adjusted and modified live.

2.1.6 Citizen Development

One term that also frequently comes up in the context of Low-/No-Code is the concept of 'Citizen Development'. Citizen Development, according to OutSystems, "is a business practice where a user — the so-called citizen developer — creates business applications for use by other people. Although they do not report directly to IT, they use tools such as No-Code and Low-Code platforms that are sanctioned by IT" [21].

Several organizations have embraced the approach of citizen development to address the problems that have plagued businesses for years and limited their capacity to innovate which are [21]:

- The growing backlogs in IT
- A Development talent shortage and expense

- The business environment is always evolving at a rate that is greater than that of the software and systems that customers depend on

Citizen development is a strategy to reduce the burden on IT by enabling citizen developers to build the business applications they require without involving IT departments. [21].

2.1.7 Advantages and Disadvantages of Using Low-/No-Code Tools

Now that the fundamentals for Low-/No-Code platforms are laid out, the next chapter will focus on the seven most important advantages and disadvantages of Low-/No-Code platforms.

At the start, the emphasis is placed on the advantages of Low-/No-Code tools.

Less Coding: At first, Low-Code and No-Code platforms accelerate the application development and deployment time, which has become crucial in the digital age, where organizations must respond rapidly to meet the demands of employees and customers or risk being outpaced by competitors [17] [4, p.510].

Democratization of IT: Additionally, these tools empower citizen developers by providing them with enhanced problem-solving capabilities, enabling them to swiftly and effortlessly create business applications that streamline their work processes [17] [22].

More Efficient Development Time: Moreover, these platforms liberate professional developers from mundane programming tasks. Development teams can utilize these platforms to quickly create applications for standard functions and then customize them further to add more value. This allows developers to focus more on creating bespoke applications or addressing other strategic initiatives that offer unique value to their organizations [17] [4, p.510].

Reduced Development Costs: According to OutSystems, Low-Code and No-Code platforms also reduce the need for traditional development teams and programming expertise [23] [4, p.509].

Although there are several advantages to using Low-/No-Code tools, the disadvantages and issues associated with these platforms should not be overlooked.

Limited Flexibility: While traditional coding allows developers to fully customize software according to their needs, Low-Code and No-Code platforms depend on the built-in functionalities and plugins they offer. If a particular functionality is not provided by the platform, the development team may face challenges in meeting related requirements [24] [4, p.511].

Vendor Lock-in: If users want to switch to another platform, they cannot modify or maintain the apps they have already produced. The users are then 'locked-in' the platform they chose. [23] [22].

Limited customization: Despite the possibility of some customization on Low-/No-Code platforms, users are generally restricted to the features and functionalities offered by the platform [23] [4, p.511].

2.2 Webflow and Shopify as Low-/No-Code Tools

In the following section, both Webflow and Shopify will be briefly introduced. In addition, an overview of the companies that are behind these tools is given.

2.2.1 Introduction to Webflow



Figure 3: Logo Webflow [25].

According to their own website, Webflow (Figure 3) is a visual editor and hosting platform that enables users to create, develop, and publish customized and professional websites, saving businesses and developers time and development resources [25]. By providing a diverse collection of customizable templates, the Webflow platform enhances the accessibility of website design. Through a drag-and-drop interface, users can customize their websites by placing elements precisely where they desire [26]. This approach enables individuals with limited website-building experience to create visually appealing and high-quality websites in a swift and efficient manner [26].

The drag-and-drop editor is the prominent feature of Webflow's interface, allowing users to assemble layouts without the need for coding knowledge. By dragging and dropping buttons, images, text boxes, menus, and other elements onto their websites, designers can customize individual components. Additionally, Webflow's editor facilitates the inclusion of dynamic interactions, such as hover effects and page animations. These features make Webflow a compelling choice for novice web designers and marketers, providing them with the ability to create high-quality designs at a fraction of the cost associated with hiring professional developers [26].

Webflow's hosting, which is directly provided within the tool, leverages the two providers Amazon Web Services and Fastly [27].

Furthermore, Webflow can be used in two ways: with a free version that has limited features, or with a paid version that unlocks various functionalities. The price for the paid versions starts at \$14 (currently approx. 12,86€) per month and increases based on the plan, with customized pricing available for larger enterprises [28].

The tool also caters to various use cases. Webflow can be utilized for creating small landing pages, developing larger CMS systems, and even building extensive corporate websites. Additionally, Webflow can be employed for crafting online shops.

Next to offering many advantages that Webflow offers, there are also some disadvantages associated with it that are often mentioned. On one hand, the tool is not considered very beginner-friendly. The learning curve is quite steep, and despite the numerous provided tutorials, it takes some time to become proficient with the user interface [29]. On the other hand, the pricing

structure is not very transparent. As a new user, it can be confusing to understand both the Workspace Pricing and the individual Site Pricings. Lastly, Webflow is currently available only in English, which can pose difficulties for users who are not fluent in this language.

The company Webflow was founded by Vlad Magdalin, Bryant Chou and Sergie Magdalin in 2012 [30]. The concept of the tool dates back to a senior thesis written by Vlad Magdalin in college, when he examined how tools that could convert drag-and-drop design into clean back-end code may decrease the entry barrier for business owners looking to create the next Facebook or Airbnb. When the team first struggled to acquire traction, Vlad Magdalin ran up credit card debt in the tens of thousands of dollars and was unable to collect money for the project on Kickstarter. Then, after seven months, Webflow began to gain traction in March 2013 in form of a viral Hacker News article [26][30]. This article led to tens of thousands of requests signing up for the beta program, so the founding team then applied for the Y Combinator, which is a startup accelerator [31] and cleared with \$2.9 million (currently approx. 2.673 million €) of funding [32].

In 2023, Webflow has 3.5 million users and 600 team members [33]. Webflow also hit the \$100 million (currently approx. 92.179 million €) annual recurring revenue and a \$4 billion (currently approx. 3.687 billion €) valuation in 2022 [34].

2.2.2 Introduction to Shopify



Figure 4: Logo Shopify [35].

According to their website, Shopify (Figure 4) is an e-commerce platform that enables users to launch, develop, and run a company [35]. With Shopify, entrepreneurs can create and personalize an online store and advertise their products via social media, online marketplaces, pop-up shops, web, mobile, in-person, brick-and-mortar locations, and other channels. Since Shopify takes care of all software and server maintenance for their users, they have the freedom to manage their company from any location in the world with just an internet connection [36].

Shopify is known for its online store builder, which, according to Shopify, requires no programming knowledge. The foundation for the shop designer lies in templates which Shopify offers in their template store. These can then be customized to meet the user's specific requirements as needed. Additionally, Shopify provides the option to integrate numerous third-party apps into the online store through the Shopify App Store. This allows users to not only sell products through the online store but also reach customers through other distribution channels, such as social media [35].

Shopify itself is written in the open-source template language Liquid which was designed by Shopify and written in Ruby. It is the foundation of Shopify themes and is utilized to load dynamic information on storefronts [37]. Furthermore, according to their own statement, Shopify uses the provider Cloudflare for the provision of their hosting services [38].

Similar to Webflow, Shopify also offers different pricing plans. In addition to a 3-day free trial version, the smallest Basic package starts at 36€ per month for small businesses and increases with advanced functionalities up to 384€ per month for medium to large businesses.

The Shopify tool clearly focuses on the development of online shops. However, the coverage of different use cases by Shopify varies based on the size of the brands that choose Shopify. Among their customers, there are both smaller, lesser-known brands and larger, well-known players such as Gymshark, Allbirds, or Heinz [39].

In addition to the advantages that Shopify offers, there are also a number of disadvantages that should be mentioned. For one, the pricing plans compared to other e-commerce website builders (such as Squarespace and Wix) are relatively expensive. And second, the ability to create blogs or engage in content marketing is significantly more limited with Shopify [40].

Tobias Lütke, a German-born entrepreneur, founded Shopify in response to his dissatisfaction with the software he was using to manage his 2004-founded online snowboard business. He created his own software, and shortly after, other merchants enquired about licensing it, resulting in the birth of Shopify in 2006 [41].

The features that set Shopify apart from other competitors at that time included order tracking, inventory management, and payment processing, including PayPal and credit card connections [42]. Shopify would grow into an e-commerce pioneer in the years that followed. The platform gave retailers more control by introducing ground-breaking solutions like its API or app store. Shopify then made its public debut on the New York Stock Exchange on May 21, 2015, with a valuation of \$2.1 billion (currently approx. 1.936 billion €), following nine years of rapid organic development. Shopify continued to expand and diversify its product offering throughout the next years. By 2019, Shopify overtook eBay to become the second-largest e-commerce platform in the United States, after Amazon, thanks to it focusing on the requirements of direct-to-consumer businesses [42].

In 2023, Shopify is being used in over 170 countries, covering 10% of e-commerce in the USA. Being utilized as a tool by millions of merchants worldwide results in a global economic activity of over \$444 billion (currently approx. 409.21 billion €) [35].

2.2.3 Assessment of Both Tools

Both Shopify and Webflow exhibit characteristics of both Low-Code and No-Code tools. Webflow self-identifies as a No-Code software [43], yet a basic understanding of HTML and CSS is required for comprehending and using the software effectively. Additionally, it allows the insertion of custom code, which aligns with characteristics of a Low-Code tool [18].

Considering the criteria defined earlier Shopify can also be labeled as a No-Code tool. However, its categorization depends on the manner in which the tool is utilized. If a user only utilizes the provided template and the limitations of that template are acceptable, then Shopify can indeed be seen as a No-Code tool. However, if a user wishes to customize the template further and make adjustments, at least a foundational understanding of HTML and CSS is necessary to comprehend the source code of the templates. Similar to Webflow, Shopify also allows the addition of custom-written code, which also aligns with aspects of a Low-Code tool [18].

Since in practice, the term "No-Code" is often used synonymously with "Low-Code" [44], throughout the remainder of this work, the term "Low-Code" will also refer to "No-Code", and vice versa.

2.3 Comparison Criteria for Low-/No-Code Tools

To draw a comparison between these two tools, it was necessary to explore various sources that deal with the comparison of Low-/No-Code tools. This work draws upon sources that are publicly accessible online. These sources include comparisons of other Low-/No-Code tools, which have been examined in scientific studies for their features. Additionally, other comparisons of Webflow and Shopify have been consulted, although these comparisons lack the depth and practical context presented in this work. In addition to the criteria mentioned in the literature, new criteria were introduced that were not previously found in the literature used. As a result, these new criteria provide an additional perspective on the comparison of both tools, augmenting the existing criteria. The main comparison criteria will now be presented in the following chapter.

Website Editor: In a first step, it is immediately apparent that Low-/No-Code tools utilize Website Editors for the interaction between users and the platform. These Website Editors allow the user to build, design, and manage the shop using various features. The sub-criteria considered in this regard are:

- Templates [45] [46] [47]
- Onboarding Process
- Graphical User Interface (GUI) [15]
- Auto-Save Feature
- Collaboration [15]

Customization: Moreover, it is crucial to assess the extent to which the created software can be customized to individual needs. For instance, Flowout considers customization and flexibility of the utilized tools as one of the pivotal criteria in their comparison between Webflow and Shopify [29]. Likewise, Alexander Sachs also briefly addresses this limitation of No-Code tools, noting that platforms like Wordpress might not accommodate highly complex designs or workflows [48, pp. 30-31]. The sub-criteria considered in this regard are:

- Customizing the Design [5] [46]
- Customized Checkout [46]
- Integrating a Custom Font
- Customized E-Mails

E-commerce Features: Since this thesis focusses on the usage of Low-/No-Code tools in the context of e-commerce and two online shops are being developed throughout this thesis, it is also important to consider how both tools perform in terms of their features in the e-commerce sector [49]. The sub-criteria assigned to this overarching criterion are:

- Transaction Fee [47]
- Inventory Management [50] [46]
- Payment Methods [51] [49]
- Discounts
- Test-Mode for the Checkout
- Mobile App

Pricing: Since, according to A. Sahay *et al.*, the price of a Low-/No-Code platform is one of the most important aspects, especially for small or medium-sized businesses [15], this overarching criterion will also be considered in the comparison of both tools. The sub-criteria assigned to this main comparison criterion are:

- Monthly Software Pricing [44] [15]
- Free Trial

Performance and Scalability: Alexander Sachs also writes about the limitations of No-Code in the book "Digitales Management und Marketing" in the chapter titled "Wie No-Code die Entwicklung von digitalen Geschäftsmodellen deutlich zugänglicher macht". In a brief section, the author discusses the boundaries of No-Code, with a broad term that arises being that of 'Performance and Scalability' [48, pp. 30-31]. These terms, for instance, are associated with the loading time of a website, which can impact user bounce rates. Additionally, it is noted that scalability, along with performance, also plays a role. This centers around factors such as how many users can visit a website or how many products can be offered in an online shop at maximum capacity [48, pp. 30-31]. The sub-criteria assigned to this main comparison criterion are:

- Amount of E-commerce Items
- Amount of CMS Items
- Loading Time [15] [48, pp. 30-31]

Development and Code: Since the literature frequently mentions the limited flexibility and customizability in code [23] [4, p.511], as well as the possible vendor lock-in with Low-/No-Code platforms [22], it is also considered important to consolidate these criteria under the main criterion of 'Development and Code.' The sub-criteria assigned to this main comparison criterion are:

- Hosting
- Code Quality [5]
- Code Export / Vendor Lock-In [44] [5]
- Adding Custom code [5]

Marketing Features: Not only should the creation of the online shop be the focus in the comparative process of both tools, but also the consideration of how the presented products can be distributed and promoted. For example, Rokas Jurkėnas includes the marketing features of both tools as a comparison criterion in his comparison of Webflow and Shopify [45]. The sub-criteria assigned to this main comparison criterion are:

- E-Mail Marketing [47]
- Social Media Marketing

All the comparison criteria of Low-/No-Code tools mentioned in this chapter are summarized in Chapter 5.1 into a comparison table, which then serves for the final comparison after the implementation process.

3 Creation of an Example Online Shop

This chapter focuses on the design process of the example online shop, which is created for the later comparison of the tools. In the first step, a persona is developed, serving as the primary target audience for the eventual shop. Subsequently, a sample brand is created, consisting of various brand elements such as colors, a font, a logo, and example products to be offered in the online shop. With this information as a foundation, the shop's design is created in the tool Figma, which was later revised with input from an e-commerce expert to avoid design and usability flaws.

3.1 Creating a Design for an Online Shop in Figma

In the next step, to adequately compare the two presented tools, it is necessary to create a store that closely resembles a real online shop. This chapter establishes a sample brand named 'Blue-Bird', comprising specific brand elements and offering various products in an online store. The creation of the brand and the subsequent process of shop design will be addressed in the following chapter.

3.1.1 Creating an Example Brand

The entire design process is carried out in the tool Figma, which is an "online design tool for riffing on, sharing, and creating the best work together" [52]. Figma also allows its users to create their own prototypes, as well as sitemaps and graphics, on a canvas. Figma offers a free educational subscription to students enrolled at a university [53], which is used in this thesis.

Persona

Creating a persona is essential in e-commerce, as the sample brand aims to sell products through a fictional online shop. A persona allows a deeper understanding of the target audience and facilitates aligning the brand with their preferences and needs.

According to their website, The Nielsen Norman Group is a "UX research and consulting firm trusted by leading organizations world-wide" [54] founded by two pioneers of User Experience, Don Norman and Jakob Nielsen. They define a persona as "[...] a fictional, yet realistic, description of a typical or target user of the product. A persona is an archetype instead of an actual living human, but personas should be described as if they were real people" [55].

For this case, the persona Max was created (Figure 5). Max represents a male in his early 20s who shops for new and used clothing online. Max prefers to explore various online stores and has a strong interest in discovering new brands. With a solid understanding of internet usage, Max seeks a simplified, efficient, and secure buying process. He actively participates in social media platforms, displaying a heightened awareness of fashion trends in the industry. Moreover, Max is willing to pay a premium price for products that exhibit superior quality.

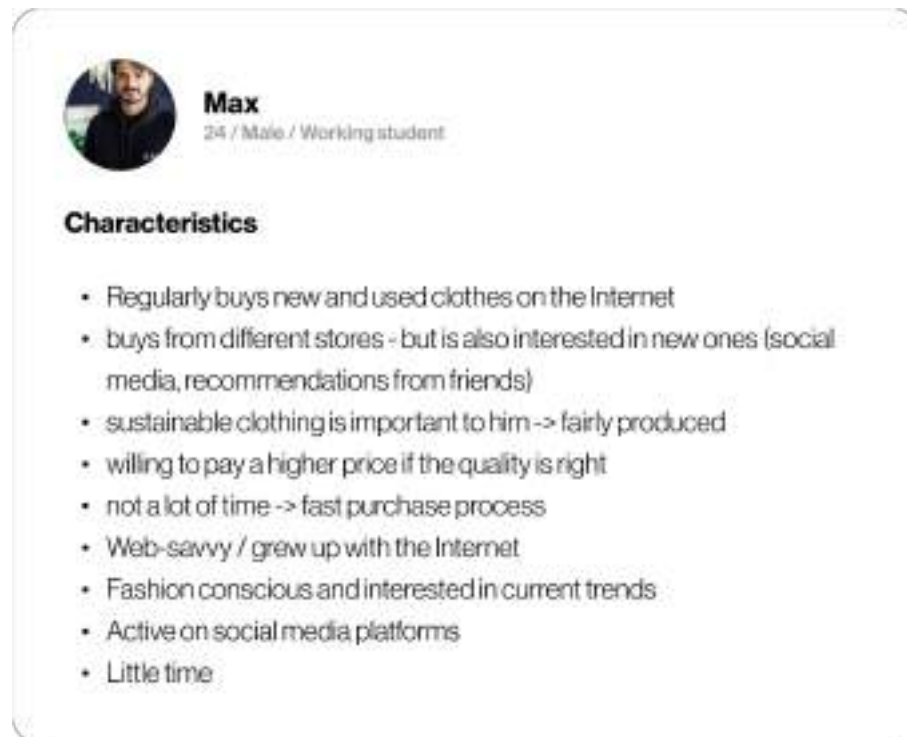


Figure 5: Screenshot of the end result of the persona Max.

Name of the Brand

In the next step, the focus shifts towards establishing the brand identity, with the initial step involving selecting a suitable brand name. The brainstorming method is employed to facilitate this process, leading to a substantial number of potential brand names being generated.

According to Aurora Harley, a Senior User Experience Specialist with Nielsen Norman Group, "brainstorming is one type of ideation, which has been formalized as a process by Alex Osborn in his book *Applied Imagination*. Osborn's guidelines for brainstorming — go for quantity, there are no bad ideas, crazy is welcomed, and encouraging building off ideas — also work well for many other ideation techniques" [56].

After applying this ideation method, many different ideas came to mind. There were seven different names for the brand; however, in the end, one was selected: 'BlueBird'.

- EthicalElephant
- MindfulMonkey
- HappyElephant
- EarthlyElegance
- BlueNinja
- EcoVibe
- **BlueBird** (selected)

Typography

Once the name of the brand is decided, it is crucial to create brand elements that are as realistic as possible. A fundamental building block of these brand elements is the typography. The typography of a brand "represents the tone and values of your brand just like colour represents a feeling or visually represents a message for your customers" [57].

The typeface Neue Haas Grotesk is a sans-serif font whose initial designs were created by Max Miedinger from 1957 to 1958. Over the years, it gained significant attention and was eventually introduced by Linotype AG as 'Helvetica'. With the increasing use of display devices, font designer Christian Schwartz decided to redesign the typeface and adapt it for screens. The result is the Neue Haas Grotesk Display Pro typeface used in this work [58]. It is characterized by its distinctive, clear design and good readability, especially on displays, making it a suitable choice for the brand design of 'BlueBird' (Figure 6).

Since the created persona Max spends much time on the internet (Figure 5), a web-readable font perfectly fits the target audience. The font's timeless and modern design also aligns with Max's interest in new trends.



Figure 6: A showcase of the used typeface Neue Haas Grotesk Display in the styles Bold and Roman.

Colors

According to M. Chandorkar, colors play a significant role in the design and impact of a brand. Branding communicates meaning, but it only becomes a unified and powerful identity when accompanied by colors [59]. Therefore, establishing 'BlueBird's' colors is essential, especially for the design of the later online shop. Since the brand name 'BlueBird' was chosen, and one does not want to confuse a potential customer, it was also crucial that the color "blue" appears somewhere in the brand design.

In the case of the example brand 'BlueBird', it was decided to use #023E8A as the primary color (Figure 7), which is a medium-dark shade of cyan blue. In order to have a secondary color that can also be used when the darker shade of blue is not suitable and to have even more design flexibility, the color #71A9F7 was introduced (Figure 7). This color code is a medium-light shade of cyan blue. In addition to these two colors, it is also important to define basic colors that can be

used, especially in the web and product design. Here, the decision was made to use a standard white (#FFFFFF) and the black shade #020300 (Figure 7).



Figure 7: Overview over the four colors used in the design process.

Logo

The next step in the brand design process was to design a logo for 'BlueBird'. A logo is crucial for a good brand design because, according to Alexander Westgarth, "the right logo says everything without saying a word. [...] It evokes a sense of connection between a brand and consumers. It establishes a bond between a company and its community of fans, friends, critics, allies and champions" [60]. Therefore, the design of the brand logo is one of the most critical steps in the design process.

Before embarking on the design of a logo, it is essential to know that there are numerous types of logos to choose from. Here, a distinction is made between lettermarks, wordmarks, pictorial marks, abstract logo marks, mascots, the combination mark, and the emblem [61] [62].

A lettermark is a typography-based logo comprising a few letters, typically the firm's initials. The lettermark emphasizes simplicity over everything. If a firm has a long name, lettermark logos can effectively shorten it by using only a few letters [61] [62]. A perfect example of a lettermark logo is the logo of IBM (Figure 8a), which stands for 'International Business Machines Corporation'.

A wordmark logo is a design primarily focused on the company's name. This type of logo works especially effectively for companies with brief and easy-to-remember names. The Google logo is an excellent example of this concept (Figure 8b). This logo significantly enhances the brand's identity and recognition when combined with impactful typography and a unique, memorable name. [61] [62].

A pictorial mark is a design that relies on icons or graphics. An excellent illustration of a pictorial mark is the Twitter bird utilized by Twitter until 2023 (depicted in Figure 8c), the Target bullseye, or the Apple logo. These logos from various organizations have reached such iconic status and brand familiarity that the symbol evokes instant recognition. Consequently, employing this type of logo can be demanding for startups or enterprises with limited brand visibility. [61] [62].

One particular kind of pictorial logo is an abstract mark. It reflects the company but is not a familiar image like an apple or a bird; instead, it is an abstract geometric shape. Some well-known

examples are the Pepsi divided circle (Figure 8d), the Adidas flower, and the BP starburst logo. Like other logo symbols, abstract marks effectively encapsulate a brand in a single visual. However, abstract logos provide the freedom to develop something entirely original to represent the company rather than being limited to a picture of a recognizable object [61] [62].

A mascot is merely a fictional character representing your firm in an illustration; consider them to be the face of the company. The Colonel from KFC (Figure 8e) and Mr. Peanut from Planters are all well-known mascots. Mascots are perfect for businesses that wish to foster a family-friendly environment by appealing to young people and families [61] [62].

A combination mark is a logo that combines a wordmark or lettermark with a pictorial mark, abstract mark, or mascot. In order to produce an image, the text and image can be combined, layered on top of one another, or set out side by side. Doritos, Burger King, or Lacoste (Figure 8f) are a few brands with well-known combination mark logos [61] [62].

The final logotype is the emblem. Consider badges, seals, and crests as examples of emblem logos containing text inside an image or symbol. These logos frequently have a classic look that can leave a lasting impression, which is why many businesses, organizations, and governmental bodies favor them. Emblem logos are also quite popular in the automotive business. Even though they have a classic look, some businesses have successfully updated the conventional emblem style with logo designs appropriate for the twenty-first century. Two perfect examples of emblem logos are Starbucks' recognizable mermaid logo (Figure 8g) or Harley-Davidson's renowned crest [61] [62].



(a) Lettermark logo - Brand IBM. Source: [61].



(b) Wordmark logo - Brand Google. Source: [61].



(c) Pictorial mark logo - Brand Twitter. Source: [61].



(d) Abstract logo mark - Brand Pepsi. Source: [61].



(e) Mascot logo - Brand KFC. Source: [61].



(f) Combination mark - Brand Lacoste. Source: [61].



(g) Emblem Logo - Brand Starbucks. Source: [61].

Figure 8: Overview of the different logo types.

For this thesis, a combination mark was chosen for the logo design. The word mark 'BlueBird' was created using the font 'Neue Haas Grotesk Display Pro' in bold style. Alongside the word mark, a pictorial mark of a bird was crafted. This bird icon was placed within a colored circle and next to the word mark. For the final touch, the circle and the wordmark were colored with the primary color, while the bird remained white (Figure 9a).

This type of logo was chosen for the brand 'BlueBird' because combination marks are very versatile in their application. The lettermark can be used independently as a text element, while the included pictorial mark can stand alone. Together, both elements create the complete logo, which can be used in various ways depending on the brand's use cases (Figure 9b). The created persona can also justify this type of logo, as the persona Max is active online and purchases

clothing. These use cases are distinct, and the brand requires a suitable, flexible logo to cater to both scenarios.

It can be noted that the color blue is not necessary. The logo can also be used in white on a dark background (Figure 9c), but also in black on a light background (Figure 9d). This provides the necessary flexibility that a brand, especially in the online and retail sectors, requires to function effectively, whether it is on printed packaging boxes or online platforms.

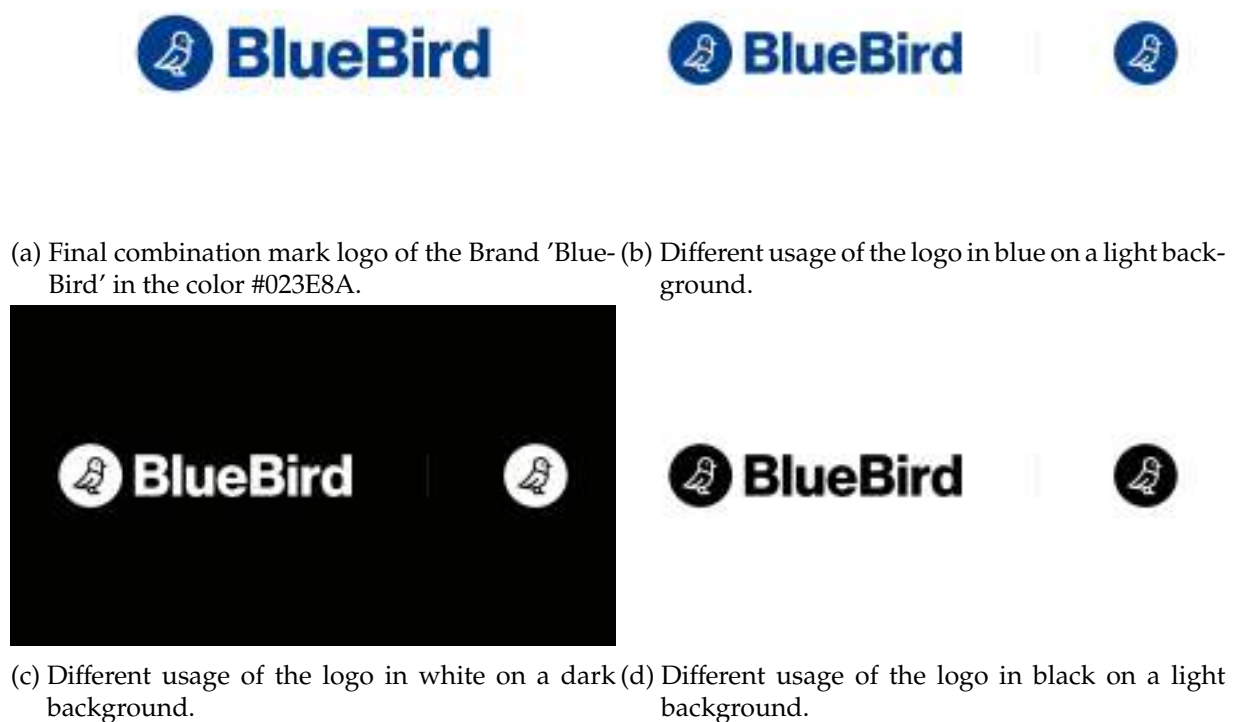


Figure 9: Different usages of the final logo of the brand 'BlueBird'.

Creating the Products

In order to make the future online shop look as realistic as possible and to provide visual material, four sample products were created. These products are each in their price category, representing many products. All sample images for the presentation of the products are created with the online platform 'PlaceIt'.

PlaceIt is a platform on which visual content can be created with templates without needing any technical or artistic skills [63]. In this way, designs created by the user can be converted into mock-ups or product photos using a drag-and-drop principle [64]. PlaceIt can also be used for commercial purposes but costs a monthly fee of \$9.99 (currently approx. 9,25€) if billed annually [65].

The first product is a t-shirt, which is available in white (Figure 10), black and blue. The price is set at 30€. The sizes that are being offered are S, M, L, and XL.



Figure 10: Overview of the three images that have been created with PlaceIt for the product 'T-Shirt' in the variant white.

The second product is a hoodie, available in grey (Figure 11) and black. The hoodie cost 49€, making it the most expensive product in BlueBird's shop. The sizes that are being offered are S, M, L, and XL.



Figure 11: Overview of the three images that have been created with PlaceIt for the product 'Hoodie' in the variant gray.

The third product is a cap, available in black (Figure 12), white, and blue colors, and offered in a single size ("one size fits all"). The cap is priced at 25€.



Figure 12: Overview of the three images that have been created with PlaceIt for the product 'Cap' in the variant black.

The fourth and final product offered is a tote bag, available in a beige color and priced at 15€ (Figure 13).



Figure 13: Overview of the three images that have been created with PlaceIt for the product 'Bag'.

All these created products align perfectly with the developed persona, Max. Firstly, there is the t-shirt, which at 30€ is the most appealing clothing item in the shop. Additionally, the hoodie offers a slightly higher-priced option, considering Max's willingness to pay for quality. Furthermore, it is important to anticipate that Max would be interested in clothing and accessories. For this purpose, the online shop offers the bag and cap, catering to Max's preferences. In conclusion, the shop covers four different price ranges, each featuring a product that fits the brand design of 'BlueBird' and is tailored to Max's persona.

3.1.2 Sitemap

According to Caddick & Cable, "a sitemap is an important document for tying all project documents together. It shows the structure of the site and how each page will be linked together

during development. It acts as a To Do list to show what wireframes need to be produced. It can be used to map out user journeys and to form the basis of content audits" [66, p.124].

A well-defined sitemap is fundamental to developing a functional and intuitive online shop. A sitemap acts as a blueprint, showcasing the hierarchical structure of web pages and facilitating navigation.

In order to accomplish this, a preliminary outline was generated in Figma, providing a visual representation of the critical web pages that should be included within the online store. The created sitemap consists of two main components. Firstly, there are the pages highlighted in blue, which are central to the following design process. This includes the Homepage, Product Pages (Product Overview, Product Detail), as well as the Shopping Cart, and the associated Checkout Process (Figure 14). Secondly, the pages highlighted in red are less significant. However, they are often part of a typical online shop structure and have therefore been included in the sitemap, for example, the About Us Page or the Legal Pages (Figure 14).

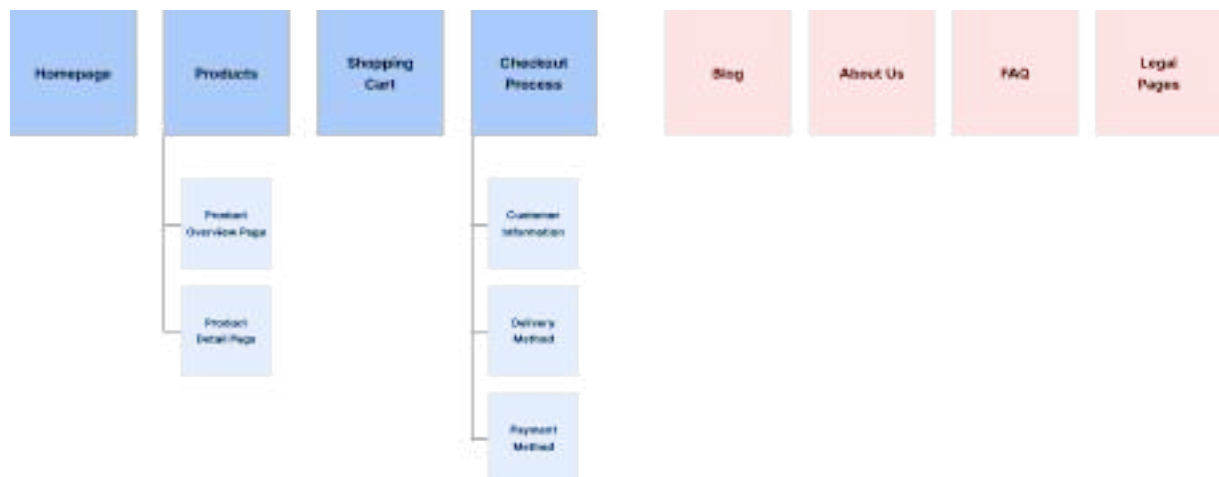


Figure 14: Sitemap of the online shop created in Figma.

3.1.3 Designing the Sites

This work's next step is to combine all the elements developed in the preceding steps and create an online shop with the pages defined in Chapter 3.1.2. For this purpose, the process began by creating essential elements and designing each page. The design tool utilized for this phase is also Figma. In order to maintain the design process focus on the online shop and the e-commerce aspect, the development of the shop centers around designing the pertinent pages. These pages encompass the Homepage, the Product Overview Page, the Product Detail Page, the Shopping Cart, and the Checkout Process. The design of these pages will be further elaborated in the upcoming chapter.

Basic Elements

In the initial phase, it is important to design elements that could appear on each of the following pages. The focus is primarily on these three elements:

- A banner for displaying potential discount promotions
- A navigation bar including a search function for finding products

- A footer

Banner: Since the created persona, Max, has limited time and moves quickly on the internet, it is necessary to create a banner. This is essential to quickly allow Max to access the current offers or discounts of the online shop. For the banner, a potential discount promotion must be easily recognizable to the user, allowing them to directly access the desired promotion and providing the option to close the banner. The primary color of the brand design is being used, along with the color white. Additionally, the selected typeface Neue Haas Grotesk Display Pro is employed in both Bold and Regular weights (Figure 15).



Figure 15: Banner element created in the tool Figma.

Navigation Bar: The navigation bar should also be kept simple, as the brand 'BlueBird' wants to keep their average user, Max, distinct from unnecessary navigation items. The goal is to provide Max with a quick overview of the shop. A typical navigation bar allows users to return to the Homepage via the website logo and navigate to the site's most important pages. Furthermore, users of an online shop should be able to search for products and access their shopping cart. Ideally, users can ascertain the number of items in their cart without clicking the Shopping Cart.

In the context of this thesis, the logo created in Chapter 3.1.1 is naturally placed in the left top corner. The pages 'All products', 'Blog', 'About', and 'Contact Us' are integrated into the navigation bar as the most critical pages. On the right side of the bar, a search option and the shopping cart display the quantity of products already included. The logo, the typeface Neue Haas Grotesk Display Pro in Bold and Regular weights, and the Primary and Black colors are used. The icons visible used here are freely available in the Phosphor Icon Library [67] (Figure 16).



Figure 16: Navigation bar element created in the tool Figma.

Footer: The final general element that is designed is the footer. The footer appears at the bottom of every website page and provides links to selected online shop pages, as well as legal pages such as the privacy policy. In this case, any available social media channels can also be linked. The footer of 'BlueBird' has a clean design. It includes the logo, homepage links, and critical page navigation links (All products, Blog, About, Contact us). Additionally, users can access the privacy policy, terms and conditions, and the imprint. Finally, links to the brand's social media channels, which are not created in this context, are also present. The elements used here include the logo, the typeface Neue Haas Grotesk Display Pro, and social media icons from the respective platforms (Figure 17).



Figure 17: Footer element created in the tool Figma.

Home Page

One of the most critical pages of an online shop is the Homepage, as potential users have their initial encounter with the online shop here. As discussed in the preceding part of this work, this page also consists of several distinct individual elements, which are detailed below:

- Banner (discussed in Chapter 3.1.3)
- Navigation bar (discussed in Chapter 3.1.3)
- Hero section
- Product slider
- Newsletter signup
- Blog article slider
- Footer (discussed in Chapter 3.1.3)

Hero Section: The hero section of a website is the first area users see upon entering the page. Therefore, it is crucial to establish a highly appealing appearance to minimize the bounce rate. This study created an aesthetically pleasing background image using the PlaceIt tool (Chapter 3.1.1), along with a call-to-action allowing users to subscribe to the offered newsletter. The option to sign up for a newsletter through an input field was integrated into the design, as the persona Max is interested in staying updated on the latest trends and can be informed about them through the newsletter. Additionally, a subtle gradient overlay was applied to the image to enhance the readability of the displayed text (Figure 18).

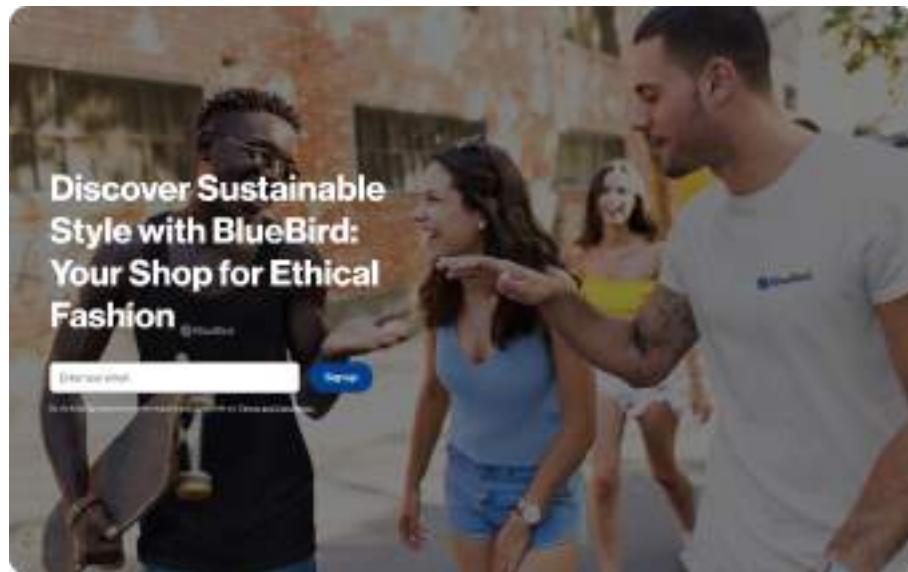


Figure 18: Hero section element created in the tool Figma.

Product Slider: A product slider appears when the user starts scrolling to provide the target audience, especially the created persona Max, with quick access to the offered products. In this case, since four products were created, they can be displayed within this slider. Key considerations included providing an option to jump to the product overview page (blue button at the top right) and displaying the product name, price, and available colors for each respective product. Furthermore, clicking on a product must lead to its detailed product page (Figure 19).

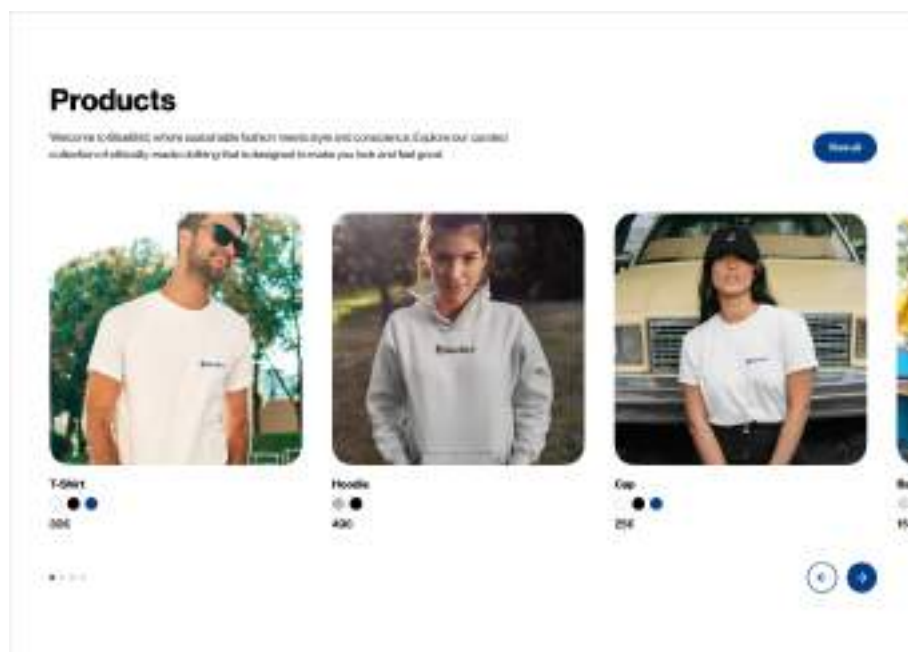


Figure 19: Product slider element created in the tool Figma.

Newsletter Signup Element: The newsletter signup element can offer users an additional means of staying connected with the online shop. Here, users can register with their e-mail addresses to receive regular newsletters. This element is also essential for the typical user, Max. He wants to stay informed about the latest trends and offers from the shop and, therefore, needs a way to subscribe to a newsletter. To add a touch of visual appeal to the minimalist design, a slightly lighter shade of the primary color was used, with the primary color employed as the text color. An input element was also integrated, allowing users to submit their information by clicking the button (Figure 20).

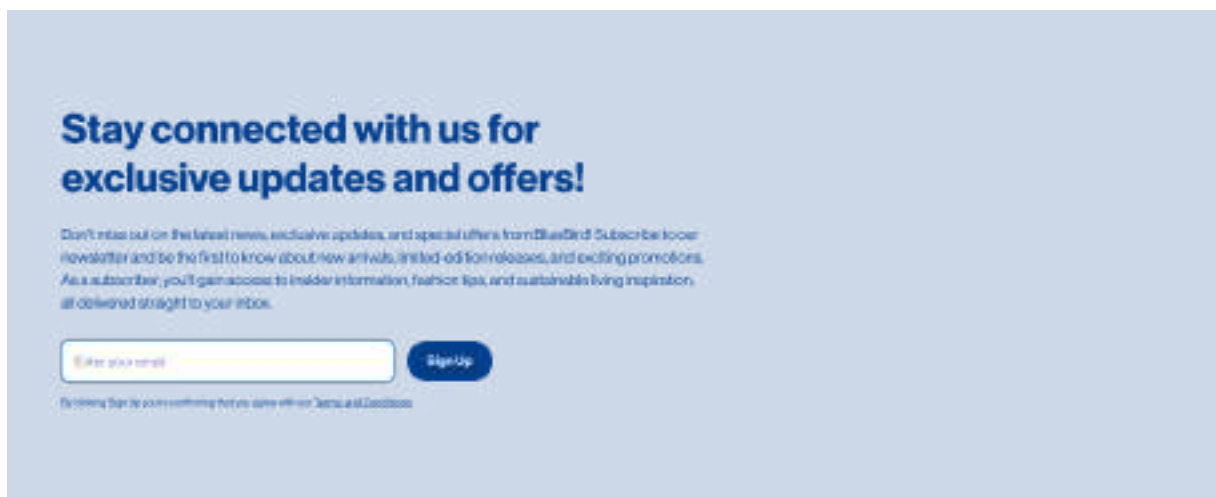


Figure 20: Newsletter signup element created in the tool Figma.

Blog Article Slider Element: Since users are encouraged to purchase products on the website and engage with the brand through other content, a blog was integrated into the shop, as mentioned in Chapter 3.1.2. In this section, employees of the brand can contribute articles on various topics of interest. Similar to the product slider, it is imperative to provide a link to the blog article overview page (top-right button) and display an image, title, and concise blog post description. Furthermore, the author's image, name, publication date, and estimated reading time can be displayed. Naturally, clicking on the desired blog article should lead to the detailed blog article page (Figure 21).

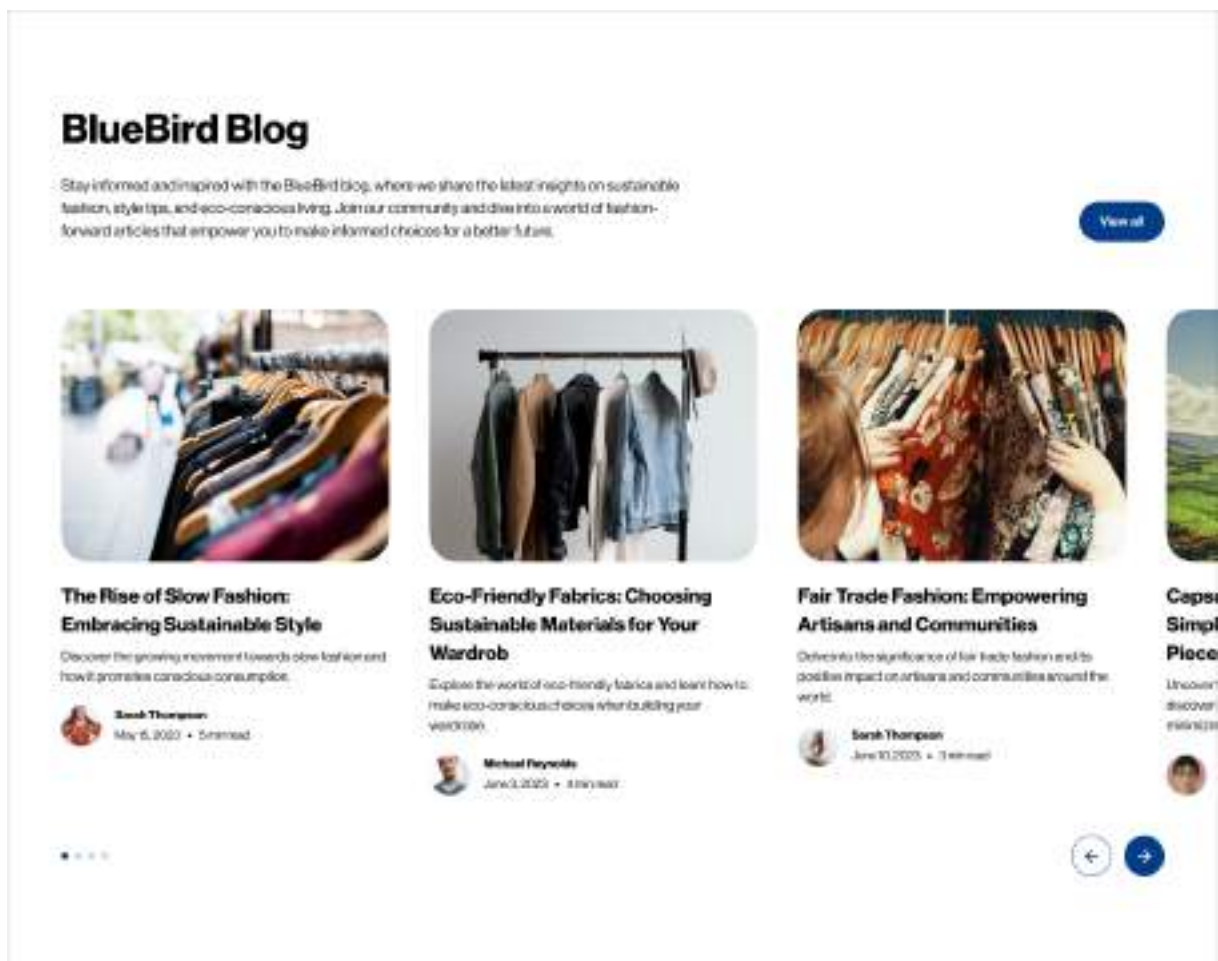


Figure 21: Blog article slider element created in the tool Figma.

Product Overview Page

Having a product overview page is essential to gain an overview of all available products within an online shop. The critical elements of this page are:

- Navigation bar (discussed in Chapter 3.1.3)
- Product overview including filtering options, sorting capabilities and an Add-to-Cart shortcut button
- Footer (discussed in Chapter 3.1.3)

Product Overview Element: In the product overview, it is crucial to display all the offered products of the online shop and allow users to filter the displayed products according to their preferences. For the 'BlueBird' overview page, the decision was made to offer a keyword search, size filtering, color filtering, and price filtering, along with sorting options based on price. Furthermore, the products created in Chapter 3.1.1 are presented with images, product names, prices, color variants, and the Add-to-Cart shortcut button (Figure 22).

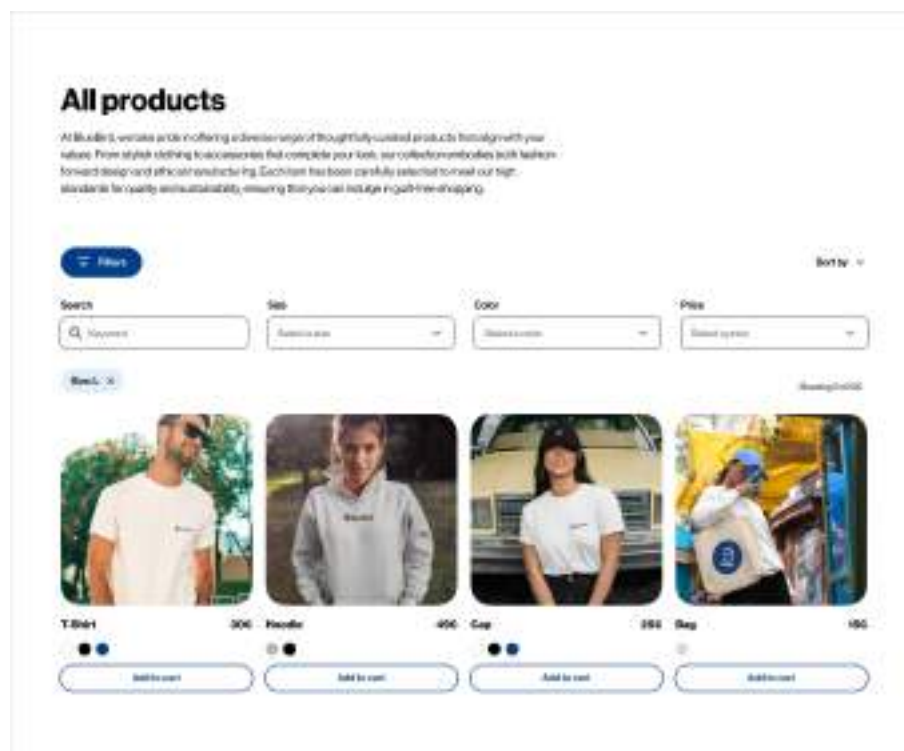


Figure 22: Product overview element created in the tool Figma.

Product Detail Page

Another essential page that must not be overlooked in the design process is the Product Detail Page, which consists of the following elements:

- Navigation bar (discussed in Chapter 3.1.3)
- Product details including breadcrumb, image display, variant selection, quantity selection, and Add to Cart button
- Footer (discussed in Chapter 3.1.3)

Product Details Element: Presenting the product detail page requires attention to several aspects. Firstly, users should be allowed to navigate back to the product overview page using a breadcrumb element (Figure 23, top left). Additionally, all relevant product information must be presented to users, along with a means to add the product to their cart. In this case, provided product images are displayed on the left side. The product name, price, product description, variant, and quantity selection with the Add-to-Cart button are located on the right side of the design. Moreover, an accordion-style display on the right side contains additional information about the selected product, such as shipping and return details (Figure 23).

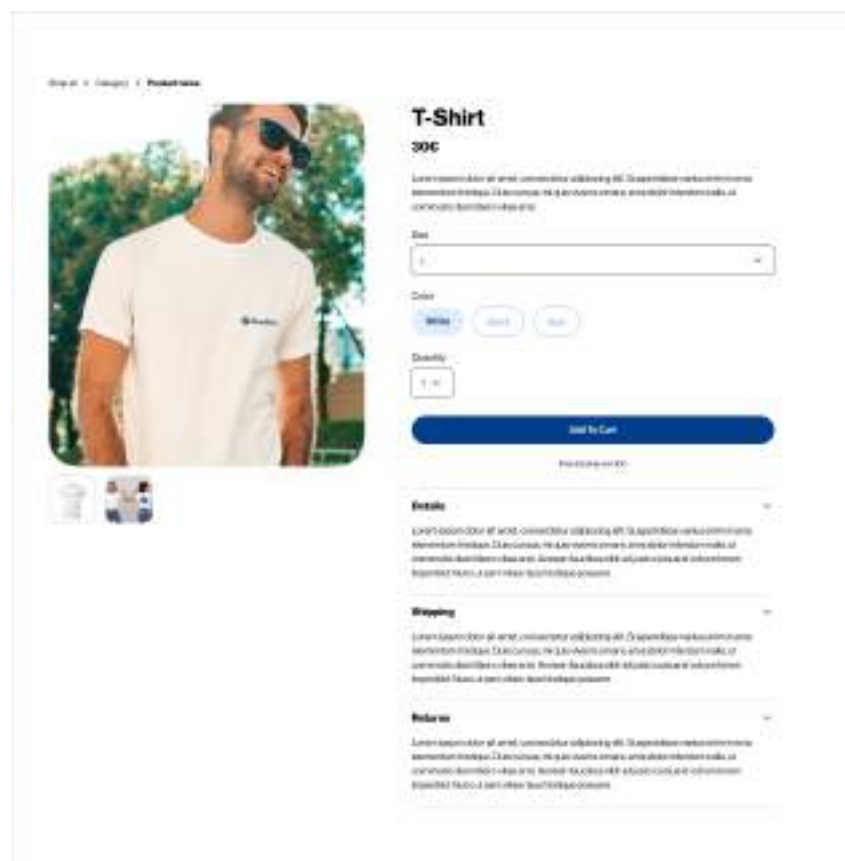


Figure 23: Product detail element created in the tool Figma.

Shopping Cart

With the Product Detail Page designed, the next step was to focus on the Checkout Process. This started with the Shopping Cart, which contains the products the customer has added and directs them to the checkout process. This page includes the following elements:

- Navigation bar (discussed in Chapter 3.1.3)
- Shopping cart element (Products, Price, Checkout Button)
- Footer (discussed in Chapter 3.1.3)

Shopping Cart Element: The shopping cart should give the user an option to return to the product, which is ensured through the link under the heading. Additionally, the user can see their current checkout step through the breadcrumbs in the upper left corner. However, the most critical aspect of the shopping cart design is the presentation of the products the user has added, along with the price, size, quantity, and an option to remove the items. Finally, a total price needs to be provided, along with a button that guides the user through the checkout process (Figure 24).

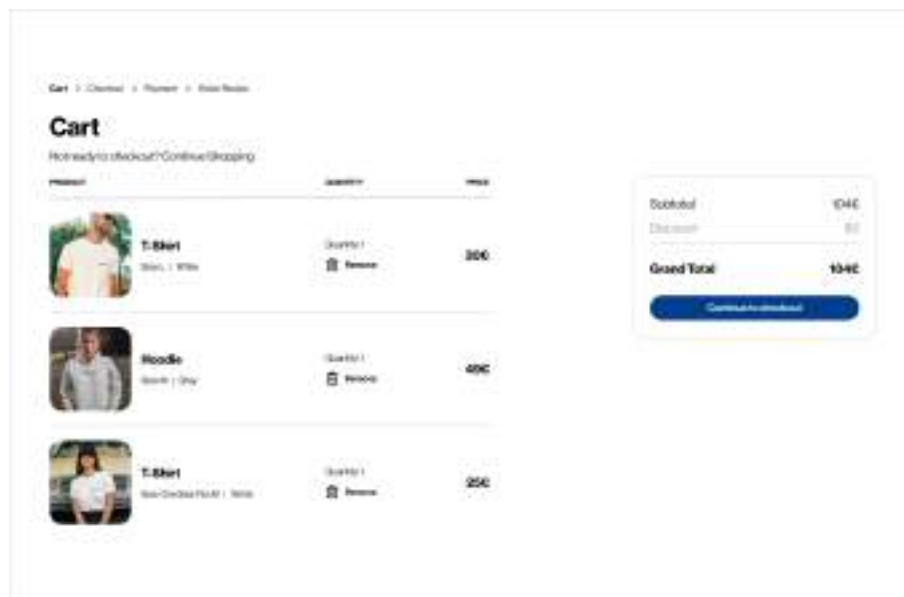


Figure 24: Shopping cart element created in the tool Figma.

Checkout Process

The last pages designed for the online shop of 'BlueBird' are the Checkout Pages. These consist of the Checkout (entering buyer information), Payment (entering payment details), and Order Review Pages. The checkout process design is also kept simple and straightforward to save time for the persona Max during the checkout process.

Checkout Element: When entering buyer information, it is essential to ensure that all important details such as name, e-mail address, and shipping address are entered and that the shipping method can be selected. Furthermore, an option should always be on the right side to provide an overview of the shopping cart, the total price, and the shipping costs. Additionally, there must be an option to enter a discount code, and a button should be present to proceed to the next step in the checkout process (Figure 25).

Figure 25: Checkout element created in the tool Figma.

Payment Element: When entering payment information, it is essential to ensure that locally preferred payment methods can be selected while also providing an overview of the shopping cart, total price, and shipping costs. Additionally, there should be an option to enter a discount code, and a button must be available to proceed to the next and final step in the checkout process (Figure 26).

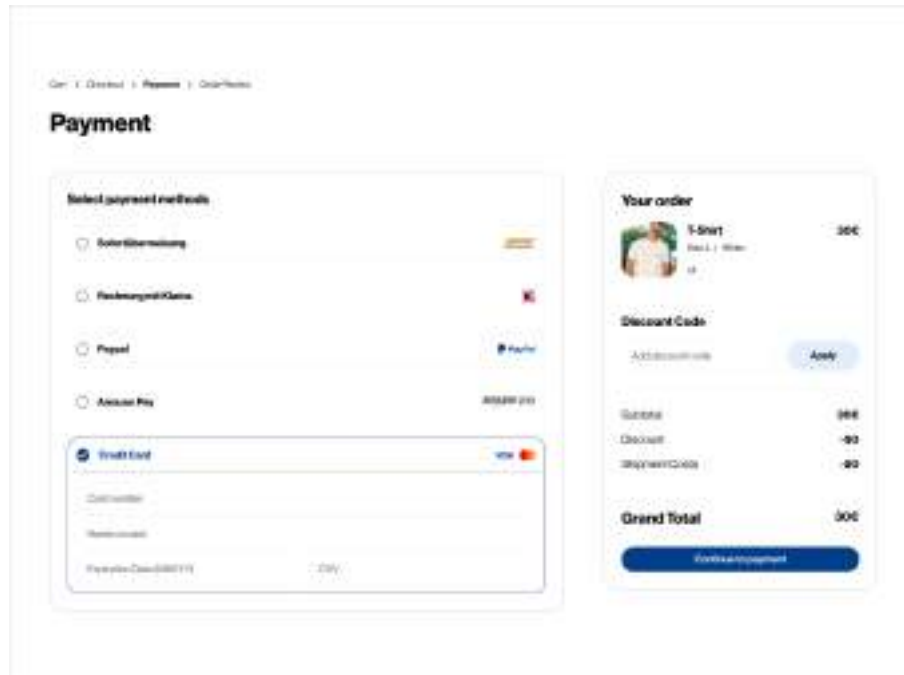


Figure 26: Payment element created in the tool Figma.

Order Review Element: The final step in the checkout process is the Order Review. Here, it should be ensured that the information entered in the previous steps can still be edited. This includes the buyer's information, billing address, delivery address, and payment method. An overview of the shopping cart, total price, and shipping costs should be visible, too. Additionally, there should be an option to enter a discount code, and a button must be available to complete the user's purchase. (Figure 27).

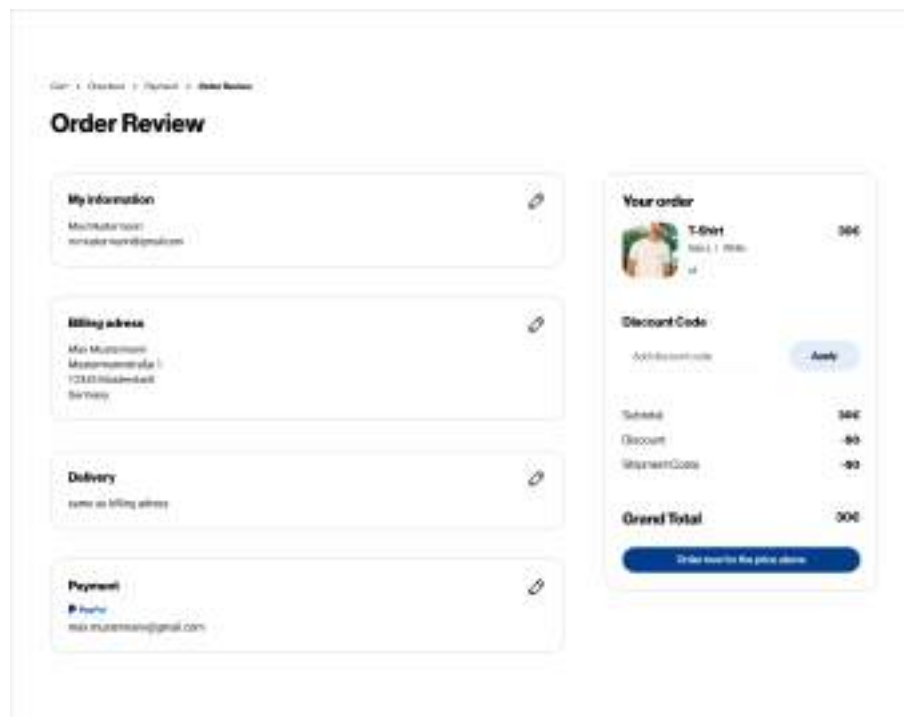


Figure 27: Order review element created in the tool Figma.

3.1.4 Writing Example Texts with ChatGPT

Since the online shop is created as a representative example of a brand, it is necessary to write various texts for the website. This includes headings, description texts, and blog articles and their authors to be used as texts for the later development process. This approach aims to make the shop appear as realistic as possible, avoiding using placeholder texts, typically Latin-based Lorem Ipsum texts.

To expedite the process of generating these texts without the need to write them manually, a highly efficient method is to leverage a text-based AI model like ChatGPT.

ChatGPT is an AI-powered large language model (LLM) that was released in November 2022. It has been trained on massive datasets of text in multiple languages and is capable of generating text responses that closely resemble human-like language [68]. Developed by OpenAI (OpenAI, L.L.C., San Francisco, CA, USA), ChatGPT is based on the generative pre-trained transformer (GPT) architecture and is connected to a chatbot. This chatbot is capable of interpreting and generating responses through a text-based interface [68, 69].

To obtain texts for the sample shop, an input was provided to the AI chatbot with the following content:

"I created this persona: Name: Max Age: Early 20s Occupation: Working student Shopping habits: Regularly buys new and used clothes online, looks for sustainable clothing, values fair manufacturing, prefers a quick buying process Online behavior: Web-savvy, grew up with the internet, buys from various online shops, open to new recommendations from friends or social media Purchase motivation: Quality over price, willing to pay a higher price if the quality is right Special requirements: Transparency in size presentation on the website is important to minimize returns.

I created a new brand called 'BlueBird' with separate apparel buying options. Write a Headline 1 for the online shop of 'BlueBird'."

The output generated by ChatGPT was (Figure 28):

"Discover Sustainable Style with BlueBird: Your One-Stop Shop for Ethical Fashion."



Figure 28: Screenshot of the usage of ChatGPT for writing example texts

Upon careful review, a minor error was identified in the previous content generated by ChatGPT, specifically the inclusion of the phrase 'one-stop'. Due to its length and incongruity with the overall context, this portion was subsequently removed. However, the remaining content was repurposed as a Heading 1 on the homepage of the 'BlueBird' website.

This method is used for all the text content visible on the online shop of the brand 'BlueBird'.

3.2 Expert Review and Incorporating the Design Feedback

After the detailed first high-fidelity prototype has been designed and conceptualized as described in the previous section, it is also essential to incorporate the opinion of an e-commerce expert into the design.

A senior UX/UI designer focused on e-commerce was found at the digital agency TEAM23 in Augsburg, who agreed to do an expert review of the first version of the high-fidelity prototype.

An expert review "usually expand[s] on heuristic evaluations by assessing the design not only for compliance with heuristics, but also against other known usability guidelines, principles of usability-related fields such as cognitive psychology and human-computer interaction, and

the reviewer's expertise and past experience in the field. The emphasis on the reviewer's past experience and knowledge of usability principles is why this type of design review is often referred to as an expert review" [70]. The main outcomes after conducting such expert reviews are a list of usability strengths, a list of usability problems and recommendations for fixing each usability problem [70].

After applying this method in the review process, the outcomes were reviewed and prioritized. In the next step, some aspects were incorporated into the final design, which are now described in the following section.

On one hand, the navigation bar in version 1 was still far too dominant in the checkout process and distracted the user from what was necessary (Figure 29).



Figure 29: Navigation bar in the checkout process - Version 1

A suggested improvement to this problem was to reduce the navigation to the essentials, namely the brand logo (BlueBird) and a link back to the shop (Figure 30).



Figure 30: Navigation bar in the checkout process - Version 2

Another factor that came up in the discussion was that the font's legibility, i.e., the accessibility of the checkout, is not given in the design of version 1. A shade of grey was used to design the input fields that were too light and could not be read by a person with impaired vision who accessed the online shop. Furthermore, the border radius of the input fields was the same as that of the buttons, which could confuse the user (Figure 31).



Figure 31: Input fields in the checkout process - Version 1

A simple but efficient solution to this problem was to increase the contrast, i.e., the opacity of the color. This step was quick and easy, as some grey tones had already been created in the design in Figma, so the process only took a few clicks. Also, to provide more clarity between the buttons and the input fields, the border radius of the input field was significantly reduced, creating a clear distinction from the buttons (Figure 32).



Figure 32: Input fields in the checkout process - Version 2

On the other hand, some aspects were not even considered in the first design. For example, the design of version 1 did not yet include a confirmation page that summarises all the essential information for the user after placing the order.

For several reasons, the order confirmation page is significant in web design:

- Customer pleasure: By confirming the specifics of their purchase and delivering a receipt or invoice, the order confirmation page assists in giving clients a sense of completeness and satisfaction [71].
- Purchase tracking information, such as a tracking number or estimated delivery date, may also be made available to clients on the order confirmation page [71].
- Opportunities for upselling and cross-selling related products may be found on the order confirmation page by displaying additional products that the consumer might find interesting [71].
- Customer engagement: By including buttons for social networking, a newsletter sign-up form, or other promotional offers, the order confirmation page may also be utilized to draw in customers [71].
- Brand reaffirmation: By displaying a recognized and consistent design that fits the rest of your website, the purchase confirmation page gives you the chance to reaffirm your brand identity [71].

As a result, a confirmation page was created that contains the following important components in the design of the version (Figure 33):

- a personal address of the customer
- a notice that an e-mail has been sent
- an order number
- the time at which the order was placed
- the option to print out the order confirmation
- the delivery address
- the billing address and method
- the shipping method
- a summary of the products ordered
- the summary of the costs



Figure 33: Final design of the order confirmation page.

The most important point raised by the e-commerce expert was the clear structuring and display for the user in which process step of the checkout he is now. This was considered in version 1 of the design but was lost in many texts (Figure 34).



Figure 34: Visibility of system status - Version 1

According to heuristic 1 'Visibility of system status' of Jakob Nielsen's 10 Usability Heuristics for User Interface Design, "the design should always keep users informed about what is going on, through appropriate feedback within a reasonable amount of time. When users know the current system status, they learn the outcome of their prior interactions and determine next steps. Predictable interactions create trust in the product as well as the brand" [72]. To achieve this, the process step of the checkout was incorporated in a more transparent and obvious way in the design of version 2 (Figure 35).



Figure 35: Visibility of system status - Version 2

After the adjustments suggested by the e-commerce expert were integrated, the result is the final design of the online store of the brand 'BlueBird' (Figure 97, Figure 98, Figure 99, Figure 100, Figure 101, Figure 102, Figure 103, Figure 104). This final design is now used in the next step as the basis for the implementation in Webflow and Shopify.

4 Practical Implementation of the Example Design in Webflow and Shopify

This chapter provides a detailed account of the practical implementation process for creating the online shop designed in Chapter 3 on the two distinct platforms, Webflow and Shopify.

4.1 Implementation in Webflow

This section itself presents the implementation process in Webflow. It starts with providing general information about the development process. In the next step, the base for the implementation process in Webflow is introduced: the Relume Webflow Library. The section then proceeds to cover the e-commerce checklist, managing the shop, instructions on using the Webflow Designer, utilizing components in Webflow, and finally, implementing the separate pages.

4.1.1 General Information about the Development Process in Webflow

With the shop design already finished in Figma, the next step is to initiate the development process in Webflow. To do this, the user must have a Webflow account. The Google login feature is utilized for this project, enabling users to sign in to Webflow using their Google account.

After registration, a well-thought-out onboarding process follows in Webflow, which asks the new user several questions about the reasons for using the tool (Figure 36). Additionally, a tutorial-like process unfolds, gradually explaining the Webflow GUI to the user and revealing more features step by step. This kind of tutorial is valuable, as it gradually provides the user with bite-sized learning and understanding opportunities, preventing immediate overwhelm (Figure 37).

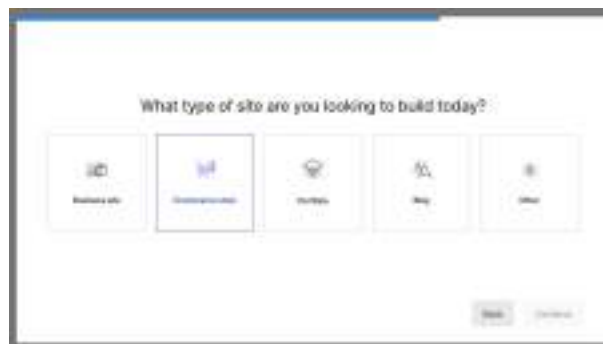


Figure 36: Screenshot of the onboarding questions for first time users in Webflow.



Figure 37: Screenshot of the tutorial for first time users in Webflow.

Furthermore, the editing area of the Webflow page is divided into three major sections: *Site Settings*, *Designer*, and *Editor*.

In the *Site Settings*, adjustments to the website can be made, such as the site's name, the appearance of the favicon, and even the number of team members allowed to work on the site. Additionally, the domain on which the website will be published later can be set under the Publishing Settings. If necessary, settings for SEO, website forms, external fonts, integrations, and custom code can also be added here (Figure 38).

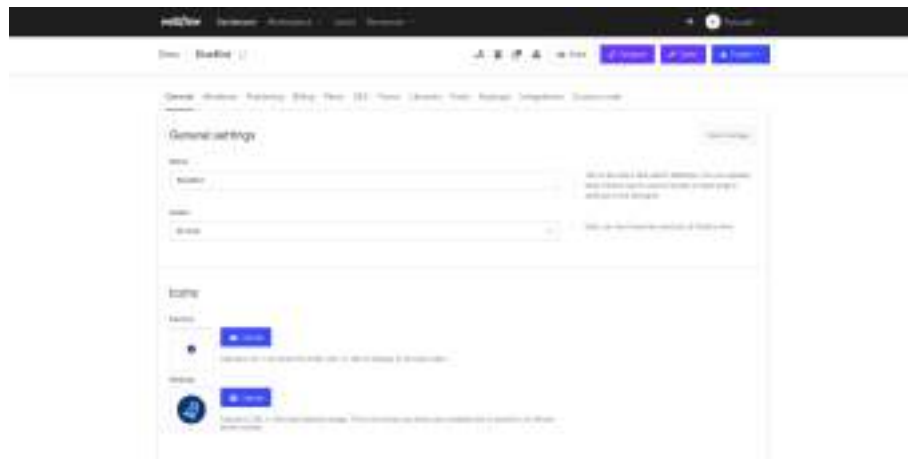


Figure 38: Screenshot of the Site Settings in Webflow.

The *Website Designer* is the core of Webflow, where all sorts of adjustments to the website can be made. Design, layout, and content adjustments are possible here. Product management and adapting the website to different breakpoints are also done here. One of the most crucial features is, eventually, publishing the website (Figure 39).



Figure 39: Screenshot of the Webflow Designer in Webflow.

The *Webflow Editor* is a streamlined version of the Website Designer. Content adjustments like text or image changes can be carried out here. It also provides an overview of orders received in the shop, and the respective products can be managed from there. This view is perfect for users responsible for website content maintenance who are not involved in design and layout modifications (Figure 40).



Figure 40: Screenshot of the Webflow Editor in Webflow.

For the development of the online shop in Webflow, the 'Ecommerce Standard' package was chosen, which is billed on a monthly basis and costs \$42 per month.

4.1.2 Introducing the Relume Webflow Library

In order to simplify the implementation process in Webflow considerably, various tools can be used. On one hand, it is possible to use so-called prefabricated templates from Webflow or templates that community members have created. On the other hand, there is the possibility of using so-called 'libraries' presented by Webflow in 2023 and created by members of the Webflow community. In the case of this thesis, the so-called Relume Webflow Library is used (Figure 41), which makes the most sense for this use case after considering the final design.



Figure 41: Screenshot of the thumbnail of the Relume Webflow Library. Source: [73].

The free version of the Relume Webflow Library allows users to initiate Webflow projects with unstyled components. It offers 50 free Webflow components that everyone can use to build a website or an online shop within minutes [74].

4.1.3 E-commerce Checklist by Webflow

For setting up a shop with Webflow, this platform also offers a very well-prepared onboarding process for setting up a store. Here, the user can quickly work through a checklist from Webflow itself, which contains all the essential steps to develop a complete and functioning shop in Webflow.

These consist of:

- Set up your store
 - Add a business adress
 - Review currency settings
 - Add a product
 - Add a shipping method
 - Review tax settings
- Design your store
 - Add and design your cart
 - Design the product page
 - Design the checkout page
 - Design e-commerce e-mails
- Launch your store
 - Add payment provider
 - Add hosting
 - Enable SSL
 - Enable checkout

This checklist is used for the development process of the online shop to include all the crucial steps in the creation and not to forget anything.

4.1.4 Managing the Shop

Before commencing the development of the shop, configuring basic shop settings is crucial, involving putting in all relevant information for the products offered.

To add new products to the online shop in Webflow, the user clicks on the shopping cart icon on the left side of the Webflow Designer. This leads to all the e-commerce settings that can be customized in Webflow. Under the Setup Guide, the user selects the products tab, which provides an overview of all the products offered in the online shop. To add a new product, the user clicks on the new product button in the upper right corner, leading to a new page (Figure 42).

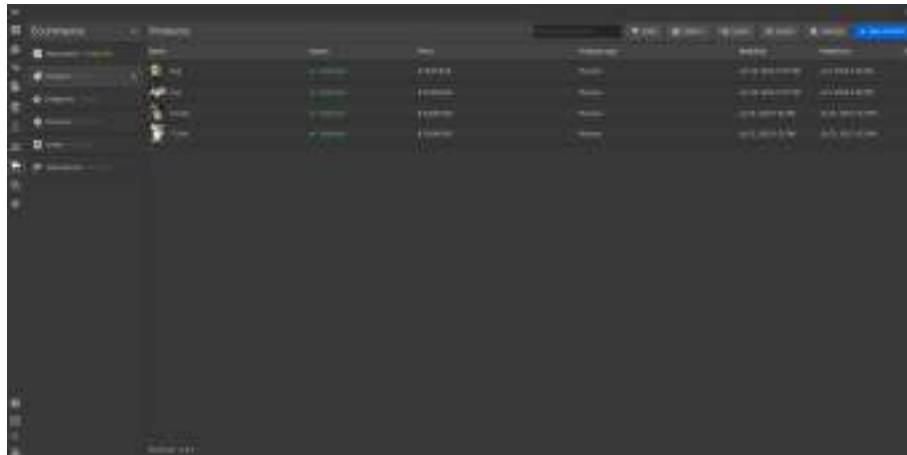


Figure 42: Screenshot of the e-commerce product overview in Webflow.

On this page, new properties can be assigned to the product. It starts with selecting the product type, which, in this case, is a physical product (clothing). The following fields are for the product name, the URL link for the online shop, a product description, and a selection of categories the user can create. Next, the product can have media added to it. Following that, the product can be assigned a price and a compare-at price. Additionally, the product tax class can be specified.

Then, the user can choose whether to track the inventory and provide shipping information, including the product's weight, length, width, and height. Finally, there is the option to add different variants to the product. For example, the variants 'Size' and 'Color' were created in this work.

In the case of this work, the products outlined in Chapter 3.1.1 were then created within this GUI (Figure 43). The generated texts were put into their respective fields (product name, description, price). Additionally, for testing purposes, two sample categories were created: 'Apparel' and 'Our Favorites'. The created images were added in the next step, and the switch labeled 'Requires Shipping' was set to 'YES'. In the lower section, variations for individual products can also be established. In this scenario, these include variations in size and color. Based on these different variations, Webflow automatically generates all the possible combinations of these variations. The quantity of these combinations can then be tracked. This tracking is crucial for determining whether a specific size or even an entire product is completely sold out, thereby necessitating the placement of replenishment orders with the manufacturer.

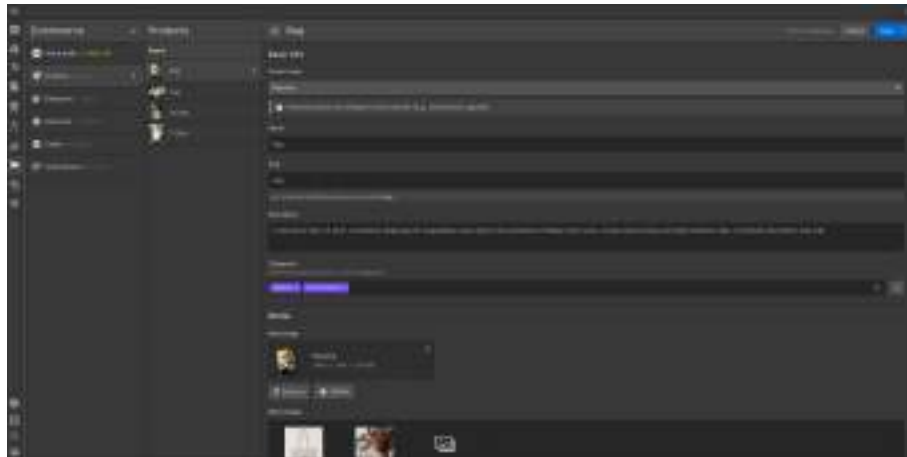


Figure 43: Screenshot of the product detail view in Webflow.

4.1.5 How to Use the Webflow Designer

As discussed in Chapter 4.1.1, the Webflow Designer is the cornerstone of Webflow. It is where all adjustments to the shop can be made. This chapter takes a closer look at the fundamental operation of the Webflow Designer.

The Webflow Designer is fundamentally comprised of 5 major sections [75]:

- The Left Toolbar
- The Canvas
- The Top Bar
- The Navigation Breadcrumb Bar
- The Right Panels

The Left Toolbar: This Left Toolbar provides users with various options for structuring their page. Firstly, the *Main Menu* offers links to the dashboard, the editor, or site settings. Following this is the *Add Panel*, which allows users to add elements or elements from pre-built libraries to the page. Next is the *Component Panel*, which displays all user-created components and enables their addition to the online shop. The *Navigator* comes next, showing users a hierarchy in a format reminiscent of HTML markup. Users can select and rearrange these elements as needed. The *Pages Panel* gives users an overview of all created and existing shop pages. The *CMS Panel* allows users to create, display, or modify new CMS collections. The *Logic Panel* permits users to establish and manage automated workflows, while the user panel facilitates creating and managing user groups and accounts. Subsequently, the *E-commerce Panel* allows users to create new products, manage them, view orders, and make adjustments. It also allows for fundamental shop settings adjustments, such as language customization or shipping costs. The *Assets Panel* displays all uploaded files, including documents, images, and animations. The *Settings Panel* then presents users with general setting options [75] (Figure 44).

The *Audit Panel* displays any accessibility-related issues, and the *Quick Find Panel* offers a search feature that can streamline the users' workflow. The *Video Tutorials* are available to assist users during development, and the *Help Settings* serve the same purpose, located beneath the tutorials [75] (Figure 44).



Figure 44: Screenshot of Webflow Designer - Left Toolbar.

The Canvas: The largest area of the Webflow Designer is occupied by the *Canvas* in the center. This Canvas allows the user to interact with individual elements, add them, and customize them as needed [75] (Figure 45).



Figure 45: Screenshot of Webflow Designer - Canvas.

The Top Bar: The Top Bar provides the user with additional important information and tools. For example, the *Site Indicator* can be found here, which shows the user which page they are currently on. Additionally, there is the *Preview Button*, which takes the user to a preview of the website as it would appear live. The *Breakpoints* in the middle allows the user to switch between different breakpoints and see how the page behaves on different devices. Next, there is an *Undo and Redo Button* for reversing or repeating changes. The *Save Indicator* indicates whether the

page has been saved. Work in the Designer is automatically saved, so clicking anywhere to save the created work is unnecessary. The *Export Code Button* allows the user to export the code of the current page as a ZIP file, the *Share Button* creates a preview link to show the work to clients, for example, and the *Publish Button* allows the user to publish and make the website live on one or multiple domains in its current state [75] (Figure 46).



Figure 46: Screenshot of Webflow Designer - The Top Bar.

The Navigation Breadcrumb Bar: The *Navigation Breadcrumb Bar* provides users with additional information about the hierarchy of the selected element, along with its parent elements, in addition to the Navigator on the left side [75] (Figure 47).



Figure 47: Screenshot of Webflow Designer - The Navigation Breadcrumb Bar.

The Right Panels: The final area of the Webflow Designer is the Right Panels. These consist of the *Style Panel*, where users can customize all CSS properties that are adjustable for this element according to their preferences. Once these properties are adjusted, the changes are immediately visible in the element on the Canvas. This again highlights the WYSIWYG (Chapter 2) approach that Webflow follows in their Designer. On the other hand, *Element Settings* allow various other settings to be made for the selected element. The *Style Manager*, in turn, provides users with a complete list of used classes and combo classes. Here, the user can instantly see which classes have already been assigned, rename them, or delete unused classes. Finally, there is the *Interactions Panel*, which enables the user to create animations or interactions and assign them to the page or a specific element [75] (Figure 48).



Figure 48: Screenshot of Webflow Designer - The Right Panel.

4.1.6 Components in Webflow

The development process of software and, thus, of websites can also be simplified immensely with the use of so-called components. Components are parts of a website or software that are repeatedly used in the same or similar way. One of the best examples of creating a component is a navigation bar or a footer.

One of the foundations of the shop design created in Figma is the selected font, Neue Haas Grotesk Display Pro. To integrate this font, made available for free through Adobe Typekit, into the Webflow shop, the user must add the Adobe API token in the *Site Settings* of Webflow under the *Fonts* tab. This token is provided by Adobe. Once this token is entered, the font can be used within the Webflow project.

Navigation: To create the navigation bar, the 'rl-navbar2-component' element from the Re-lume Webflow Library was utilized (Figure 49a).

In place of the 'Logo', the logo of 'BlueBird' was inserted, and the links in the middle were adjusted to match the desired design (Figure 49c). As the button in the right corner did not allow for the shopping cart's representation, the cart button had to be manually inserted from the Add Panel on the left side. The old button was thus removed. With the content now final in the navigation bar, the next step was to style it. In this process, special attention was given to changing the font from the system UI to the defined Neue Haas Grotesk Display Pro. Additionally, the correct color had to be assigned to the links in the middle. The following step involved adapting the shopping cart button. Unfortunately, the icon cannot be replaced, as Webflow pre-determines it. Nonetheless, aspects like font, color, and the border radius of the button could be adjusted, rendering the button almost identical to the desired design (Figure 49c). Finally, the border that outlined the navigation bar completely was removed.

However, a slight compromise in the navigation bar customization pertains to the display of the search function. This function can only be represented with a visible input field, label, and search button. As a result, the decision was made to exclude this feature from the navigation bar (Figure 49b).

To transform the finalized styled element into a reusable component, it is necessary to right-click on the element and then select the option 'Create component'. Once this is done, the navigation bar, as designed, can now be used as a reusable element on all pages.

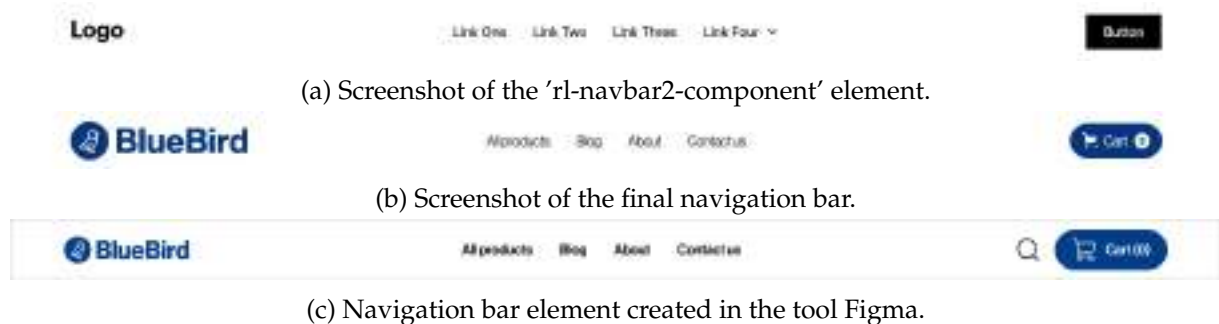


Figure 49: Overview of the navigation implementation in Webflow.

Footer: To create the footer component, the 'rl-footer4-component' element was utilized (Figure 50a).

In place of the logo, the 'BlueBird' logo was inserted, and the links were adjusted to align with the desired design (Figure 50c). Once the text was substituted, the design process could commence. Like the navigation bar design, careful attention was given to changing the font from the default System UI to Neue Haas Grotesk Display Pro. Furthermore, distinct font weights were assigned to the text, and the links were linked to their respective pages. In the final step, the font color was adjusted from regular black to the black value defined in Chapter 3.1.1.

No compromises were required for this element, and all modifications were seamlessly executed (Figure 50b compared to Figure 50c). To transform the finalized styled element into a reusable component, it was also necessary to right-click on the element and then select the option 'Create component'. Once this is done, the designed footer can also be used as a reusable element on all pages.



(a) Screenshot of the 'rl-footer4-component' element.



(b) Screenshot of the final footer in Webflow.



(c) Footer element created in the tool Figma.

Figure 50: Overview of the footer implementation in Webflow.

Banner: To create the banner component, the 'rl-banner10-component' element was employed (Figure 51a).

Only minor adjustments were necessary here, including changing the background color from white to blue, aligning the text to match the two components above it, and inserting and adjusting the link. No compromises were required for this element either, and all modifications were executed (Figure 51b compared to Figure 51c).

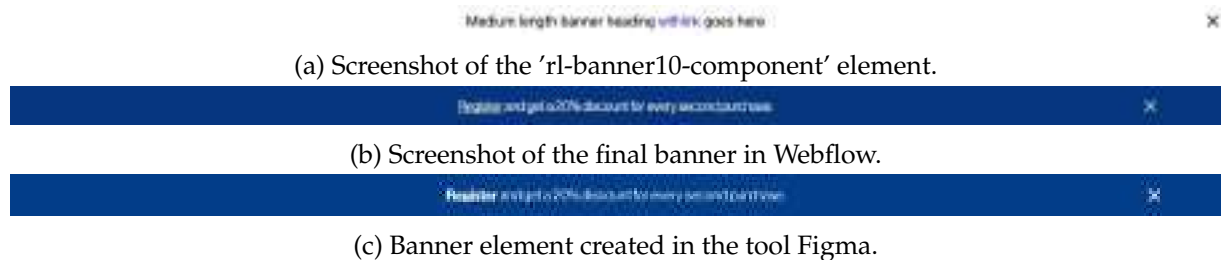


Figure 51: Overview of the banner implementation in Webflow.

Checkout Navigation Bar: The final component created in Webflow was the checkout navigation bar. This simplified version of the main navigation bar contains only a back button and the 'BlueBird' logo in the center. The foundation for this element was also the 'rl-navbar2-component' element from the Relume Webflow Library, but several elements had to be removed, such as the links in the center and the shopping cart on the right. A back button was also integrated, and the logo was moved from the left to the center. No compromises were needed for this element either, and all adjustments were made seamlessly (Figure 52a compared to Figure 52b).

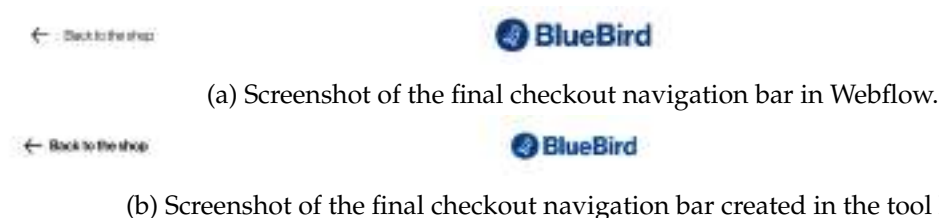


Figure 52: Overview of the checkout navigation bar implementation in Webflow.

4.1.7 Implementing the Shop

With the components and products ready for the online shop development, implementing the online shop in Webflow could commence. The focus is crafting the Homepage, Product Overview, Product Detail, and the Checkout Process pages. Finally, the result of the implementation process in Webflow will be presented.

Homepage

As described in Chapter 3.1.3, the Homepage of the 'BlueBird' online shop consists of the following elements:

- Banner
- Navigation Bar

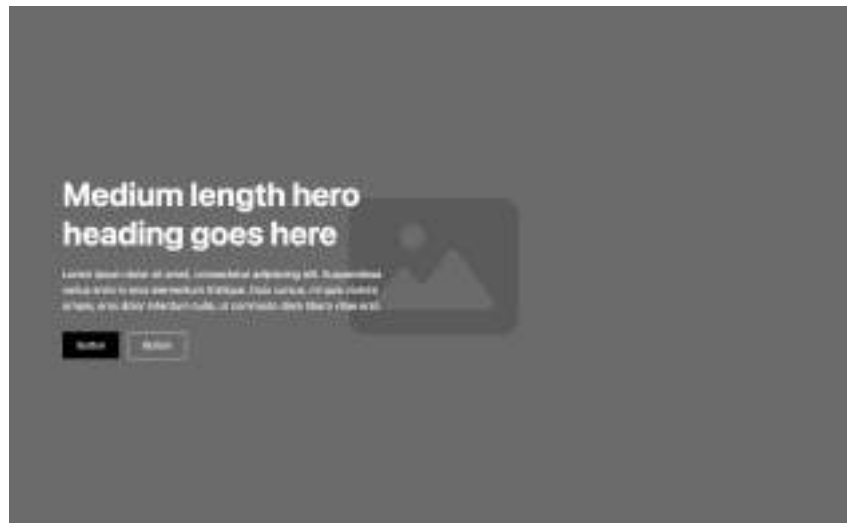
- Hero Section
- Product Slider
- Newsletter Signup
- Blog Article Slider
- Footer

Since the banner was already created as a component, it only needed to be dragged and dropped from the component library into the canvas, just like the navigation bar.

Hero Section: To develop the Hero Section as closely as possible to the design, a suitable element was sought in the Relume Webflow Library. The element that was then customized is called 'rl-section-header5' (Figure 53a).

Several steps were taken to achieve the desired design (Figure 53c). Firstly, the buttons and text had to be removed. An e-mail input field needed to be created, and a small text element was inserted. Additionally, the heading provided by ChatGPT was placed in the text field. Lastly, the background photo was replaced with the one created in Chapter 3.1.1. Once the content was in place, design adjustments were made using the right panel in Webflow. This involved setting the correct font (Neue Haas Grotesk Display Pro), assigning the proper color to the input field and button, and adjusting the corners of the hero area.

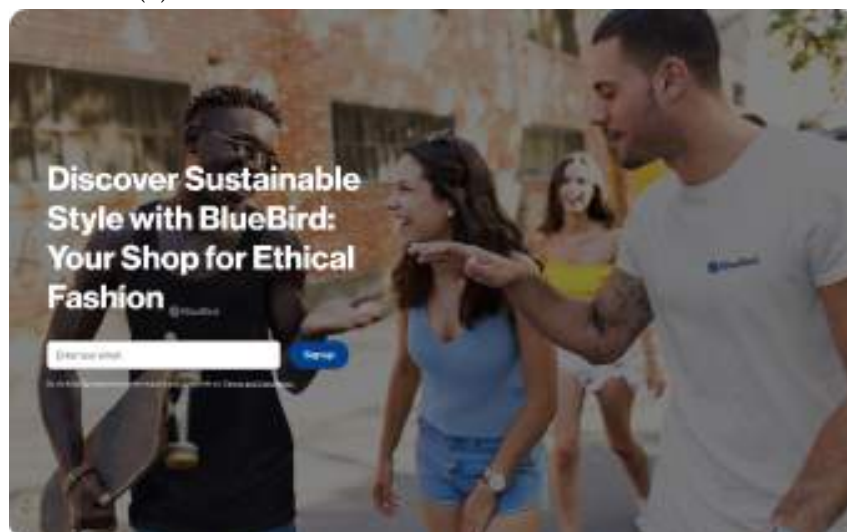
Looking at the result, it is evident that no compromises were made in the design of this element (Figure 53b) either, and it resembles the desired design (Figure 53c).



(a) Screenshot of the 'rl-section-header5' element.



(b) Screenshot of the final hero section in Webflow.



(c) Hero section element created in the tool Figma.

Figure 53: Overview of the hero section implementation in Webflow.

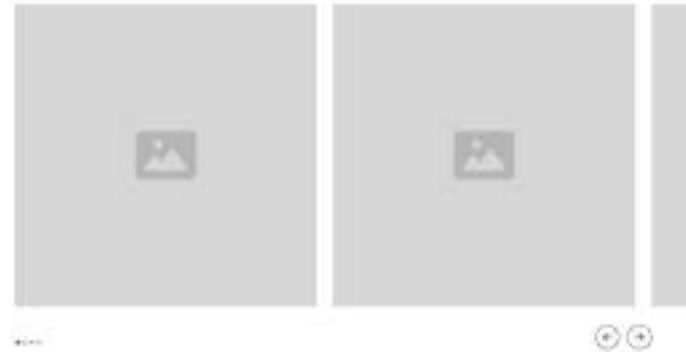
Product Slider: The next element to develop was the product slider. To implement it as closely as possible to the design, a suitable element was sought in the Relume Webflow Library. The element that was then customized is called 'rl-section-gallery21' (Figure 54a).

In order to get as close as possible to the desired design, the following steps were taken: Firstly, the texts in the upper section were replaced. Then, a Collection List was added to each slide element from the Add Panel, to which a product from the 'products' collection was assigned. Subsequently, two text fields and small circles were inserted into the slides, automatically filled with content from the respective products. In the next step, the element's link was set to the respective product detail page. At this point, the design process could begin. Fonts and colors were adjusted as before. The final step adds a border radius to the images to create rounded corners.

Despite requiring slightly more effort for element adjustments, the design was nearly replicated 1:1 in Webflow (Figure 54b compared to Figure 54c).

Image Gallery

Low-resolution image placeholder



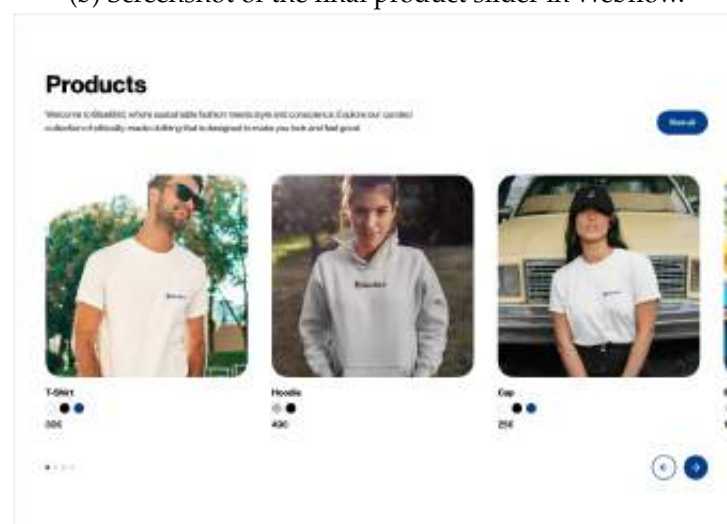
(a) Screenshot of the 'rl-section-gallery21' element.

Products

Welcome to our store! We're excited to have you here and we hope you'll love our products. We're committed to providing you with the best possible shopping experience.



(b) Screenshot of the final product slider in Webflow.



(c) Product slider element created in the tool Figma.

Figure 54: Overview of the product slider implementation in Webflow.

Newsletter Signup: The next element that needed to be developed was the Newsletter Signup Element. For this purpose, the 'rl-section-cta26' element was used (Figure 55a).

To achieve the final result, which is now also visible in the final shop, the texts provided by ChatGPT needed to be inserted first. Additionally, the layout had to be adjusted so that the entire content was left-aligned instead of centered. Then, the correct color and font values were assigned to the elements.

As can be seen, no compromises were made in the design for this element either (Figure 55b compared to Figure 55c).

Medium length heading goes here

Connect with us for all the latest news, exclusive updates and special offers from Bluebird. Subscribe today and be the first to know about new arrivals, limited edition releases, and exciting promotions. As a subscriber, you'll gain access to insider information, fashion tips, and sustainable living inspiration, all delivered straight to your inbox.

By clicking Sign Up, you are confirming that you agree with our Terms and Conditions.

(a) Screenshot of the 'rl-section-cta26' element.

Stay connected with us for exclusive updates and offers!

Don't miss out on the latest news, exclusive updates, and special offers from Bluebird. Subscribe to our newsletter and be the first to know about new arrivals, limited edition releases, and exciting promotions. As a subscriber, you'll gain access to insider information, fashion tips, and sustainable living inspiration, all delivered straight to your inbox.

By clicking Sign Up, you are confirming that you agree with our Terms and Conditions.

(b) Screenshot of the final newsletter signup in Webflow.

Stay connected with us for exclusive updates and offers!

Don't miss out on the latest news, exclusive updates, and special offers from Bluebird. Subscribe to our newsletter and be the first to know about new arrivals, limited edition releases, and exciting promotions. As a subscriber, you'll gain access to insider information, fashion tips, and sustainable living inspiration, all delivered straight to your inbox.

By clicking Sign Up, you are confirming that you agree with our Terms and Conditions.

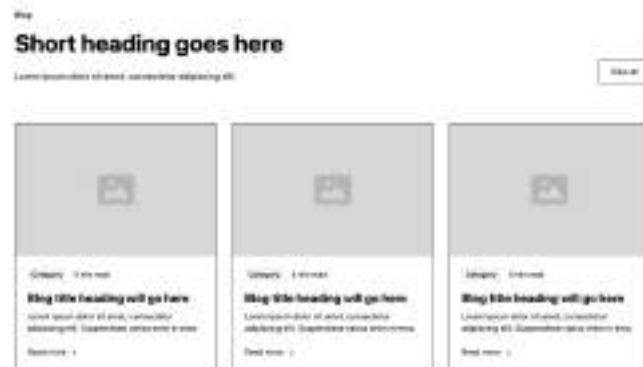
(c) Newsletter signup element created in the tool Figma.

Figure 55: Overview of the newsletter signup implementation in Webflow.

Blog Slider: Finally, the Blog Article Slider needed to be developed on the homepage. For this purpose, the 'rl-section-blog44' element was used (Figure 56a).

To come as close as possible to the desired design and to make the content easily manageable, a new CMS Collection was created first, where all the essential data of a blog article could be maintained. ChatGPT also generated the text content of these three sample articles, and the images were sourced from the copyright-free image database Unsplash. Once the data was entered, the three blog article elements had to be connected with the CMS. Elements such as the category and the reading time, displayed at the top, were removed. Furthermore, the article's frame was removed, and the text was aligned to the left. After assigning the correct color and font values and adding a border radius to the images, the Blog Article Element was completed.

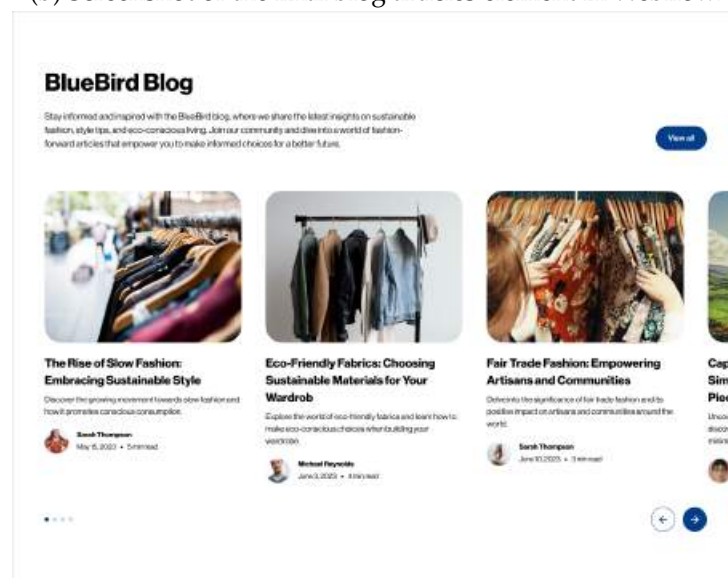
When comparing the created element with the desired design (Figure 56c), it is noticeable that the result in Webflow is not a slider. This is because, currently, Webflow does not offer a native solution to create a slider populated with data from the created CMS. Hence, a small compromise had to be made in this case (Figure 56b).



(a) Screenshot of the 'rl-section-blog44' element.



(b) Screenshot of the final blog articles element in Webflow.



(c) Blog articles element created in the tool Figma.

Figure 56: Overview of the blog articles element implementation in Webflow.

Last but not least, the footer which was also created as a component, could be easily dragged and dropped into the canvas.

Product Overview

As described in Chapter 3.1.3, the Product Overview page of the 'BlueBird' online shop consists of the following elements:

- Navigation Bar
- Product Overview including filtering options, sorting capabilities and an Add-to-Cart shortcut button
- Footer

Since the navigation bar and footer were already created as components, they only needed to be dragged and dropped from the component library into the canvas.

Product Overview Element: The basis for displaying the products on the Product Overview page was also the 'rl-section-gallery21' element (Figure 54a). However, this had to be extensively modified. The slider and button were removed entirely. Once all the text was inserted, a Collection List with all the products was added. Similar to the homepage, information such as the product name or price could be presented to the user here. Despite these efforts, some design compromises had to be made. For example, the "Add to Cart" shortcut button could not be integrated with Webflow precisely as desired in the design (Figure 57b).

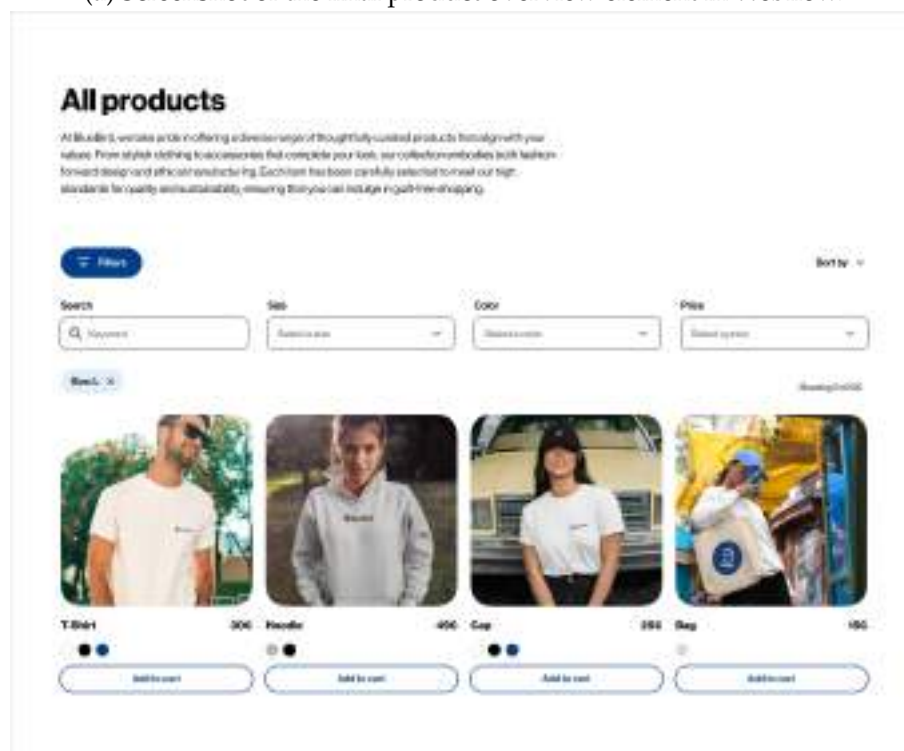
The most significant compromise, however, is that Webflow does not natively provide a way to incorporate filters. As a result, the ability to filter products had to be entirely omitted on the Product Overview page (Figure 57a).

All products

At Bludell we pride ourselves on offering a diverse range of thoughtfully curated products that align with your values. From stylish clothing to accessories that complete your look, our collection embodies both fashion-forward design and ethical manufacturing. Each item has been carefully selected to meet our high standards for quality and sustainability, ensuring that you can relax and guilt-free shopping.



(a) Screenshot of the final product overview element in Webflow.



(b) Product overview element created in the tool Figma.

Figure 57: Overview of the product overview implementation in Webflow.

Product Detail

As described in Chapter 3.1.3, the Product Detail page of the 'BlueBird' online shop consists of the following elements:

- Navigation Bar
- Product Details including breadcrumb, image display, variant selection, quantity selection, and Add to Cart button
- Footer

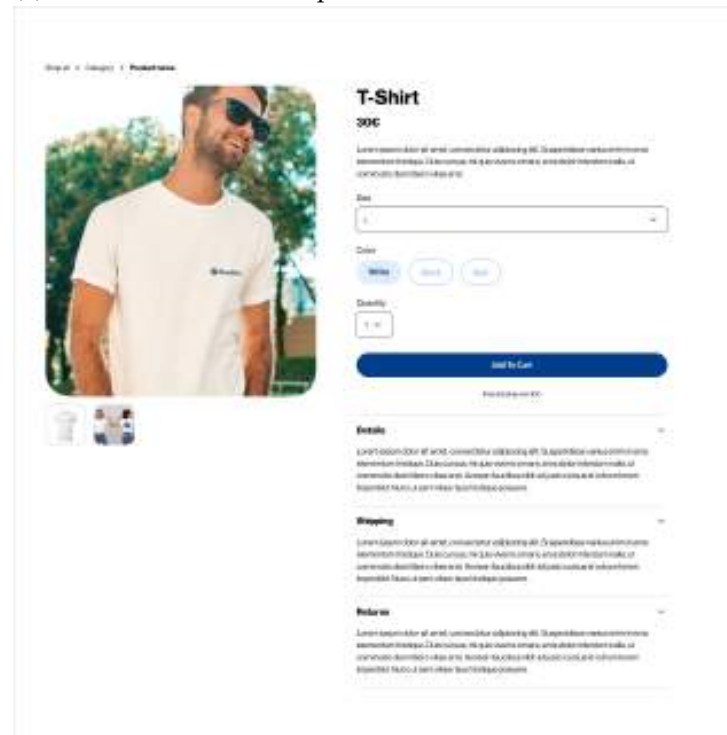
Since the navigation bar and footer were already created as components, they also only needed to be dragged and dropped from the component library into the canvas on this page.

Product Details Element: Since a suitable element for the designed layout could not be found in the Relume Webflow Library, this page had to be created entirely from scratch. However, since this is a template page and the user only needs to build one page and apply it to all products, this process still saves time. Nevertheless, the development process for this page includes inputting an image lightbox gallery, various blocks of text, an FAQ component from another Relume Library element, and finally, adding an "Add to Cart" button along with a quantity selection. Once these elements were placed correctly, the design of these elements was undertaken. This involved assigning the correct color and font values, adjusting font sizes, and setting the border radius of the images.

Only a minor compromise had to be made for this element, as the color selection could not be developed as clickable buttons as intended; instead, it had to be solved through a dropdown input. Otherwise, the Product Detail page is very close to the desired design (Figure 58a compared to Figure 58b).



(a) Screenshot of the final product details element in Webflow.

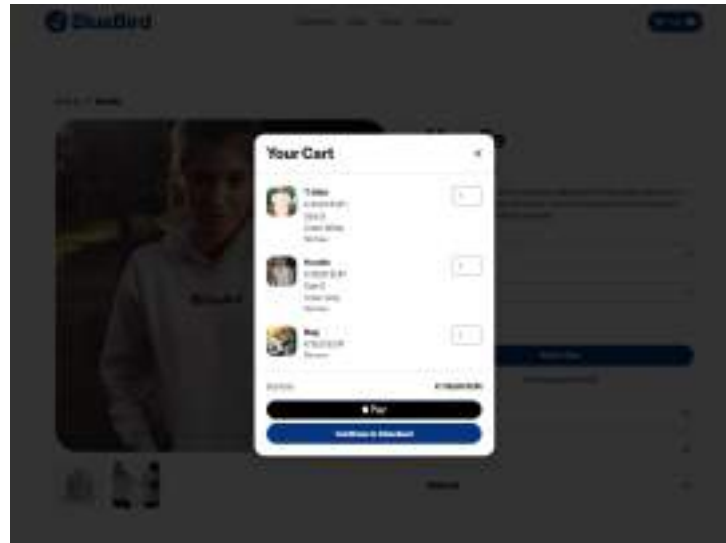


(b) Product details element created in the tool Figma.

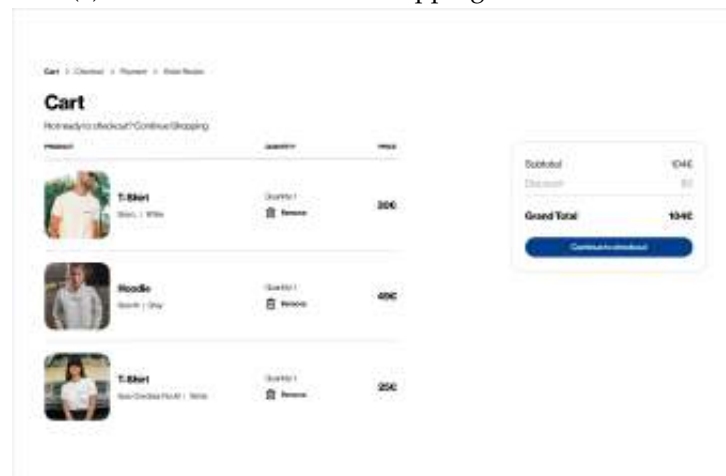
Figure 58: Overview of the product detail implementation in Webflow.

Shopping Cart

The development of the shopping cart in Webflow revealed a limitation of the platform. Currently, the shopping cart can only be implemented as a popup (centered, left, or right) or as a dropdown from the navigation bar. The desired design could not be achieved using these options, so the decision was made to implement a centered shopping cart as a popup in the style of 'BlueBird'. However, this differs significantly from the intended design (Figure 59a compared to Figure 59b).



(a) Screenshot of the final shopping cart in Webflow.



(b) Shopping cart created in the tool Figma.

Figure 59: Overview of the shopping cart implementation in Webflow.

Checkout Process

Checkout Steps: The final significant step in the development was creating the checkout process. Here, it became apparent that there is a limitation to the platform. While the checkout page can be adjusted slightly to meet the desired design requirements, it is impossible to split the process into three separate steps. All customer information (Shipping Address, Shipping Method, Payment Info) must be entered on the same page. As usual, the simplified navigation bar and footer could be easily added using drag and drop. Nevertheless, it is evident here that while Webflow offers design freedom in the checkout, it does not provide complete control over the entire checkout process (Figure 60a compared to Figure 60b).

The screenshot shows a checkout page with the following sections:

- Customer Info:** Fields for Email, Password, and Confirm Password.
- Shipping Address:** Fields for Address, City, State, and Zip.
- Shipping Method:** Radio buttons for Free Shipping, Regular Shipping, and Express Shipping.
- Payment Info:** Fields for Card Number, Expiry Date, and Cardholder Name.
- Billing Address:** Fields for Address, City, State, and Zip.
- Review Order:** A summary of the order with a 'Place Order' button.

(a) Screenshot of the final checkout page in Webflow.

The screenshot shows a checkout page with the following sections:

- Shipping Address:** Fields for Full Name, Email Address, Phone Number, Street Name and House Number, City, and Postal Code.
- Shipping Method:** Radio buttons for Free Shipping, Regular Shipping, and Express Shipping.
- Your order:** A summary of the order with a 'Place Order' button.

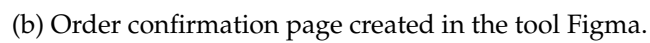
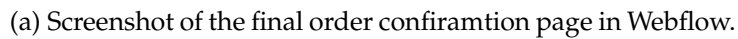
The 'Your order' section shows a breakdown of costs:

Item	Amount
Subtotal	300
Discount	-50
Shipping Costs	-50
Grand Total	200

(b) Checkout page created in the tool Figma.

Figure 60: Overview of the checkout implementation in Webflow.

Order Confirmation Page: The order confirmation page, however, could be implemented almost exactly as it was designed in the Expert Review. Although all the elements had to be added entirely new and by themselves, there are hardly any noticeable differences. First, the reduced navbar and footer were dragged and dropped onto the page as components. Then, the new elements (ordered products, delivery address, payment information, and shipping method) had to be added to the page. Once all the information was included, as in every step, the next step was to assign the correct color and font values to the respective elements. The only compromise that had to be made here was that no display of the order timestamp was possible, and the page could not be directly printed via a link (Figure 61a).



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Final Result

To avoid unnecessarily extending the presentation of the development process, additional pages were also created, but they are not relevant for the comparison. The complete online shop can currently be accessed using the link provided. Visitors can navigate through the website, view the products, and examine the final result of the implementation process in Webflow.

As a note, it must be added that due to the absence of a checkout test mode in Webflow and the lack of a Stripe or PayPal business account, the checkout process can not be tested on the live site.

Link to the shop: [BlueBird Online Shop - Webflow](https://bluebird-e9e00e.webflow.io/)⁴

4.2 Implementation in Shopify

This chapter presents the implementation process in Shopify. It begins with providing general information about the development process. Next, the focus shifts to exploring the foundation of every Shopify online shop: the templates. Following that, attention is given to Shopify's predetermined e-commerce checklist and a detailed examination of the shop settings. Finally, the chapter includes an introduction to Shopify's Graphical User Interface (GUI), followed by the implementation of the basic elements and the whole pages.

4.2.1 General Information about the Development Process

Since the shop's design was already completed, it is time to begin the development process in Shopify. To do this, one needs to have an account with Shopify. In the case of this work, the Google login feature is utilized, allowing users to sign in to Shopify using their Google account. Following this, Shopify offers a well-thought-out onboarding process, where fundamental questions about the shop being created are asked (Figure 62).



Figure 62: Screenshot of the second step in the onboarding process of Shopify.

⁴<https://bluebird-e9e00e.webflow.io/>

Once the onboarding process is completed, users find themselves on Shopify's main dashboard. The dashboard features a left-side navigation bar, allowing users to access and modify the essential shop settings. These main navigation points can be seen in Figure 63.

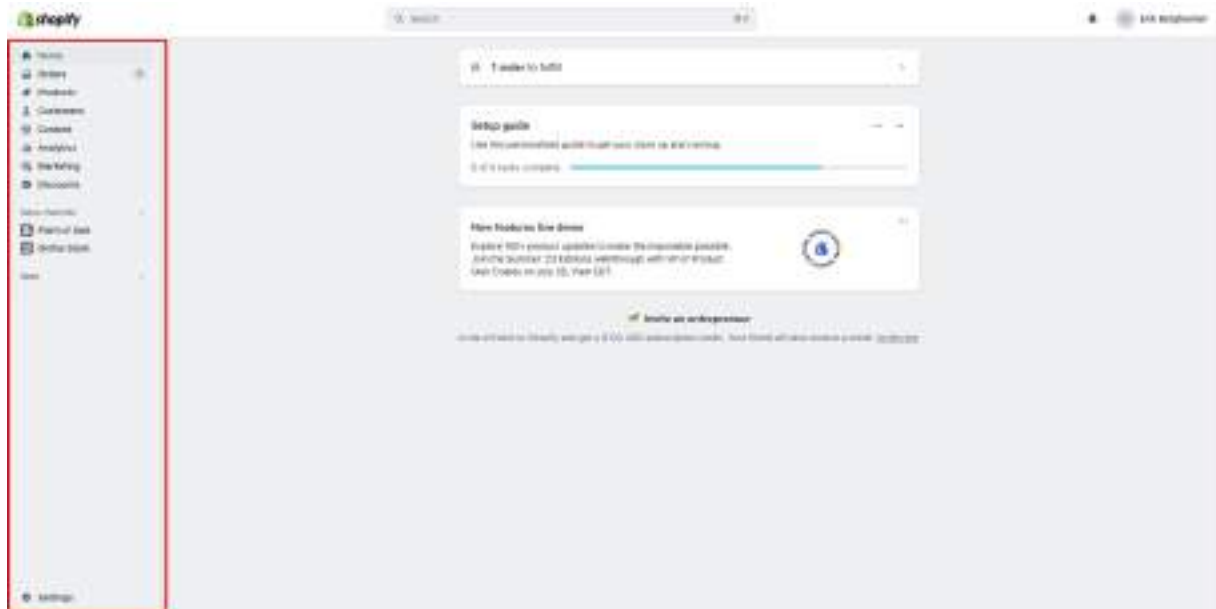


Figure 63: Screenshot of the dashboard and menu structure in Shopify.

Furthermore, Shopify offers a 14-day trial period. While this was utilized for this work, it was insufficient for the final development, leading to the adoption of the Basic Plan, which costs 1€ per month for the first three months and 27€ for every ongoing month.

4.2.2 Templates

During the initial development and research stages, one noticeable aspect was that Shopify utilizes templates as the foundation for every shop, which can then be customized to suit specific purposes. A Template Store is available, offering 150 templates (Figure 64).

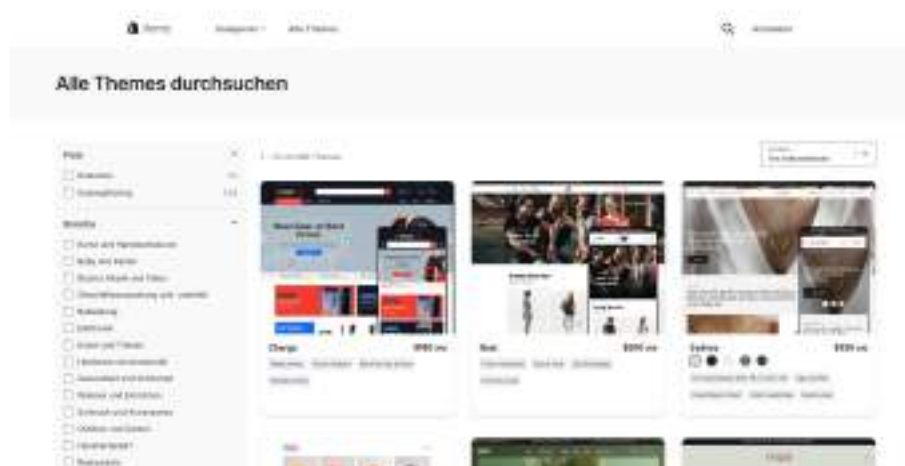


Figure 64: Screenshot of the Shopify theme store [76].

However, it is worth mentioning that most of these templates come with additional high costs, starting at \$170 (currently approx. 156,70€). In this case, a free template called 'Dawn' was opted for, ensuring quality and design without compromise (Figure 65).

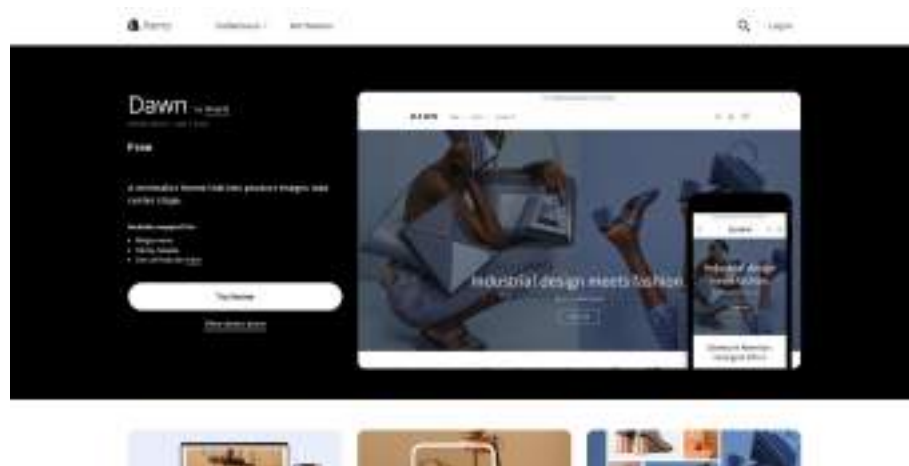


Figure 65: Screenshot of the Shopify Theme "Dawn" [77].

The 'Dawn' template stands out primarily due to its minimalist design, a media-forward product page, and advanced customization options [77], which was ultimately why this template was chosen for this work.

4.2.3 E-commerce Checklist by Shopify

For setting up a shop with Shopify, this tool also offers a very well-prepared onboarding process. Here, the user can quickly work through a checklist from Shopify itself, which contains all the essential steps to develop a complete and functioning shop in Shopify.

These consist of:

- Set up your online store
 - Add your first product
 - Customize your online store
 - Add pages to your store
 - Organize your store's navigation
 - Add a custom domain
 - Create a professional logo
- Store settings
 - Set up Shopify Payments
 - Review legal requirements for Germany
 - Manage orders with the Billbee app
- Launch your online store
 - Place a test order

This checklist is used for the development process of the online shop to include all the important steps in the creation and not to forget anything.

4.2.4 Managing the Shop

Before starting the development of the shop, it is essential to configure basic shop settings, primarily involving inputting all relevant information for the products offered.

Product management in Shopify can be found under the 'Products' navigation tab, leading to a dedicated subpage (Figure 66).

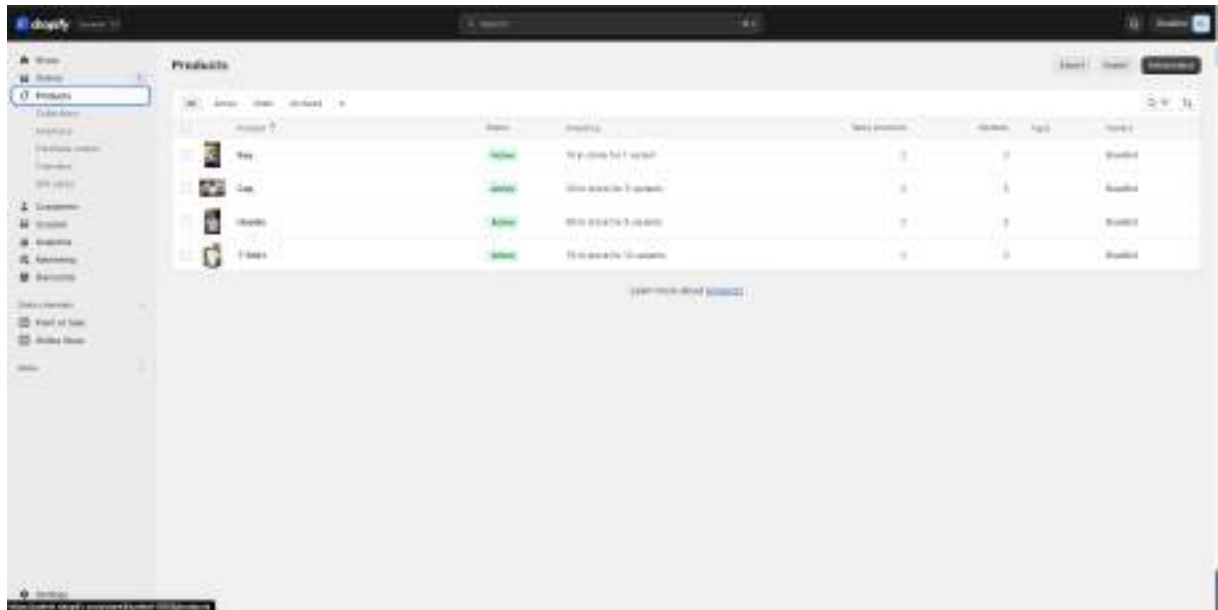


Figure 66: Screenshot of the Product Overview in the Shopify dashboard.

Following this, the project progressed to generating new products. Commencing with the product 'T-Shirt' as outlined in Chapter 3.1.1, names and descriptions were assigned accordingly. Subsequently, multiple images were added to the product, which were generated using the tool PlaceIt. Adding images to a product is a straightforward process in Shopify, using a drag-and-drop interface.

Furthermore, different variants can be added to the product. In the case of the 'T-Shirt', various variants were created, encompassing sizes (S, M, L, XL) and colors (white, black, blue). Once these variants were added, an overview of all the different variants of the 'T-Shirt' product is displayed at the bottom of the page. For instance, a variant could be named 'T-Shirt, S, white' (Figure 67). Shopify also allows users to assign a quantity to each variant, indicating the current availability of the product in the shop. Additionally, Shopify tracks the inventory even after purchases, allowing users to maintain a comprehensive overview of their product inventory.

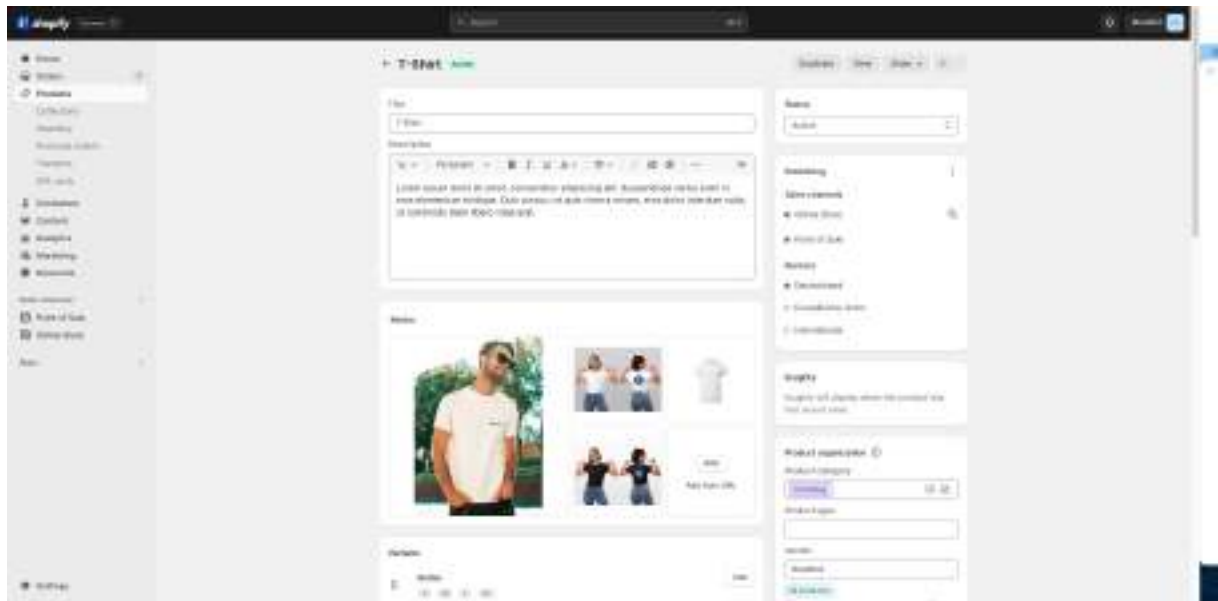


Figure 67: Screenshot of the Product Detail of the T-Shirt in the Shopify dashboard.

Finally, at the end of the page, there is a sample representation of the Search Engine Result Page for this product. This feature gives users an idea of how the product would appear in Google search results.

Once all the products are successfully added, the product overview page in Shopify provides a clear view of all the created products. Furthermore, users can easily access an overview of their inventory, which can be adjusted as needed (Figure 66).

4.2.5 How to Use the Shopify Designer

The cornerstone of Shopify for customizing the displayed online store is the Website Designer. There, any adjustments to the template can be made, and individual elements can be customized to the user's needs. Also, from there, the user can access the source code of the selected template. This chapter takes a deeper look at the basic features of the Shopify Website Designer.

Similarly to the Webflow Designer, the Shopify Website Designer is also divided into different areas:

- The Left Toolbar
- The Top Bar
- The Canvas
- The Right Panel

The Left Toolbar: The Left Toolbar allows users to view an overview of the page structure in the *Sections Panel*. This displays all elements on the website, which can be rearranged or have new elements added to them. Below that are the *Theme Settings*. These include basic settings and adjustments that can be made to the theme without the need for code. These settings may include logo input, color input, button settings, brand settings, or checkout customizations.

Below the theme settings are the so-called *App Embeds*. Here, third-party apps can easily be integrated into the shop, and an overview of the embedded apps is displayed (Figure 68).



Figure 68: Screenshot of Shopify Designer - The Left Toolbar.

The Top Bar: The top bar includes a *Back Button* that takes the user to the dashboard (Figure 63). Additionally, the name of the selected template is displayed, and there is a *Dropdown Menu* that allows the user to customize the theme code and access the theme documentation. In the middle of the top bar is a *Language Switch* that allows the user to edit the shop in different languages, as well as a *Page Indicator* that shows which page of the online shop is currently being edited. A *Dropdown Menu* also makes easy navigation between the pages possible. On the right side, there is an *Inspector button* that provides more information about an element when clicked, as well as *Breakpoint Buttons* that display the page on three different breakpoints. Finally, *Undo and Redo Buttons*, as well as a *Save Button*, are displayed. Unlike Webflow, Shopify requires manual saving of changes by clicking the *Save Button* (Figure 69).

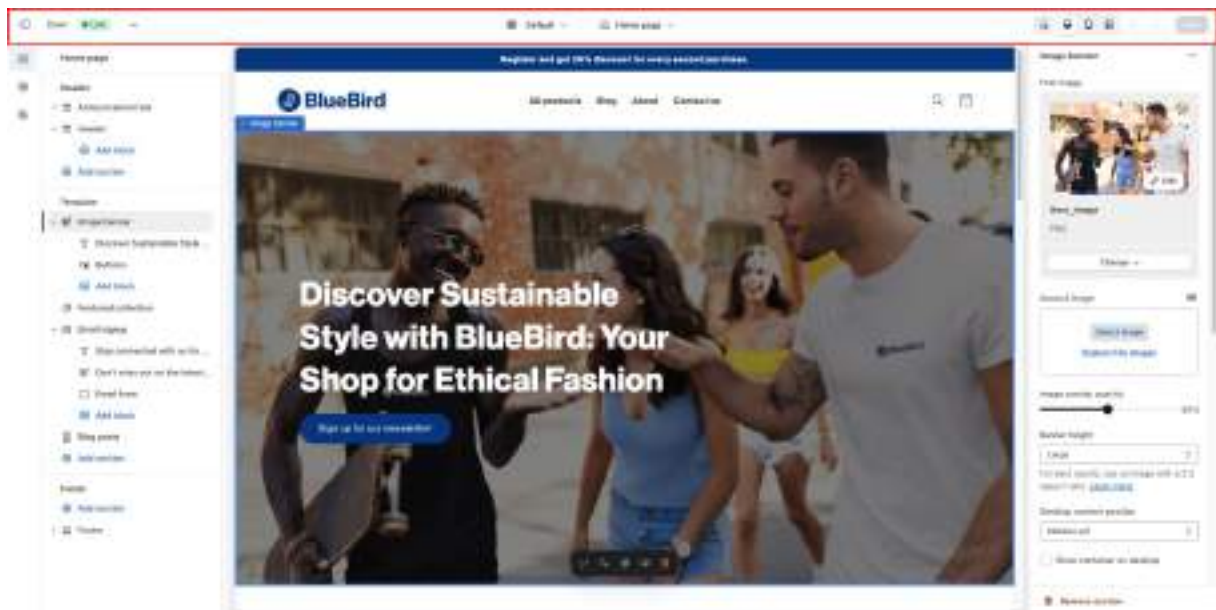


Figure 69: Screenshot of Shopify Designer - The Top Bar.

The Canvas: Like in Webflow, the *Canvas* in Shopify is located in the center of the Website Designer and provides a live view of the online shop. Elements can be selected, moved, or deleted within the Canvas (Figure 70).



Figure 70: Screenshot of Shopify Designer - The Canvas.

The Right Panel: Users can edit specific elements within the framework of *Theme Settings* in the Right Panel once a particular element is selected. This includes changing photos, adjusting text, or adding custom CSS. Shopify also follows the WYSIWYG principle mentioned in Chapter 2, so changes made in the Right Panel are immediately visible in the Canvas (Figure 71).



Figure 71: Screenshot of Shopify Designer - The Right Panel.

4.2.6 Basic Elements in Shopify

The basic elements that should exist on every page of the shop include, for example, the navigation bar, the footer, and the banner element. In Shopify, there are no component settings for these elements; instead, all three are automatically displayed at the beginning or end of each page. The following section describes how to customize these elements.

One of the foundations of the shop design created in Figma is the Neue Haas Grotesk Display Pro font. To integrate this font, made available for free through Adobe Typekit, into the Shopify shop, the user needs to add a link created by Adobe to the template's metadata. Additionally, it is necessary to override the font variables provided by Shopify in the CSS with those of the selected font from the Adobe Typekit project.

Navigation Bar: The navigation bar was developed using the provided navigation from the 'Dawn' template (Figure 72a).

First, a new main menu with the desired subpages had to be created in the Online Store Settings. After the four menu items were set up and the logo was placed in the navigation bar, the four menu items were centered in the template's source code (Figure 72b).

When comparing the final design in Shopify with the one created in Figma (Figure 72c), it is evident that there are some differences. For instance, custom icons cannot be added to the navigation bar. Additionally, the appearance of the shopping cart cannot be customized as intended in the design. However, a pre-built search function can be used. Nevertheless, compromises in design are noticeable.



(a) Screenshot of the navigation bar element provided by the Template 'Dawn'.



(b) Screenshot of the final navigation bar in Shopify.



(c) Navigation bar element created in the tool Figma.

Figure 72: Overview of the navigation implementation in Shopify.

Footer: The footer was also developed using the provided footer from the 'Dawn' template (Figure 73a).

Firstly, the e-mail signup element had to be removed from the footer. Next, the main menu, which was already created, was integrated into the footer through a simple click, and the logo was also placed using the drag-and-drop approach. However, the effort for customization was relatively high in this case. An element needed to be hidden, and then a text element was added to the template's source code to contain the social media icons. Finally, the checkbox for 'Show Policy Links' was enabled to allow access to the legal pages from the footer.

When examining the footer, it becomes evident that the two footers are not exactly identical (Figure 73b and 73c). Due to the use of custom code, the social media icons are blue-colored, which could not be adjusted further. Additionally, all legal sites are now displayed, which amounts to six on the live site. When developing a custom-designed footer in Shopify, it is important to be prepared for slight compromises in design and a relatively extensive adjustment effort.



(a) Screenshot of the footer element provided by the Template 'Dawn'.



(b) Screenshot of the final footer in Shopify.



(c) Footer element created in the tool Figma.

Figure 73: Overview of the footer implementation in Shopify.

Banner: The banner was implemented using the provided announcement bar element from the 'Dawn' template (Figure 74a).

Initially, the banner was assigned the appropriate background color, and the relevant text was inserted. Subsequently, a link to a product could be effortlessly added via the Theme Settings.

When comparing the banners (Figure 74b and 74c), there are hardly any differences in design. The only notable distinctions are the text formatting, as desired in the design, and the option to close the banner using a close icon.

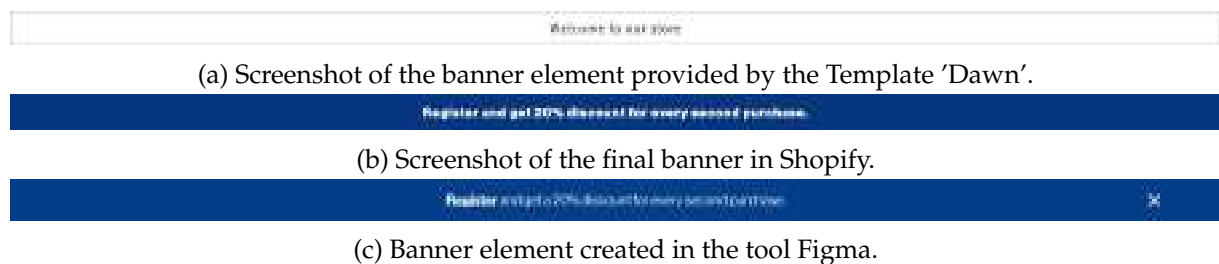


Figure 74: Overview of the banner implementation in Shopify.

4.2.7 Implementing the Shop

With the basic elements and products ready for the online shop development, implementing the online shop in Shopify could start. The focus here is on crafting the 'Homepage', 'Product Overview', 'Product Detail', and the 'Checkout Process' pages. Finally, the result of the implementation process in Shopify will be presented.

Home

As described in Chapter 3.1.3, the homepage of the 'BlueBird' online shop consists of the following elements:

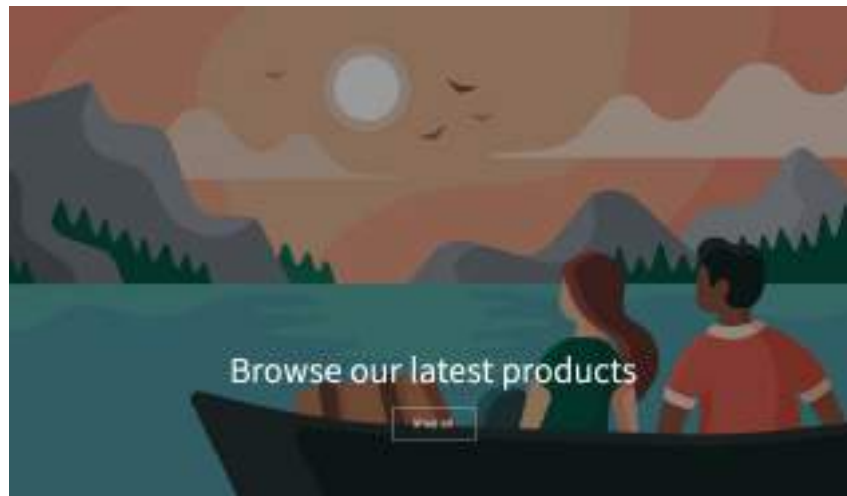
- Banner
- Navigation Bar
- Hero Section
- Product Slider
- Newsletter Signup
- Blog Article Slider
- Footer

Since the banner and navigation bar were already discussed in the previous step, these were already visible on the page and adjusted.

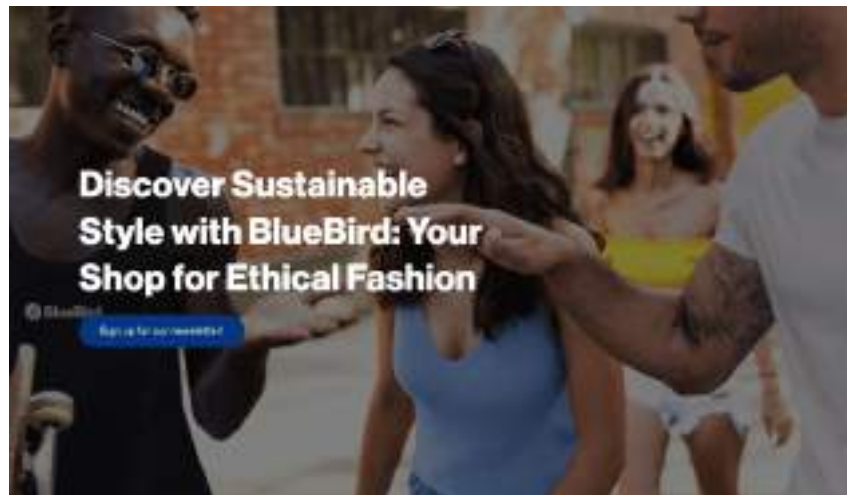
Hero Section: The next step in the implementation was to work on the representation of the Hero Section. For this purpose, the 'Image banner' element from the 'Dawn' template was employed (Figure 75a).

To portray the appearance as faithfully as possible, settings for button presentation were adjusted in the Theme Settings. Furthermore, the desired background image and text were inserted. Subsequently, the positioning of the heading and the button was fine-tuned. Lastly, the left alignment had to be adjusted in the source code using CSS to align the design as closely as possible with the desired shop design from Figma.

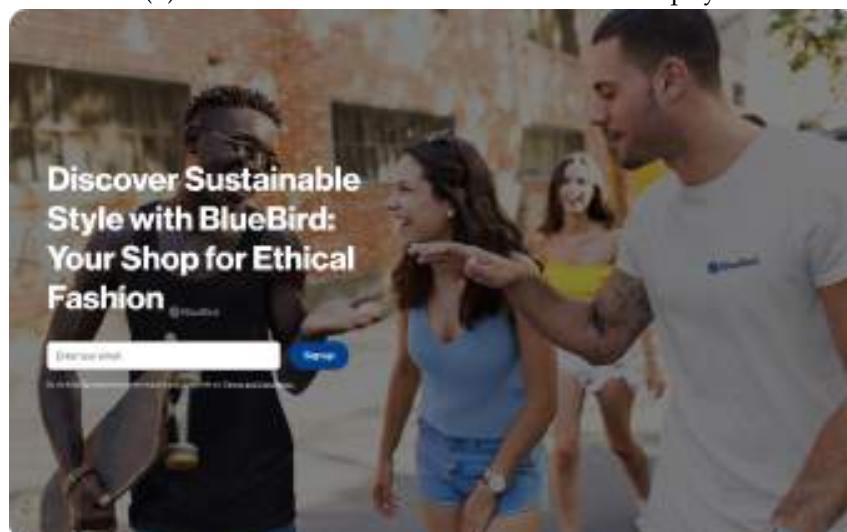
Comparing the final element (Figure 75b) created with the intended design (Figure 75c), the only noticeable discrepancies are the absence of an e-mail input field and the inability to replicate the border radius of the image.



(a) Screenshot of the hero section element provided by the Template 'Dawn'.



(b) Screenshot of the final hero section in Shopify.



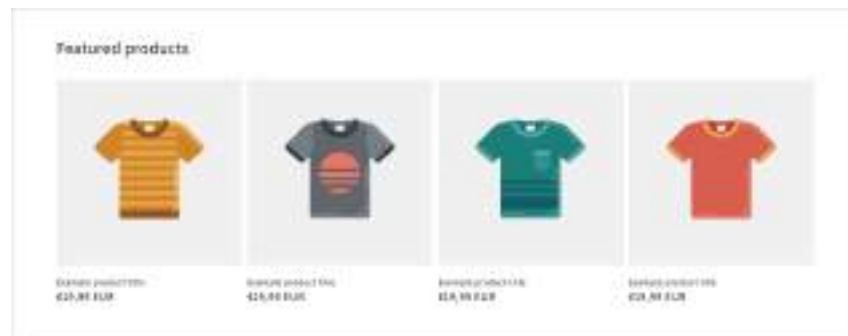
(c) Hero section element created in the tool Figma.

Figure 75: Overview of the hero section implementation in Shopify.

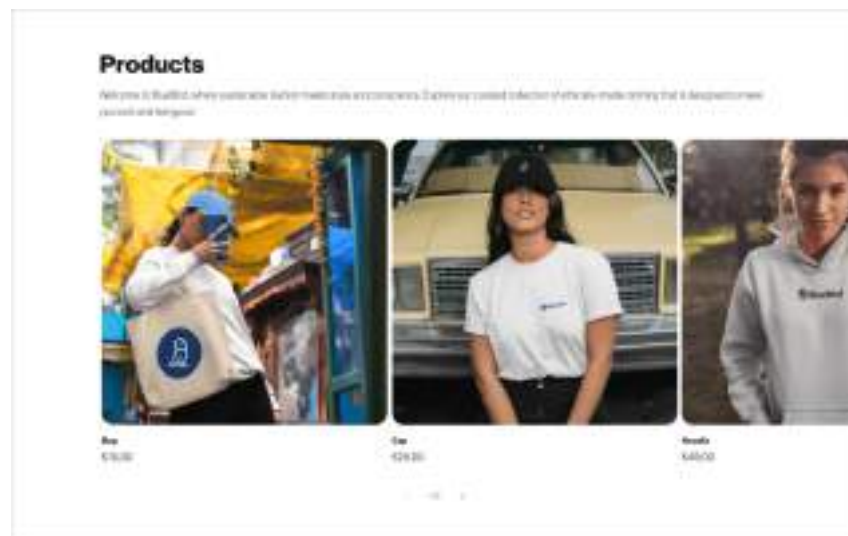
Product Slider: The next step in the implementation process in Shopify was to develop the Product Slider. The foundation for its development was the 'Featured Collection' element from the 'Dawn' template (Figure 76a).

Since all the products were already added to the inventory, they could easily be presented in this slider. In the Element Settings on the right side, the corresponding texts were inserted, and the element was transformed into a slider. Finally, in the template's source code, the product images' border-radius was adjusted to match the design in Figma.

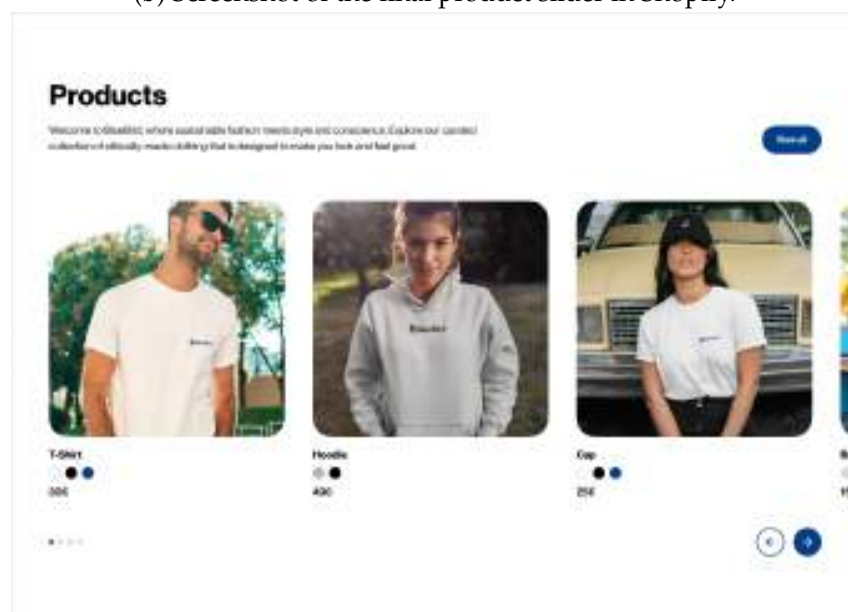
Comparing the created version of the Product Slider in Shopify (Figure 76b) with the desired design (Figure 76c), it is noticeable that they do not differ significantly. Here, too, the most important adjustments could be made. The only differences are in the button linking to the product overview, the display of the available product colors, and the presentation of the slider arrows, which could not be developed 100% faithfully to the design.



(a) Screenshot of the product slider element provided by the Template 'Dawn'.



(b) Screenshot of the final product slider in Shopify.



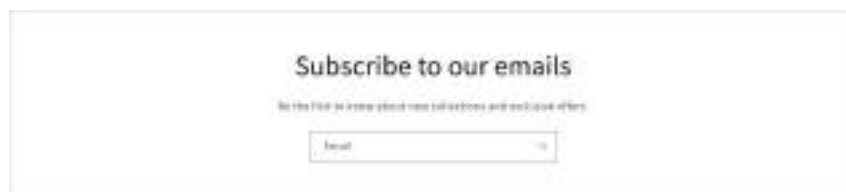
(c) Product slider element created in the tool Figma.

Figure 76: Overview of the product slider implementation in Shopify.

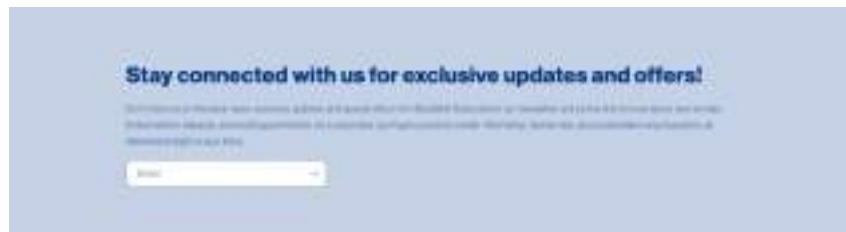
Newsletter Signup: The next step in the implementation process in Shopify was to develop the Newsletter Signup. The foundation for its development was the 'Email signup' element from the 'Dawn' template (Figure 77a).

In order to accurately represent the element, all textual content was initially entered into the element. This included the headline and the short description text below it. Then, the color scheme had to be adjusted, changing the background to the light blue color and assigning the primary color to the font and accents. In the final step, in the template's source code, the element's alignment was changed to left alignment, and the border radius of the input element was modified.

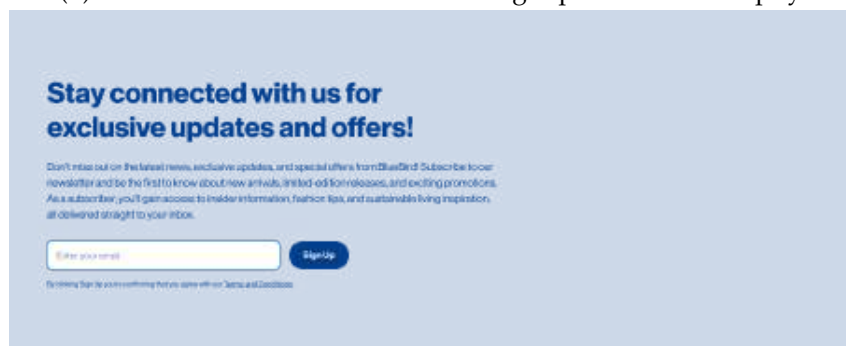
When comparing the final element in Shopify (Figure 77b) with the desired design created with Figma (Figure 77c), it becomes clear that they are not precisely identical. Particularly, the width of the text could not be elegantly and efficiently solved in the Shopify website designer and the source code, resulting in the text spanning the entire width of the element. Additionally, the button could not be integrated since the template for subscribing to a newsletter list does not include buttons. Therefore, the user's subscription is triggered by pressing the Enter key or clicking the arrow icon in the input field.



(a) Screenshot of the newsletter signup element provided by the Template 'Dawn'.



(b) Screenshot of the final newsletter signup element in Shopify.



(c) Newsletter signup element created in the tool Figma.

Figure 77: Overview of the newsletter signup implementation in Shopify.

Blog Article Slider: The next step in the implementation process in Shopify was to develop the Blog Article Slider. The foundation for its development was the 'Blog posts' element from the 'Dawn' template (Figure 78a).

Similarly, for the last element that needed to be developed on the homepage, it was important to adjust the element from the template to match the desired outcome closely. For this purpose, blog articles had to be created in the *Blog Panel* within the Shopify Dashboard. These articles consisted of text content generated by ChatGPT and images from the copyright-free image library Unsplash. Once these articles were created, the design process could begin.

After providing a title to the element and setting the number of blog posts to 3, checkboxes for 'Show Date' and 'Show Author' were selected in the Theme Settings to display the date and author on the preview. Additionally, the border radius of the images was adjusted in the template's source code. Finally, a 'View All' button was integrated at the bottom of the blog element.

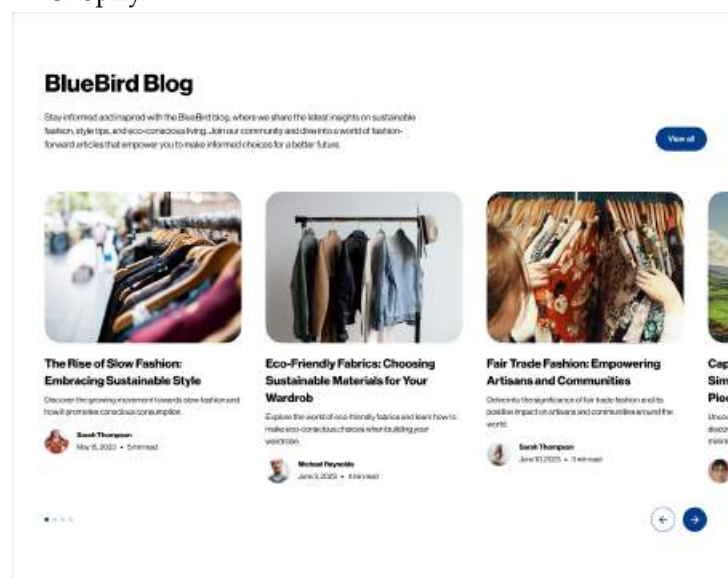
When comparing the final result in Shopify (Figure 78b) to the desired design (Figure 78c), it is immediately noticeable that there is no slider in Shopify. This is due to the limitations of the 'Dawn' template, which does not offer the option to display blog articles as a slider. Furthermore, a description text could not be placed under the heading, and the author's images and reading time were not visible. Lastly, the 'View All' button is present but located beneath the posts rather than in the top-right position as desired. The chosen template constrains all of these aspects and makes it challenging for users to make individual design adjustments.



(a) Screenshot of the Blog Article Slider element provided by the Template 'Dawn'.



(b) Screenshot of the final Blog Article Slider element in Shopify.



(c) Blog Article Slider element created in the tool Figma.

Figure 78: Overview of the Blog Article Slider implementation in Shopify.

Product Overview

As described in Chapter 3.1.3, the Product Overview page of the 'BlueBird' online shop consists of the following elements:

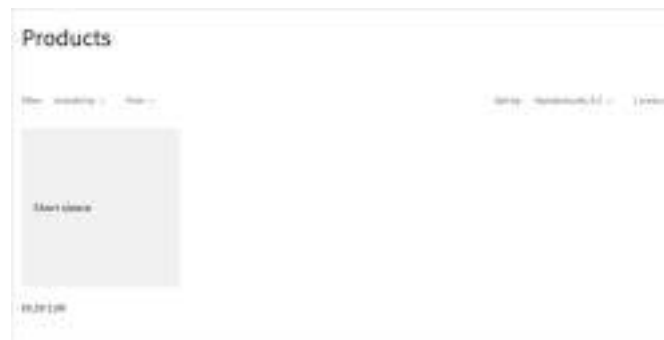
- Navigation Bar
- Product Overview including filtering options, sorting capabilities and an Add-to-Cart shortcut button
- Footer

The navigation bar and footer were already created and viewable on the site.

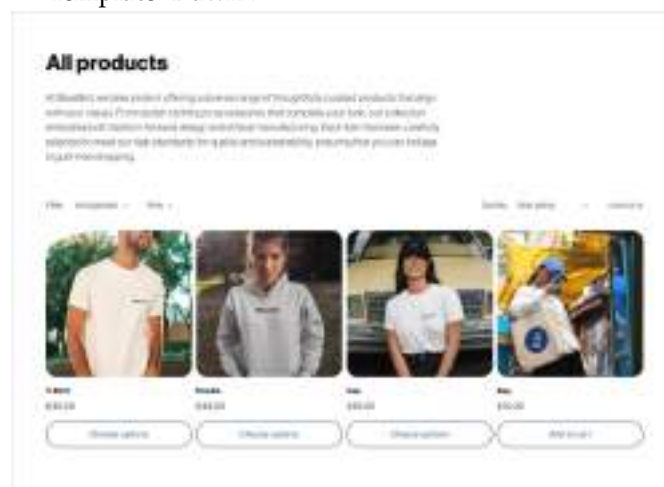
Product Overview Element: For the development of the Product Overview, the predefined Product Grid from the 'Dawn' template was also used (Figure 79a).

Since, as previously mentioned with the Product Slider, all the products were already set up, this step focused on presenting the products on this page. On one hand, the example texts generated with ChatGPT were inserted. On the other hand, the product photos needed to be converted to a square layout, and the border radius of the images was adjusted in the source code.

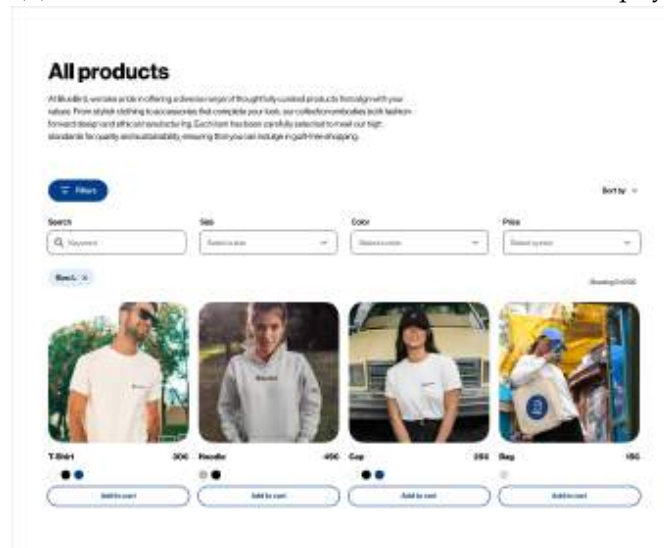
Comparing the desired design (Figure 79c) with the developed element in Shopify (Figure 79b), it is evident that the display of filters could not be replicated 100%. The template settings do not allow customization of these filters in any way, which is why the default version of the filters was used. Nonetheless, it can be seen that the Add-to-Cart button was integrated into the Product Overview, though the color representation could not be fully achieved.



(a) Screenshot of the Product Overview provided by the Template 'Dawn'.



(b) Screenshot of the final Product Overview in Shopify.



(c) Product Overview created in the tool Figma.

Figure 79: Overview of the Product Overview implementation in Shopify.

Product Detail

As described in Chapter 3.1.3, the Product Detail page of the 'BlueBird' online shop consists of the following elements:

- Navigation Bar
- Product Details including breadcrumb, image display, variant selection, quantity selection, and Add to Cart button
- Footer

The navigation bar and footer were already created and viewable on the site.

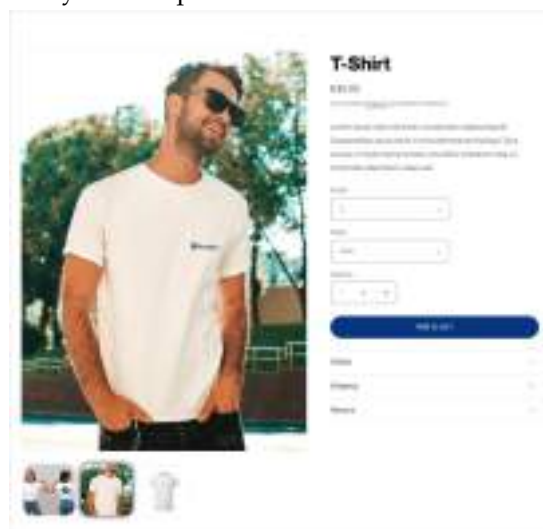
Product Details Element: As in the previous steps, the development of the Product Detail element also relied on the predefined 'Product Information' element from the 'Dawn' template (Figure 80a).

Since the products were already set up, development could start immediately. The initial layout was chosen to be side by side, and the media width of the image was set to medium. Additionally, the Share Button was removed, and three collapsible rows were inserted to accommodate frequently asked questions. This step did not require any adjustments in the template's source code.

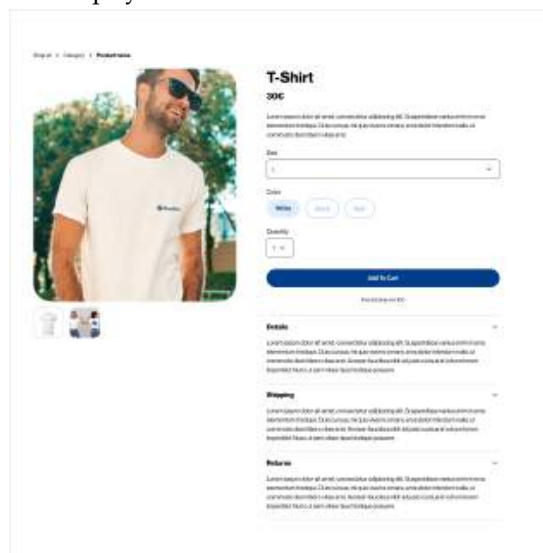
Comparing the desired design (Figure 80c) with the final Product Detail view (Figure 80b), it is immediately noticeable that the two views are pretty similar. Unfortunately, achieving a square image layout and the desired presentation of variant selection could not be realized as desired due to the template's limitations.



(a) Screenshot of the Product Details provided by the Template 'Dawn'.



(b) Screenshot of the final Product Detail in Shopify.



(c) Product Details created in the tool Figma.

Figure 80: Overview of the Product Details implementation in Shopify.

Shopping Cart

For the development of the shopping cart, the existing page of the 'Dawn' template was also used (Figure 81a).

The setting could display the shopping cart as an utterly standalone page. However, only minor adjustments could be made. Rounding the quantity input, adjusting the border radius of the product image, and horizontally centering the product information were all done in the template's source code.

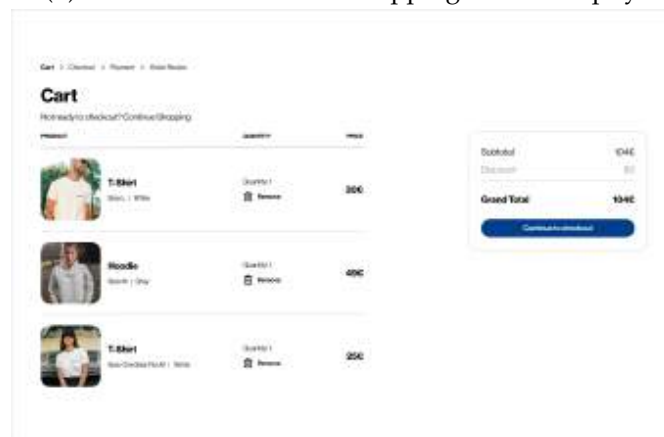
When comparing the Shopping Cart (Figure 81b) with the desired design in Figma (Figure 81c), it becomes evident that the two representations are quite different. Here, there is a limitation of the platform, as the customization options of the template are so limited that any customization desires need to be added through custom code.



(a) Screenshot of the shopping cart provided by the Template 'Dawn'.



(b) Screenshot of the final shopping cart in Shopify.



(c) Shopping cart created in the tool Figma.

Figure 81: Overview of the shopping cart implementation in Shopify.

Checkout Process

Checkout Steps: The development of the checkout steps in Shopify revealed one of the most significant limitations of the platform. The checkout steps are the same for every template and are provided by Shopify itself, with only minimal customization options available. These options include adding a logo, changing the background color, selecting one of Shopify's predefined fonts, and specifying the color codes for accents, buttons, and error messages.

When comparing the final result in Shopify (Figure 82a) with the desired design in Figma (Figure 82b), it is immediately apparent that the two views are entirely different. Unfortunately, achieving the desired development outcome was not possible in this aspect.

BlueBird

Cart > My account > Orders > Payment > Review

Shipping method

PayPal **Pay**

OK

Contact

Email

Enter my email to receive offers

Shipping address

Company/Region
Company

First name Last name

Address

Apartment, Suite, etc. (optional)

Postal code City

Save this information for next time

Continue to shipping

Order summary

Subtotal 415.00

Shipping 0.00

Total 415.00

(a) Screenshot of the final Checkout Process in Shopify.

Cart > Checkout > Payment > Confirmation

Checkout

Shipping Address

Full name*

Email address*

Phone number*

Street name and house number*

City*

Postal code*

Shipping Method

☐ Free shipping 7.50

☐ Regular shipping 9.00

☒ Express shipping 22.25

Your order

Subtotal 300

Discount -40

Shipping Costs -60

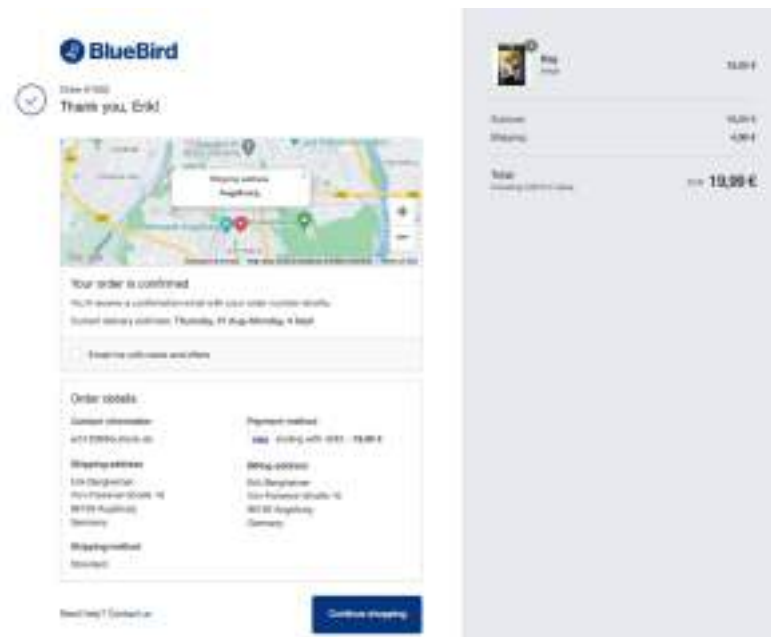
Grand Total 200

(b) Checkout created in the tool Figma.

Figure 82: Overview of the Checkout implementation in Shopify.

Order Confirmation Page: The development of the order confirmation page also highlights a significant limitation of the Shopify platform. Just like in the checkout steps, the order confirmation page is also provided by Shopify itself and can only be customized to a minimal extent. This customization includes adding the logo, selecting the desired background color, choosing from Shopify's predefined fonts, and defining color codes for accents, buttons, and error messages.

When comparing the final result from Shopify (Figure 83a) with the desired design in Figma (Figure 83b) , it is immediately apparent that the two views are vastly different. Unfortunately, it was impossible to make adjustments aligning the default view provided by Shopify with the desired design.



(a) Screenshot of the final Order Confirmation Page in Shopify.



(b) Order Confirmation Page created in the tool Figma.

Figure 83: Overview of the Order Confirmation Page implementation in Shopify.

Final Result

To avoid unnecessarily extending the presentation of the implementation process, additional pages were also created, but they are not relevant for the following comparison. The complete online shop can currently be accessed using the link provided. Visitors can navigate through the website, view the products, and examine the final result of the implementation process in Shopify.

Link to the shop: [BlueBird Online Shop - Shopify](https://bluebird-6806.myshopify.com/)⁵

⁵<https://bluebird-6806.myshopify.com/>

5 Limitations and Problems with the Use of Webflow and Shopify

This section focuses on the limitations and challenges encountered in the study, highlighting the specific areas where each tool demonstrates its unique advantages. By examining these limitations and problems, a deeper understanding of the complexities associated with utilizing the Low-/No-Code tools Webflow and Shopify in the context of e-commerce is gained.

5.1 Definition of a Comparison Table

To make a comparison between the two tools, it is crucial to define a table with various comparison criteria that serve as the basis for the comparison. For this purpose, the overarching criteria mentioned in Chapter 2.3 are used, to which appropriate sub-criteria are assigned. This results in a comparison table with seven overarching criteria, which is presented below (Table 1).

Criteria	Webflow Standard	Shopify Base
<u>Website Editor</u> Templates Onboarding Process Graphical User Interface (GUI) Auto-Save Feature Collaboration	many free templates + libraries multi-step process + tutorial complicated to learn, more settings ✓ 3 staff accounts	few free templates multi-step process easy to learn, restricted in the settings ✗ 2 staff accounts
<u>Customization</u> Customizing the Design Customized Checkout Integrating a Custom Font Customized E-Mails	easy customization of the design just one step, but design is customizable possible through e.g. Adobe API moderately customizable	custom code required pre-build by Shopify custom code required moderately customizable
<u>E-commerce Features</u> Transaction Fee Inventory Management Payment Methods Discounts Test-Mode for the Checkout Process Mobile App	2% per order ✓ Stripe, PayPal, Apple Pay and Google Pay ✓ ✗ ✗	depends on the payment method ✓ PayPal, AmazonPay, Credit-Card, Klarna and many more ✓ ✓ ✓
<u>Pricing</u> Monthly Software Pricing Free Trial	≈26,16€/mo billed yearly ✓	27€/mo billed yearly ✓
<u>Performance and Scalability</u> Amount of E-commerce Items Amount of CMS Items Loading Time	500 2,000 PageSpeed Insights: 83/100	no product limit no limit PageSpeed Insights: 90/100

<i>Development and Code</i>		
Hosting	hosting with webflow or code export	hosting at shopify
Code Quality (Markup Validator)	6 errors	46 errors
Code Export / Vendor Lock-In	code export possible -> no vendor lock-in	code export not possible -> vendor lock-in
Adding Custom Code	easily possible	complicated
<i>Marketing Features</i>		
E-Mail Marketing	only with third-party apps	natively possible
Social Media Marketing	only with third-party apps	natively possible

Table 1: Final Comparison Table.

5.2 Comparing the Tools

In the following chapter, each of the comparison criteria is addressed, and subsequently, the differences or similarities between the two tools in each criterion are detailed. Additionally, preliminary conclusions can be drawn for specific criteria at this stage.

5.2.1 Criterion 1: Website Editor

Templates

First and foremost, templates play a significant role in the development process using Low-/No-Code tools. They serve as a foundation for creating websites and online shops, enabling users to save a substantial amount of time.

Shopify, for instance, relies on templates as a basis for the design process. Users have access to 150 different templates in the template store [76], or users can create a completely custom template, which would require advanced development skills. Also, in the development of the online store of 'BlueBird', a template called 'Dawn' was used. Nevertheless, it is noticeable in the practical application that basic knowledge of CSS is necessary to adapt the template to one's needs for more significant customization wishes.

In contrast, Webflow offers users multiple options: users can use templates from the community, utilize libraries with pre-built components, or create a completely original design and layout. Although Webflow may not demand advanced development skills, it does require more time compared to using existing templates. Moreover, in the development of the store with Webflow, a kind of template was used. Here, a so-called library was used, which provides individual elements for use. When using this library, it is immediately noticeable that adjustments can be made very quickly without code. Almost all design wishes can be implemented in Webflow, and the elements of the library can thus be adapted to one's wishes.

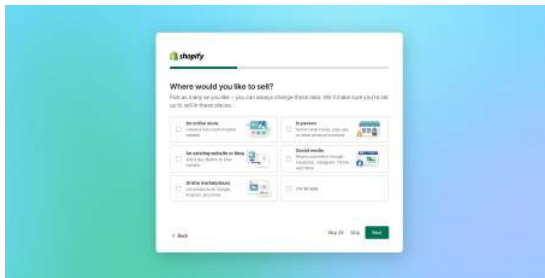
From this perspective, it becomes evident that Shopify relies entirely on templates, and if users wish to make extensive customizations, they would need to modify the template's source code. Webflow however provides greater flexibility, allowing users to use templates if desired, while also providing the option to easily implement their own designs or use specific components from libraries. It can be clearly said that in this criterion, Webflow provides significantly better possibilities in the customization of a template compared to Shopify, as no custom code adjustments are necessary for these.

Onboarding Process

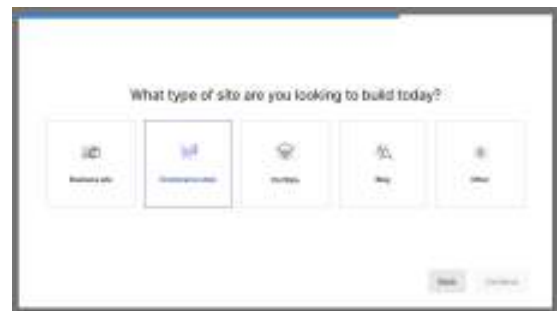
Another important aspect that is significant when comparing both tools is the onboarding process. This involves examining the process that occurs when a new user opens the software. The key factors here include, for example, ensuring that the user is introduced to the basic functions of the system and can navigate it with ease.

In the case of Shopify, the user undergoes a multi-step onboarding process, allowing both experienced users and newcomers to start developing an online shop. Questions "Where would you like to sell?" are answered during this process, and the user is then immediately shown the e-commerce checklist, indicating the necessary steps to begin selling their own products (Figure 84a).

Webflow follows a similar onboarding process, initially asking about the specific use case the user intends to address with the tool (Figure 84b). It then gradually reveals the features of the Webflow Designer to avoid overwhelming the new user.



(a) Screenshot of the onboarding process in Shopify.



(b) Screenshot of the onboarding process in Webflow.

Figure 84: Different onboarding processes provided by Shopify and Webflow

Both tools excel in guiding new users by helping them define their use cases and requirements for the tool. In the case of Webflow, it also provides self-explanatory software guidance. Additionally, both tools ensure that no steps are overlooked during the shop setup process through an e-commerce checklist. Therefore, they are on par in this aspect.

Graphical User Interface (GUI)

The Graphical User Interface (GUI) in both tools is structured very similarly. Using the WYSIWYG principle (Chapter 2) and the drag-and-drop mechanism, a website can be created step by step. However, the two interfaces differ in the level of customization abilities of individual elements.

On one hand, Shopify allows the user to have an overview of the page elements and make certain basic settings for the shop, which are, however, tied to the template's setup. Additionally, if a user wants to make customizations beyond what the template covers, they need to navigate to the source code through the menu and make the desired adjustments using custom code. However, if a user looks at the structure of the GUI in Shopify, it is very clear and structured. Anyone without prior knowledge of this tool can learn and understand how to use it.

On the other hand, Webflow provides users with more options in this regard. Firstly, the user has the Navigator to gain an overview of the elements. Moreover, it is also possible to make various CSS adjustments, within the limits of the selected element, using an intuitive

user interface on the right side. However, if a user looks at the structure of the GUI in Webflow, he will immediately notice that it is similar to common image editing programs. Despite a good onboarding process, the learning curve in this tool is very high [29], and familiarization with Webflow can take a long time with no prior knowledge.

In summary, Webflow's GUI offers more customization options for design adjustments and requires less programming experience than Shopify. Unfortunately, the Shopify GUI is limited by the boundaries of the template and does not always grant users complete freedom in design and implementation. However, looking at the structure of the GUI and the learning curve, Shopify's GUI is significantly faster to learn than Webflows. So, in this criterion, there is no advantage for either tool.

Auto-Save Feature

As the development of the website and its associated individual work steps also need to be saved, there are two different approaches in the presented tools to handle this.

Webflow for example relies on an auto-save feature throughout the entire development process, allowing users to delegate the task of saving their work entirely to the tool and focus solely on implementation.

Shopify requires the user to click the save button to confirm each respective step of their work.

This highlights that Webflow offers a much more user-centered GUI in this aspect, enabling users to focus entirely on implementation. In contrast, using Shopify may lead to issues where users forget to click the save button, resulting in certain steps not being saved. So, considering this criterion, it is clear that Webflow holds a significant advantage over Shopify in the implementation process.

Collaboration

Another crucial aspect that should be considered when comparing the two tools is the ability to collaborate with multiple users on the online shop. Providing companies with the option for their employees to work together with colleagues on the same website at the same time is essential. However, it should be noted that the information for this criterion is derived from the product descriptions of both tools, not from the practical shop implementation.

On one hand, Webflow allows the user to create up to three staff accounts, enabling either clients or internal team members to make changes to the online shop [28].

On the other hand, Shopify only provides users with the option to create two staff accounts [78].

Given that the difference here is only one account, it is evident that Webflow and Shopify do not differ significantly in this aspect.

Conclusion: Website Editor

When considering this criterion to compare both tools, the comparison becomes a bit nuanced. Both Webflow and Shopify offer their users the ability to achieve quick results in the development process using templates. Additionally, the onboarding process is well-handled in both tools, although Webflow's comprehensive explanation of its website editors makes it slightly more user-friendly. The interfaces themselves are well-designed in both tools, but Webflow

holds a slight advantage here as it allows more non-code customization within its website designer. The auto-save feature in Webflow also sets it apart from Shopify, which still requires a manual click on the save button to preserve work progress. Lastly, both tools allow a similar number of team members to collaborate on the online shop. Taking these factors into account, Webflow has a slight edge over Shopify in this criterion.

5.2.2 Criterion 2: Customization

Customizing the Design

As mentioned in Chapter 2.3, the degree to which the online shop can be customized within each tool plays a significant role. This point was particularly evident in the context of this work, as it only required placing the desired design from Figma alongside the two created shops to observe the differences between them.

Looking at the homepage in Webflow, it is immediately noticeable that the page is almost identical to the design created in Chapter 3.1. By using the Relume Webflow Library, new elements can easily be added to the website by dragging and dropping. These elements can then be customized and individually designed as described in Chapter 4.1 (Figure 85b).

As described in Chapter 4.2, customizing the design in Shopify requires significantly more effort, as the design options in the Website Editor are limited. In this case, users must resort to the source code provided by Shopify, which is written in the Liquid programming language, so making the desired adjustments in Shopify involves modifying the CSS code. Thus, at least basic CSS knowledge and quick comprehension are required to customize an online shop in Shopify according to individual preferences. Thus, looking at the result in Shopify, it is evident that the design does not perfectly match the one created in Chapter 3.1. Upon closer inspection, one can observe, for example, that the hero section and navigation bar differ in some aspects from the design in Figma (Figure 85a).



(a) Screenshot of the Homepage of the 'Blue-Bird' online shop in Shopify. (BlueBird Online Shop - Shopify)



(b) Screenshot of the Homepage of the 'Blue-Bird' online shop in Webflow. (BlueBird Online Shop - Webflow)

Figure 85: Different Homepages implemented in Shopify and Webflow.

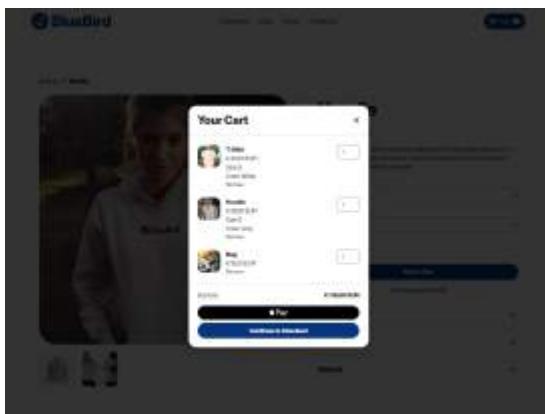
When considering this criterion, it can be clearly stated that Webflow has an advantage over Shopify. Webflow empowers its users to customize designs and even templates to meet their needs without requiring code. In contrast, Shopify users are confined to the limitations of the template. If they need to make adjustments beyond the template's constraints, they have to modify the source code, which involves at least basic knowledge of HTML and CSS.

Customized Checkout

One aspect that is particularly relevant in the context of e-commerce is the customization of the checkout process. There are several differences between Webflow and Shopify in this regard.

Webflow, on one hand, offers its users the option to choose from four different shopping cart display styles. It can either be a dropdown, a column on the left or right side of the screen, or a popup. While this display can be customized in terms of design, creating a separate page for it is not possible (Figure 86a).

On the other hand, when using Shopify, the users are reliant on the shopping cart display style provided by the template. In the case of the 'Dawn' template used here, the shopping cart is presented on a separate page. With modifications in the source code, its appearance can also be adjusted (Figure 86b). So, in terms of the shopping cart display, Shopify's implementation is closer to the desired outcome than in Webflow.



(a) Screenshot of the Shopping Cart of the 'BlueBird' online shop in Webflow. (Blue-Bird Online Shop - Webflow)



(b) Screenshot of the Shopping Cart of the 'BlueBird' online shop in Shopify. (BlueBird Online Shop - Shopify)

Figure 86: Different Shopping Carts implemented in Shopify and Webflow.

However, looking at the checkout process itself, it becomes evident that Shopify significantly limits customization for users. Shopify relies on its standard checkout, which looks nearly the same in every shop created in that tool. Here, the user can only insert the desired logo, adjust the background color, and select from a few predefined fonts for the checkout process. Changing the order or appearance of input fields or checkout steps is not possible in Shopify (Figure 87a). The presentation of the order confirmation page cannot be customized either, and Shopify provides it as is, which is substantially different from the desired design (Figure 88a). Therefore, it is clear that the checkout process created in Shopify differs significantly from the desired outcome.

In contrast, Webflow allows for much more customization of the checkout process. While the process can't be split into multiple subpages and all buyer, shipping, and payment information needs to be entered on a single page, these input fields and their appearance can be adapted as desired to match the design created in Figma. The outcome might vary slightly from the desired design, but it resembles it much more closely compared to Shopify (Figure 87b). The order confirmation page in Webflow can also be customized according to individual needs. It is evident that Webflow provides users with significantly more possibilities to adapt the design in this aspect. The order confirmation page created in Webflow, as seen, only slightly deviates from the desired design (Figure 88b).

(a) Screenshot of the Checkout of the 'BlueBird' online shop in Shopify. (BlueBird Online Shop - Shopify)

(b) Screenshot of the Checkout of the 'BlueBird' online shop in Webflow. (BlueBird Online Shop - Webflow)

Figure 87: Different Checkouts implemented in Shopify and Webflow.

(a) Screenshot of the Order Confirmation Page of the 'BlueBird' online shop in Shopify. (BlueBird Online Shop - Shopify)

(b) Screenshot of the Order Confirmation Page of the 'BlueBird' online shop in Webflow. (BlueBird Online Shop - Webflow)

Figure 88: Different Order Confirmation Pages implemented in Shopify and Webflow.

Looking at this criterion, it is evident that Webflow holds an advantage over Shopify. Despite minor limitations in the checkout process and shopping cart structure, Webflow enables users to customize nearly every element within the checkout process to their preferred design. However, Shopify users are constrained to the pre-defined checkout structure with limited customization options.

Integrating a Custom Font

To achieve the highest level of individuality in designing a shop, it is also important to consider how users can integrate custom fonts, such as those from third-party providers. In the case of the online shop 'BlueBird', the font 'Neue Haas Grotesk Display Pro' from the Adobe Typekit platform was used.

When integrating this font into a Webflow project, users simply need to navigate to the *Site Settings* and access the Fonts tab. There, the Adobe API token can be entered, which is easily found in the Adobe account. Once this token is entered, different Typekit projects can be selected. In this scenario, the 'BlueBird' project containing the desired font was created. This project can then be chosen within the Webflow *Site Settings*. After a short loading time, the fonts will be available in the Webflow Designer and can be easily utilized (Figure 89a).

However, Shopify makes it more challenging for its users to integrate such third-party fonts. To incorporate fonts from Adobe Typekit, a link tag for the metadata needs to be generated within Adobe initially. This tag must then be added to the shop's metadata in Shopify's template source code. Next, the font variables found under the file 'base.css' must be overwritten and replaced with the name of the desired font (Figure 89b).



(a) Screenshots of the integration process of the font in Webflow.



(b) Screenshots of the integration process of the font in Shopify.

Figure 89: Different custom font integration process in Shopify and Webflow.

Comparing the effort required for both integrations, it is clear that Webflow significantly simplifies the process of integrating custom fonts into the online shop. If a user still wants to do this in Shopify, the process involves much more effort and requires knowledge of HTML and CSS. In contrast, Webflow does not necessitate any source code adjustments for this step. So, looking at this criterion, it is evident that Webflow again holds an advantage over Shopify.

Customized E-Mails

Another crucial point that should not be overlooked is the customization of e-mails sent to customers after, for instance, placing an order.

In Shopify, for example, various templates are provided to the user (Order Confirmation, Order Cancelled, etc.) that can be edited. However, Shopify only allows the user to insert their

own logo, adjust its size, and set the accent color for call-to-action elements. All other essential information in the e-mails comes directly from Shopify and is not customizable (Figure 90).

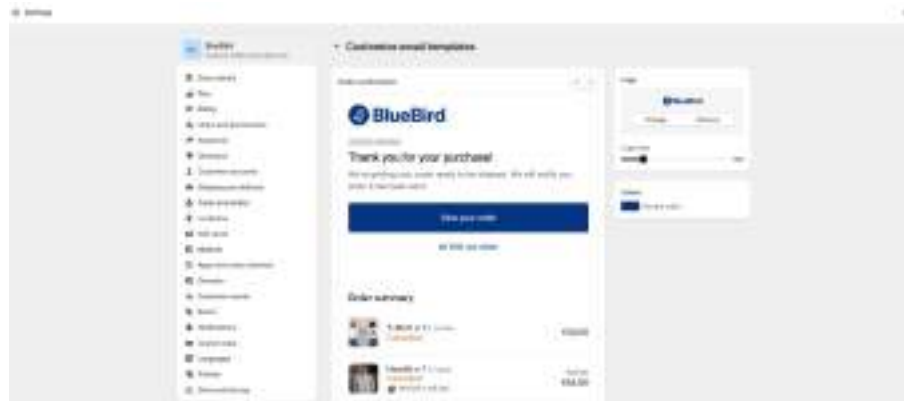


Figure 90: Screenshot of the customization of the e-mail templates in Shopify.

Webflow also offers different e-mail templates for various use cases (Order Received, Order Confirmed, etc.). However, Webflow, like Shopify, does not allow its users to completely customize these e-mail templates according to their own preferences. The desired logo can be added, the accent color for call-to-action elements and the background color can be set, and only the text content itself can be modified by the user. Nevertheless, even with the selected Webflow E-Commerce package, a Webflow branding remains visible at the bottom of the e-mail (Figure 91).



Figure 91: Screenshot of the customization of the e-mail templates in Webflow.

In conclusion, both tools do not significantly differ in this criterion. Both platforms allow minimal customization of the e-mail templates to suit the needs of the online shop. Therefore, neither of the two tools has an advantage over the other in this aspect.

Conclusion: Customization

When considering this criterion to compare both tools, it becomes evident that Webflow holds a distinct advantage over Shopify in this aspect. Firstly, in Webflow, the design can be customized down to the tiniest detail to match the user's desired needs. Secondly, even the checkout process can be tailored to fit the desired design. The integration of third-party fonts also works

seamlessly. There are slight compromises with Webflow concerning the display of error messages and e-mails. However, if a user desires an online shop that can be adjusted to their design vision as much as possible, then Webflow is the preferable choice.

5.2.3 Criterion 3: E-commerce Features

Transaction Fee

When examining the two presented tools, it is essential to consider the differences in features that are crucial for managing an e-commerce platform. Firstly, the transaction fee is worth mentioning, as it is a percentage that is applicable to every order processed through the respective tool and must be paid to the respective company. However, it should be noted that the information for this criterion is derived from the product descriptions of both tools, not from the practical shop implementation.

Webflow defines this percentage at 2% per order, which goes to Webflow. Additionally, there are costs associated with the connected payment processors (Paypal and Stripe) discussed in Chapter 5.2.3 [28].

Shopify is a bit more complex in this regard, as it offers more payment processors options compared to Webflow. For example, the Transaction Fee for payment with Sofort Überweisung is 1.4% with an additional fee of 25 cents, for payment with Klarna Pay Now it is 1.5% with an additional fee of 30 cents, for European credit card payments it is 2.1% with an additional fee of 30 cents, and for any other third-party payment providers (if not using Shopify Payment), the Transaction Fee is 2.0% of the order [78].

From this, it can be observed that Webflow keeps the percentage consistent, making it relatively clear how much the user earns with each order, considering the limited number of supported payment providers. On the other hand, Shopify calculates transaction fees in a more complex manner, as it demands both a percentage of the order and an additional fixed amount per order. Since the percentage can be lower than that of Webflow in some cases, such as for purchases using Sofort Überweisung, the user may pay lower fees with Shopify. However, for credit card payments, the user may end up paying more in fees with Shopify compared to Webflow. So, in this criterion, neither tool has a significant advantage.

Inventory Management

Another important point that should be addressed in the comparison of both tools is inventory management. Inventory management oversees and controls a company's amount of products, ensuring that the right products are available in the right quantities at the right time. This criterion includes how easily the offered products can be maintained and adjusted in the backend.

For example, in Webflow, users can easily adjust the data of each product through the *Products Panel*. This includes adding images, text, prices, and available quantities of products. Unfortunately, due to the lack of a test mode for orders and a missing business account with Stripe or PayPal, it wasn't possible to test how an order view could further enhance inventory management.

Shopify addresses this aspect in a similar manner. Here as well, product data can be easily adjusted through the *Products Panel*, including images, sizes, and quantities. Additionally, there's

a separate section called *Inventory* where the quantity of each individual product variant can be easily managed and edited.

Upon examining this criterion, it is evident that Shopify places a much greater emphasis on easy shop management. With its dedicated *Inventory* section where the quantity of each product variant can be managed separately, users are provided with extreme convenience, eliminating the need to manage these details within each individual product. This level of functionality is not available in Webflow. In Webflow, users still need to navigate into each individual product to make changes to the inventory, for instance. Thus, it is clear that Shopify has a distinct advantage over Webflow in this regard.

Payment Methods

To offer users a wide range of payment options according to their preferred payment method, various payment methods need to be integrated into the platform.

Webflow currently provides the option to integrate payment services like Stripe, PayPal, Google Pay, and Apple Pay. However, other payment methods are not currently supported.

Shopify offers a significantly broader range of payment methods. Users can pay with various credit cards, Apple Pay, Google Pay, as well as options like Klarna or Sofortüberweisung. This is achieved through the integration of their own payment gateway called 'Shopify Payments'. Additionally, Shopify allows the integration of many other payment methods, including cryptocurrencies and international payment options, based on user needs.

In this criterion, Shopify holds an advantage over Webflow as it offers users far more options for payment. If a Webflow-based shop does not support one of the four specified payment providers, the user cannot complete their purchase. In contrast, Shopify's extensive array of payment methods significantly reduces the chances of a buyer not having access to the required payment method. Therefore, Webflow has room for improvement in this aspect if it aims to compete with Shopify effectively in the long term.

Discounts

A highly significant feature for online shops in today's era is the use of discount codes. These codes can be employed for promotional purposes and enable customers to receive discounts on the offered products.

For instance, Webflow provides a way to create discount codes. Users can choose the type of discount (percentage, fixed price reduction, or free shipping) and customize the name of the discount code as desired.

Similarly, Shopify offers this feature as well. Shopify allows users to assign any name to the discount code, choose between a percentage or fixed amount discount, and set limits on the maximum usage of the discounts.

In the realm of discount codes, both Shopify and Webflow offer robust capabilities. Both tools provide users with ample options to customize these discount codes according to their preferences.

Test-Mode for the Checkout Process

One of the most important features required during the development of an online shop is a test mode for the checkout process, allowing users to go through all the steps needed to complete a purchase and identify any potential errors.

On one hand, Shopify offers such a test mode. Upon activating the test mode in the payment settings, users can simulate a successful transaction by providing credit card information from the Testing Shopify Payments website. For a successful simulation, users need to input at least two words for the cardholder's name, any future date for the expiration date, three arbitrary digits for the security code, and one of the provided test card numbers as the card number. Shopify also provides credit card numbers for simulating failed transactions, ensuring users are prepared for all scenarios [79]. After going through the checkout process step by step, the test order then appears just like a regular order but is marked as a test order in the order overview (Figure 92).

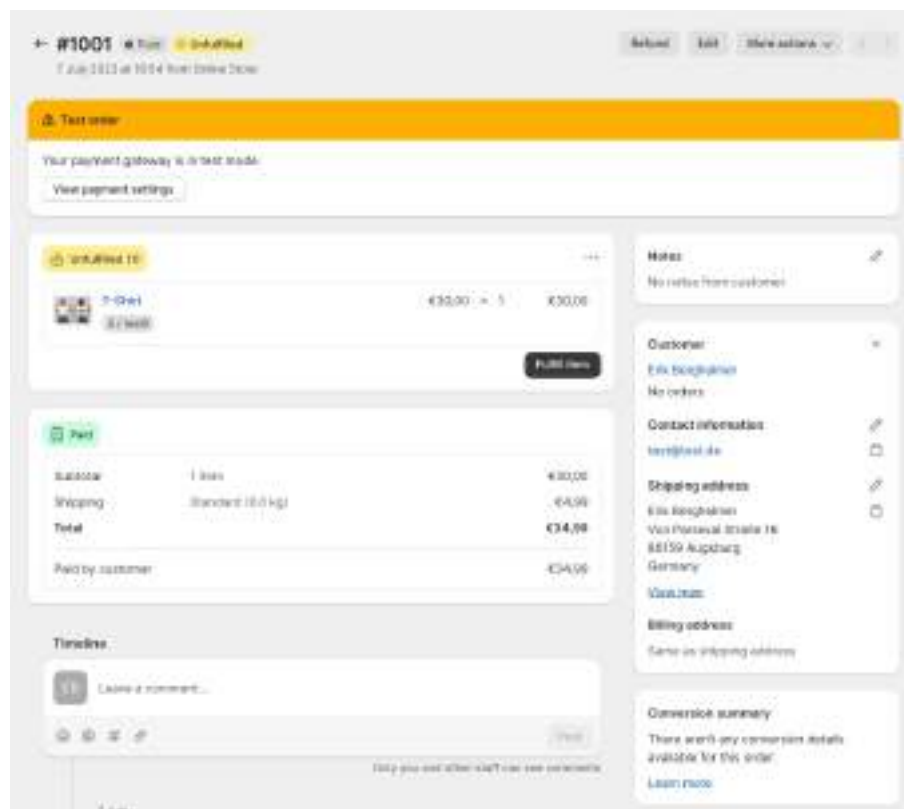


Figure 92: Screenshot of the test order in Shopify.

On the other hand, Webflow does not currently offer such a feature, despite being requested as a new feature on their wishlist back in 2018 [80]. To perform a test order in Webflow, users have to use a workaround, involving a valid credit card and a 100% discount code. However, this test mode in Webflow requires the integration of a payment provider. Since setting up a payment provider requires a company account with a valid tax number on either PayPal or Stripe, this test mode in Webflow has not been tested.

This aspect clearly highlights that Webflow has significant room for improvement when it comes to testing the created online shop. The absence of this feature has been a determining

factor for some users to not use Webflow as the technology for their shop. In this regard, Shopify stands out with its well-designed test mode, giving it a distinct advantage.

Mobile App

To have the ability to manage the online shop and orders from anywhere as a business owner, having a mobile app that provides complete shop management is advantageous. However, it should be noted that the information for this criterion is derived from the product descriptions of both tools, not from the practical shop implementation.

On one hand, Shopify offers such a mobile app that allows users to control and manage orders, shipping, inventory, and payments from their smartphones. This app is available on the Apple App Store and Google Play Store, enabling users to keep track of their shop even while on the go [81]. On the other hand, Webflow does not provide a similar app. Shop management for Webflow must be done entirely through the platform on a computer. While order notifications can be received via e-mail, processing these orders requires access to the online platform on a computer.

This aspect highlights that Shopify demonstrates more experience and innovation in terms of e-commerce features. The Shopify online shop can be more conveniently managed on the go through its mobile app, whereas Webflow relies on the online platform accessible via a computer.

Conclusion: E-commerce Features

Considering this criterion, it becomes clear that Shopify has a clear advantage over Webflow. While Shopify requires a slightly higher transaction fee per order to be paid to Shopify, Shopify shines with its easy inventory management, various payment methods, and the ability to create discount codes. The two most significant advantages, however, that set Shopify apart from Webflow are the Test-Mode, which allows testing of the checkout process, and the Mobile App, which provides shop owners with insights into their orders while on the go. In conclusion, it can be clearly stated that Shopify is much better suited for this criterion than Webflow.

5.2.4 Criterion 4: Pricing

Monthly Software Pricing

The monthly price for each tool's provision plays a crucial role, as more and more companies seek to save money in a world facing increasing inflation and rising costs. This criterion exclusively compares the costs incurred to have the tool provided by the company. However, it should be noted that the information for this criterion is derived from the product descriptions of both tools, not from the practical shop implementation.

The monthly price for the Webflow E-commerce Standard package is \$29 per month with an annual subscription, currently approximately equivalent to 26,30€ [28].

On the other hand, Shopify charges 27€ per month for the Shopify Basic package with an annual subscription [78].

It is evident from this comparison that both packages selected for this study fall into the same price category, making them highly comparable.

Free Trial

To make it as easy as possible for new users to try out their desired tool, many platforms offer users a free trial period.

Webflow provides new users with the opportunity to create a free Starter site. There, you can already try out how a potential shop would look, how it could be designed and managed in Webflow.

Shopify has a similar approach in this aspect. Shopify allows its users to try the tool with a low entry barrier. Anyone interested can test Shopify completely free for three days without providing credit card information. Additionally, Shopify occasionally runs discount promotions where users can use Shopify for the first three months at a monthly cost of 1€.

In this regard, none of the tools has an advantage as both of them enable new users to try out their tool with low initial costs.

Conclusion: Pricing

In this aspect, neither of the two tools has an advantage. Both tools are in a similar price range, and both Webflow and Shopify offer new users the opportunity to test the tool for free through a low entry barrier. It can be clearly stated that neither of the two tools has an advantage in this criterion.

5.2.5 Criterion 5: Performance and Scalability

Amount of E-commerce Items

The scalability of the online shop is, of course, a significant factor in comparing these two tools. It is crucial for the development of the online shop and the growth of a brand to consider limitations on the number of products offered.

Shopify does not have a real limit on the number of e-commerce items. However, once a shop reaches 50,000 different product variants, Shopify limits the number of variants that the user can create per day to 1000. If this limit is reached, the user must wait for 24 hours to create new variants [82].

In contrast, Webflow allows a maximum of 500 different e-commerce items to be created, including products, variants, and categories [28].

So, Shopify allows its users to create significantly more e-commerce items than Webflow. This aspect can be a crucial point for fast-growing online stores, as Webflow has a limit of 500 e-commerce items. This limitation of Webflow clearly shows that Shopify has an advantage over Webflow here.

Amount of CMS Items

Another point to consider is the number of CMS items that can be created with each tool. These items can be used for various purposes. In the case of this thesis, the CMS was used for managing and creating blog articles.

Webflow offers the possibility to create up to 2000 CMS items [28].

In contrast, Shopify has no limitations on CMS items. However, the display of blog articles on, for example, the blog overview page is limited to a number of 50.

Also, when considering this criterion, it becomes evident that Shopify holds an advantage over Webflow. The limitation in Webflow, with a cap of 2000 CMS items, contrasts Shopify's lack of CMS restrictions, particularly in blog articles. This further underscores Shopify's suitability as the right tool for highly scalable businesses with substantial demands in the blog article domain.

Loading Time

The loading time of a website is a crucial factor to consider in the context of constant accessibility. Tools like PageSpeed Insights⁶ are used to analyze and display the performance score of a website.

In this regard, Webflow scores slightly lower with an overall score of 83 (PageSpeed Insights BlueBird Webflow⁷) compared to Shopify's score of 90 (PageSpeed Insights BlueBird Shopify⁸). This difference can be attributed to Webflow's customization. Although Webflow allows uploaded images to have a maximum file size of 4MB, the platform does not automatically compress these images. In contrast, Shopify automatically compresses uploaded images, which may result in slightly reduced image quality on the website's display [83].

It is evident that there is not a definitively better tool in this aspect. Webflow offers users complete design freedom and does not automatically compress uploaded images, while Shopify focuses on faster loading times, potentially at the cost of slightly reduced image quality in the display.

Conclusion: Performance and Scalability

In the 'Performance and Scalability' criterion, it is relatively clear that Shopify has the advantage. Firstly, Shopify doesn't impose any limits on the creation of e-commerce items. Users can expand their shop effortlessly from a few products to several hundred products. Additionally, an unlimited number of blog articles can be created, providing users with extensive freedom. Compared to Webflow, performance is a minor factor. Although the difference is not significant, Shopify gains an edge through its automatic image compression, contributing to faster loading times. Webflow prioritizes higher image quality, which slightly affects loading times. In conclusion, it is evident that Shopify is better suited for this criterion than Webflow.

5.2.6 Criterion 6: Development and Code

Hosting

In the realm of development, it is also essential to examine how the hosting of the online shop created in each tool is handled.

For instance, Webflow provides the user with the option to either host the created shop entirely with Webflow itself or export all the data (separate HTML, CSS, and JavaScript files) and host it with another hosting provider.

Shopify limits the user's options and only allows hosting with Shopify itself.

⁶<https://pagespeed.web.dev/>

⁷https://pagespeed.web.dev/analysis/https-bluebird-e9e00e-webflow-io/s5008zkt0h?form_factor=desktop

⁸https://pagespeed.web.dev/analysis/https-bluebird-6806-mys Shopify-com/dspd12ks5w?form_factor=desktop

This point demonstrates that Webflow does not force its users to host with them, as they offer the flexibility to use external hosting providers. However, hosting with an external provider means losing access to the user-friendly GUI provided by Webflow, which makes shop management and administration much more straightforward.

Code Quality

When considering both tools, the quality of the generated code is a significant factor that should be taken into account. For this purpose, tools like the W3C Markup Validation Service⁹ are used to assess the markup validity of web documents, such as HTML files. To compare the codes of both platforms, the homepage links of both created online shops were input into the Markup Validator.

Upon reviewing the results of Shopify's validation (Result Markup Validator - Shopify¹⁰), it is immediately apparent that there are 46 error messages detected by the HTML validator. These errors range from improper use of style elements as child elements of div blocks to missing end tags of HTML elements (Figure 93).



Figure 93: Screenshot of the results of the Markup Validator for the online shop developed in Shopify.

In contrast, when looking at the results of Webflow's validation (Result Markup Validator - Webflow¹¹), there are only 6 error messages, which include issues such as duplicated use of IDs or an empty src attribute in an img element (Figure 94).

⁹<https://validator.w3.org/>

¹⁰<https://validator.w3.org/nu/?doc=https%3A%2F%2Fbluebird-6806.myshopify.com%2F>

¹¹<https://validator.w3.org/nu/?doc=https%3A%2F%2Fbluebird-e9e00e.webflow.io%2F>

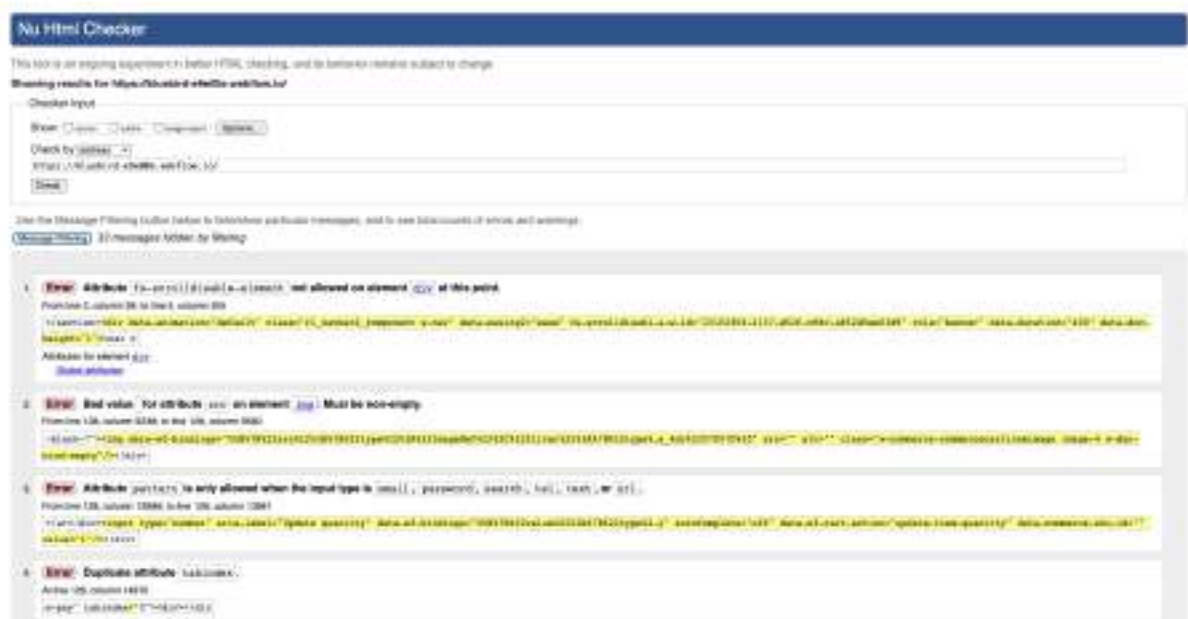


Figure 94: Screenshot of the results of the Markup Validator for the online shop developed in Webflow.

Examining the code quality of both tools, the number of errors identified by the validator stands out. Shopify generates 46 error messages on the homepage alone, while Webflow produces only 6 error messages. Furthermore, when delving into the error messages themselves, it is challenging for users with basic knowledge of HTML and CSS, who are using Low-/No-Code tools, to even comprehend the error messages from Shopify's validation. The code generated by Shopify appears convoluted and opaque, making it difficult to read. However, the generated code from Webflow has error messages that are much easier to understand, and the code itself can be comprehended with basic HTML and CSS knowledge.

In the context of this criterion, it can be easily concluded that Webflow generates significantly cleaner and more understandable code for users with basic HTML and CSS knowledge. This code results in fewer errors in markup validation tools compared to Shopify. Analyzing the example of the homepages of both created online shops, it is evident that the code quality generated by Webflow is notably higher than that of Shopify.

Code Export / Vendor Lock-In

As briefly mentioned in Chapter 5.2.6, the possibility of code export from a Low-/No-Code tool is also a significant factor when choosing a platform. Code export allows users to use other hosting providers or share the code with a software developer who can suggest changes in the source code and evaluate its quality.

Webflow indeed provides users with the option to export all files, including HTML, CSS, and JavaScript files, as well as any inserted attachments, with a simple button click. However, this feature comes with additional costs amounting to an extra \$19 (currently approx. 17,60€) per month. It is also essential to note that the exported code does not include a functional CMS, search functionality, form features, or e-commerce capabilities (Figure 95).



Figure 95: Screenshot of the code export popup in Webflow.

On the other hand, Shopify only allows users to export certain elements, such as products, as CSV files. To access the source code itself, users would need to view it through developer tools or custom code settings and then manually copy individual components.

In this regard, it can be clearly stated that if a user chooses the tool Shopify, they should stick with this tool. A possible code export is not possible, which creates a vendor lock-in with Shopify. Webflow offers a code export option to its users, providing a high level of flexibility. However, it is important to consider that their straightforward code export comes with added costs and limited usability. In conclusion, it can be said that there is no vendor lock-in with Webflow.

Adding Custom Code

Furthermore, in some use cases, it may be necessary to incorporate custom-written code into the online shop.

Webflow allows users to easily add custom code blocks to almost any part of the online shop by inserting a 'Custom Code' element. Users can do this directly within the Webflow Website Designer, which emphasizes Webflow's WYSIWYG approach. However, in Shopify, if a user wants to add custom-written code to the shop, they need to navigate from the Shopify Website Designer to the template code, where various folders containing different elements are visible. Navigating through this folder structure can be cumbersome, as it is not immediately intuitive, and it is challenging to determine if the edited source code corresponds to the element used in the shop or if it is even in use at all.

In this regard, it is evident that Webflow makes it significantly easier for users to incorporate custom-written code into the shop. Users can stay within their familiar Website Designer environment and add a custom code block almost anywhere with ease. However, Shopify makes it relatively challenging for users to add custom code to the template, requiring more coding

experience to navigate the folder structure, locate the relevant files, and make the necessary adjustments accurately.

Conclusion: Development and Code

For the criterion of 'Development and Code,' it can be clearly stated that Webflow offers its users significantly more flexibility than Shopify. With Webflow, users can choose a different hosting provider if needed. The generated code is simpler and more streamlined, and there is no vendor lock-in. Webflow also allows the user to export the source code and insert custom code blocks at almost any point within the online shop. Therefore, in this criterion, Webflow is significantly better suited than Shopify.

5.2.7 Criterion 7: Marketing Features

E-Mail Marketing

Marketing, especially e-mail marketing, is a powerful tool for online shop owners to expand their business and build customer relationships. However, it should be noted that the information for this criterion is derived from the product descriptions of both tools, not from the practical shop implementation.

Shopify on one hand offers various options in this regard. Firstly, Shopify provides the option to send automated e-mails to customers who have abandoned their checkout process. Each of these e-mails includes a link to the abandoned checkout, allowing users to complete their purchase [84]. Additionally, Shopify offers its own free app for Shopify online shops (Shopify Email) that allows users to create e-mail marketing campaigns. These promotional e-mails are free for the user up to a limit of 10,000 e-mails [85].

On the other hand, Webflow does not offer such e-mail marketing features. If a user abandons the checkout process, abandoned cart e-mails cannot be sent as they can with Shopify. Additionally, for any e-mail marketing campaigns, third-party apps need to be integrated into the shop, which do not come from Webflow itself [86].

This point clearly demonstrates that Shopify has an advantage in this criterion. Shopify provides users with various options to utilize e-mails in marketing and increase conversions. Webflow has some catching up to do in this aspect to reach the level of e-mail marketing capabilities that Shopify offers.

Social Media Marketing

In a world where more and more people are active on social media platforms like Instagram [87] or Facebook [88], it is important for online shop owners to consider social media marketing.

With Shopify, for instance, product sharing on social media platforms like Facebook, Twitter, Reddit, LinkedIn, and Pinterest can be done directly from the product inventory page using the 'Share' button. It allows products to be shared or links to products to be created for sharing across platforms. Moreover, there are numerous apps available in the Shopify App Store that can integrate social media features into the shop. Lastly, the Shopify dashboard includes a 'Marketing' section that provides a comprehensive overview of all marketing campaigns. It also enables the creation of Pinterest or Snapchat marketing campaigns that can be tracked directly within Shopify, eliminating the need for a separate tool to manage marketing campaigns (Figure 96).

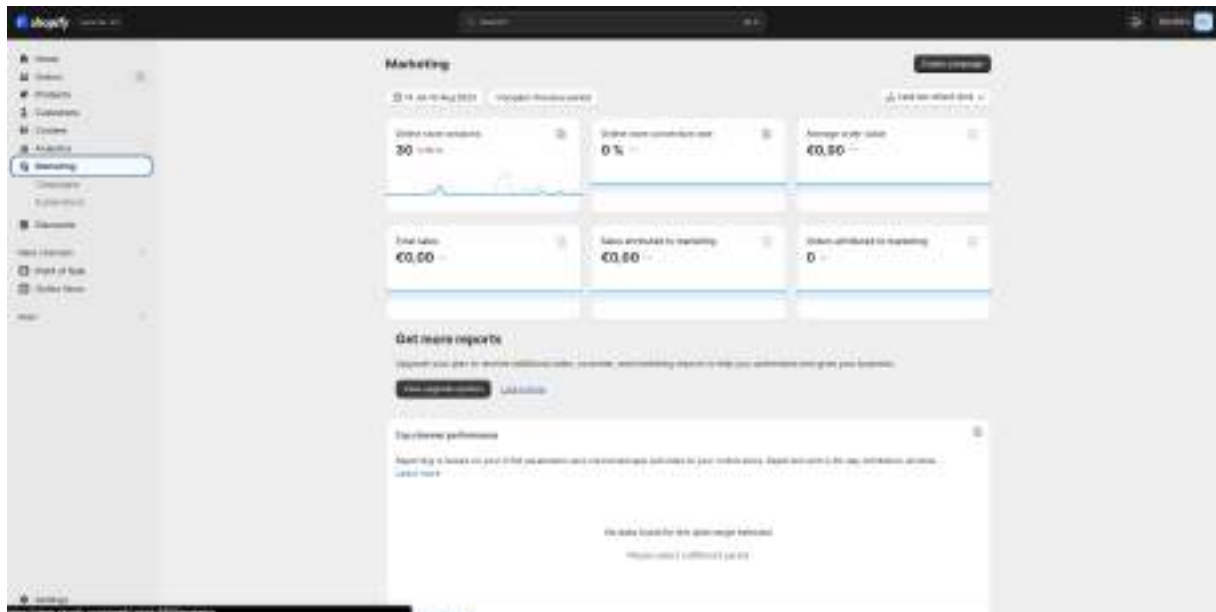


Figure 96: Screenshot of the marketing settings in Shopify.

However, Webflow does not offer any native marketing features within its shop system. Any social media marketing functionalities have to be manually integrated into the shop using third-party solutions.

Once again, in this criterion, Shopify clearly holds an advantage over Webflow in terms of social media marketing. Shopify's native integration of a marketing dashboard eliminates the need for relying on third-party apps for social media marketing. In contrast, Webflow requires third-party apps to provide such functionalities. This highlights an area where Webflow needs further development if they intend to compete with Shopify in the e-commerce sector.

Conclusion: Marketing Features

In the criterion of 'Marketing Features', it is quite evident that Shopify offers its users significantly more features and capabilities compared to Webflow. Shopify allows users to manage e-mail marketing and social media marketing natively within the platform. While these functions are also possible in Webflow, they require third-party apps, often incurring additional costs. Shopify demonstrates that it goes beyond just building and managing an online shop; it also integrates native marketing for the created shop and its products. In conclusion, it is clear that Shopify is a far superior choice to Webflow in this criterion.

6 Conclusion

This thesis aims to investigate and identify the limitations and challenges faced by the two Low-/No-Code tools, Webflow and Shopify, in the context of e-commerce. In the following and final section, recommendations for individuals who are currently employing or planning to adopt Webflow or Shopify in their ventures are presented. Furthermore, this section explores the potential and possibilities that emerge from leveraging Low-/No-Code tools. Additionally, it addresses lingering research questions and proposes avenues for further exploration. Lastly, the limitations of this study will be acknowledged, recognizing areas that would benefit from future research and investigation.

6.1 Recommendations for Practice

Upon comparing both tools and considering the applied comparison criteria, it becomes evident that Webflow and Shopify cater to different use cases.

Webflow, being a relatively new tool in the e-commerce market, stands out with its exceptional customization options. Almost every element, except for the shopping cart, can be designed and adjusted according to the user's preferences. Webflow is ideal for users who aim to create a completely unique and custom-designed online shop, but may not prioritize the extensive e-commerce features that are not as well-developed in Webflow compared to Shopify. For instance, it lacks a test mode for the checkout process and integration with payment providers other than Stripe or PayPal.

In contrast, Shopify is an established player in the e-commerce tools market, existing since 2006. It excels in providing extensive e-commerce features, ranging from integration with numerous payment providers to a test mode for the checkout process and even a mobile app for order management. All these functionalities demonstrate that Shopify places a strong focus on e-commerce features. If a user's priority is to create an online shop that follows best practices in the checkout process, and is willing to use templates as a foundation, then Shopify is the right choice. However, for users seeking a highly personalized online shop without coding skills in areas like Liquid, Ruby or CSS, it may be best to avoid using Shopify. Minor customizations often require changes in the template's source code, which can be dependent on the chosen template.

There is no clear recommendation for practical use. Both tools have their advantages and disadvantages, and the decision to use a particular tool should be based on the specific use case and requirements.

6.2 Potentials and Development possibilities of No-Code Tools

When considering Webflow and its planned future features, it becomes clear that there will be significant changes in the development process, overall usability, and workflows. One noteworthy feature is the 'Figma to Webflow' plugin, which allows users to seamlessly transfer designs from Figma to Webflow using copy-and-paste actions [89]. Additionally, Webflow has plans to incorporate artificial intelligence (AI) into their tool, enabling users to translate text content, reveal hidden settings in the interface to new users, and even create and design entire components using AI [90]. In 2023, Webflow also intends to integrate a function for adding multilingual capabilities to their tool [91]. These anticipated enhancements are indicative of

Webflow's commitment to continuously improving and expanding its platform's capabilities to meet the evolving needs of users.

Shopify is also keen on continuously integrating new innovative features into their platform. The so-called 'Shopify Magic' feature is already available within the tool. This artificial intelligence enables users to significantly expedite their writing process. For instance, blog posts, product descriptions, and e-mails to users can be artificially generated [92]. Additionally, in August 2023 available for early bird access, another new feature named 'Sidekick' will be integrated into the development process. This feature is also an artificial intelligence designed to serve as an e-commerce assistant in setting up, operating, and expanding a business on Shopify. Its purpose is to for example enhance the quality of the shop and streamline time-consuming tasks [92].

Indeed, it is evident that both companies are working diligently to make their tools even more user-friendly and accessible, thereby enticing a broader audience of citizen developers to utilize their platforms for software creation.

6.3 Limitations

In addition to the insights highlighted in this study, several limitations can be identified that were not considered within the scope of this work. For instance, the focus during the creation of the online shop was on the desktop view, excluding a detailed examination of how the online shops adapts to mobile breakpoints. Additionally, there are other providers for creating online shops, such as WooCommerce or Wix, which were not examined in this work. Furthermore, the goal of this study was not to identify the perfect tool for every use case. Rather, the aim was to introduce both tools, define their strengths and weaknesses, and provide recommendations for prospective users who are in the process of deciding on which platform to choose for developing their online shop.

6.4 Open Research Questions and Further Studies

Following the findings of this thesis, it would be interesting to explore how and whether the insights gained from the development process apply to mobile devices. Furthermore, a future study could investigate how an online shop with real users and customers can be managed and adapted using the two presented tools, Webflow and Shopify, during ongoing operations.

Moreover, it is certainly possible to expand the comparative process by including other relevant tools with a substantial user base, such as WooCommerce or Wix, thus broadening the scope of the comparison.

Potential research questions could be:

- How does the development process using the two Low-/No-Code tools, Shopify and Webflow, affect the mobile display of an online shop?
- How do the two tools, Webflow and Shopify, differ in terms of adaptability and maintenance while real customers interact with the online shop?
- To what extent do the tools Webflow, Shopify, WooCommerce, and Wix differ from each other, and what practical recommendations can be derived from these differences?

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Appendices

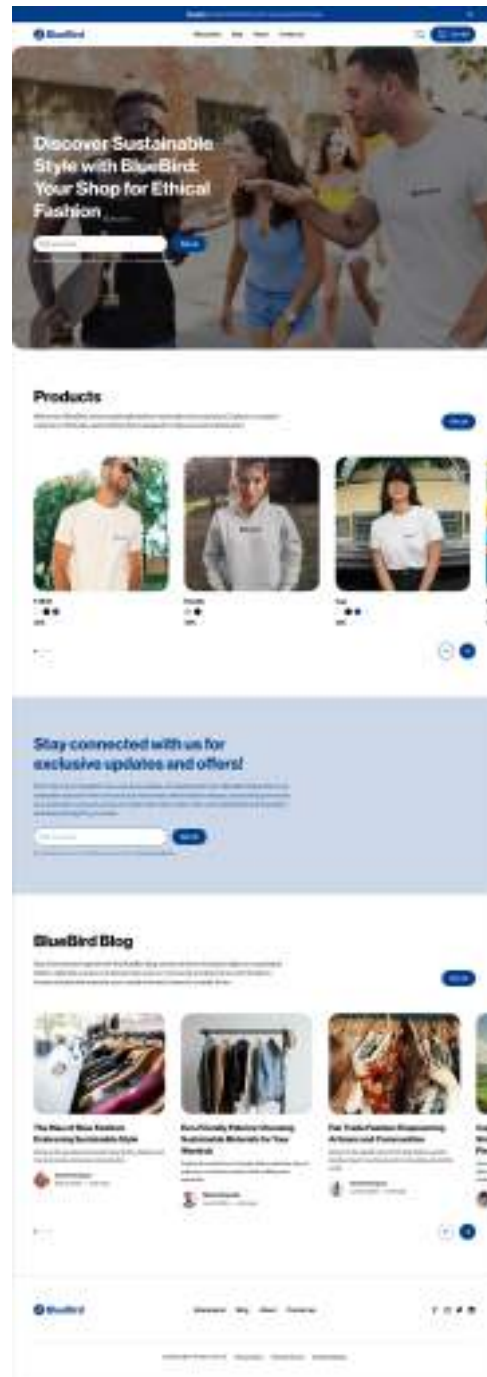


Figure 97: Final designed Homepage of the online shop of 'BlueBird'.

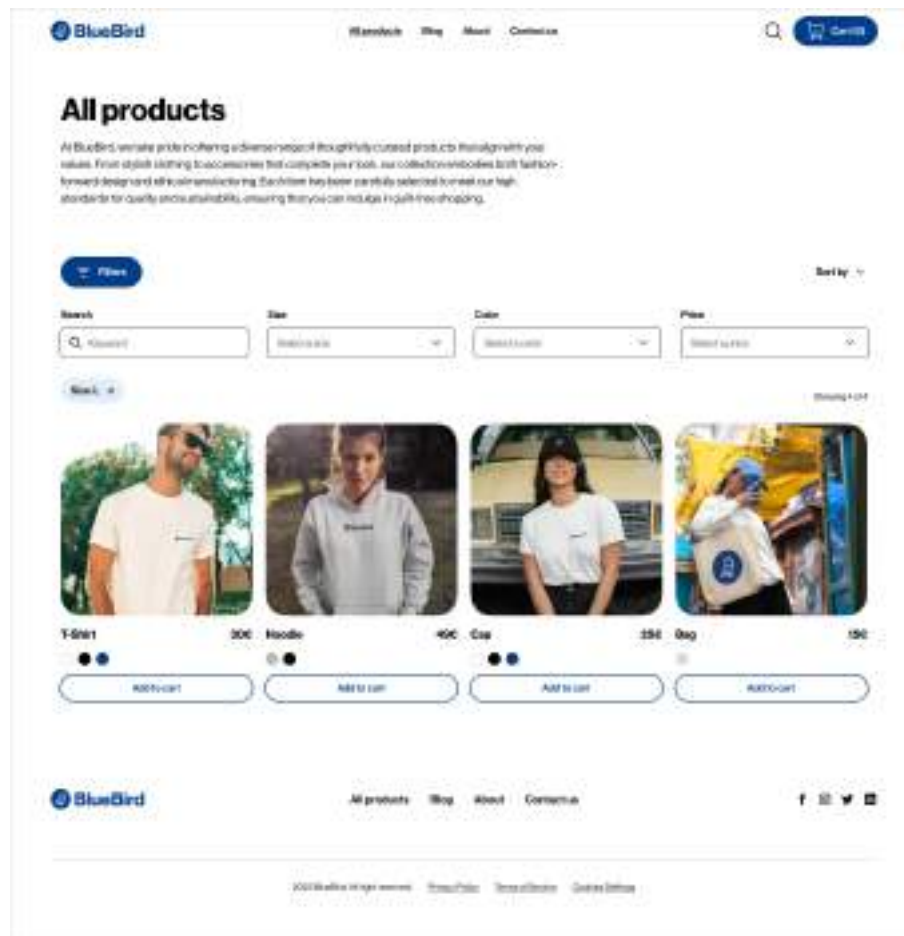


Figure 98: Final designed Product Overview page of the online shop of 'BlueBird'.

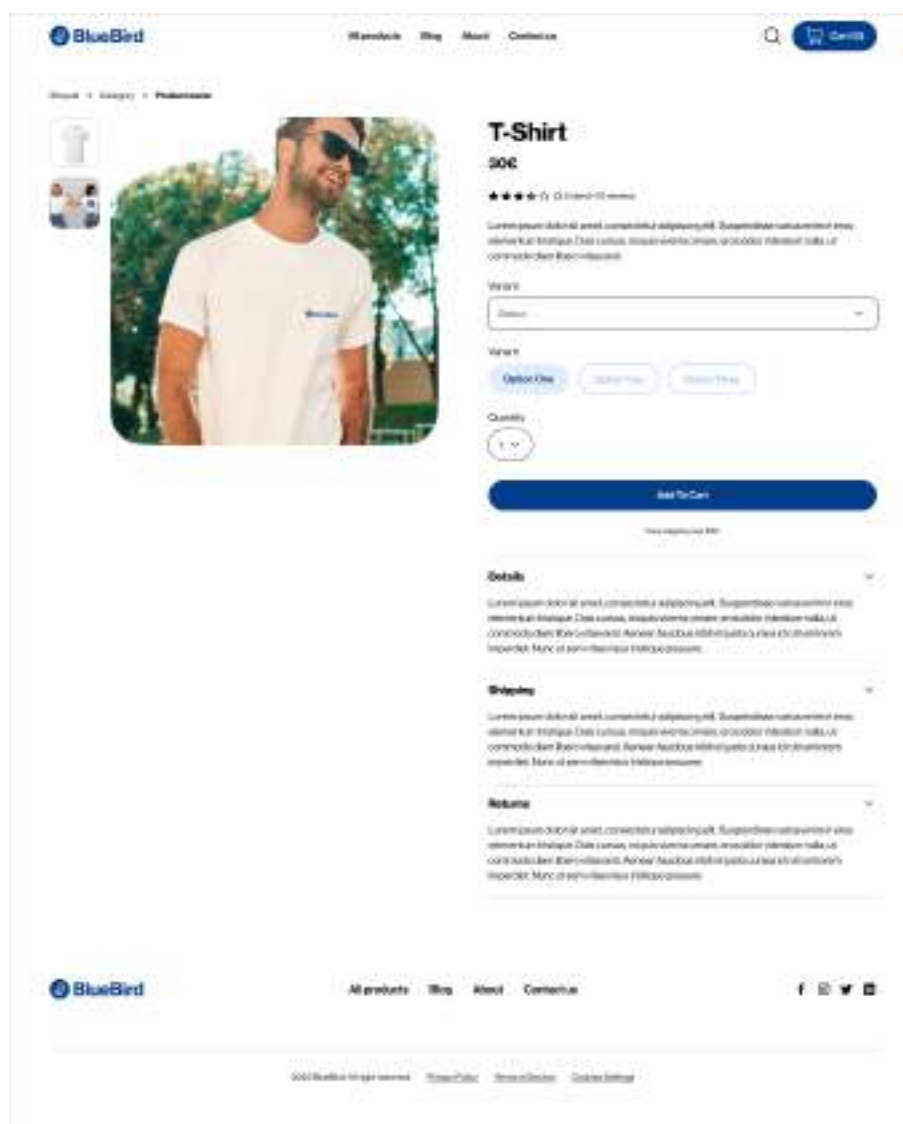


Figure 99: Final designed Product Detail page of the online shop of 'BlueBird'.

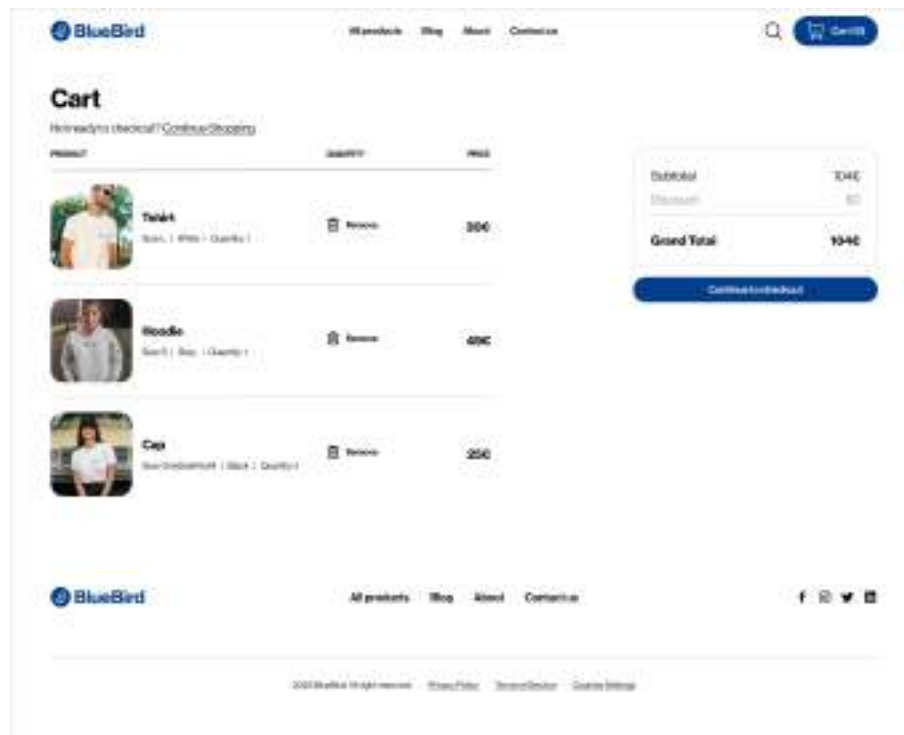


Figure 100: Final designed Shopping Cart page of the online shop of 'BlueBird'.

[← Back to Shopping](#)



1 Cart

2 **Checkout**

3 Review

4 Complete

Checkout

Shipping Address

Customer

Street address

(Continuation of your street address)

Postal Number

+49

Maximum valid house number*

(if applicable)

Phone Number

City*

Region

Shipping Method

☐ Free shipping
7-10 business days **0€**

☐ Regular shipping
3-4 business days **8€00**

☒ Express shipping
1-2 business days **88€00**

Your order

 **T-shirt**
Size: S | Color: Blue | 1 x **39€**

 **Hoodie**
Size: M | Color: Grey | 1 x **49€**

 **Cap**
Size: One Size | Color: Black | 1 x **29€**

Discount Code

[Apply](#)

Subtotal	104€
Discount	-8€
Shipping Costs	-8€
Grand Total	104€

[Continue to payment](#)

[All products](#) [Shop](#) [About](#) [Contact Us](#)[f](#) [ig](#) [tw](#) [in](#)[© 2021 BlueBird - All rights reserved](#) [Privacy Policy](#) [Terms of Service](#) [Contact Us](#)

Figure 101: Final designed Customer Information page of the online shop of 'BlueBird'.

← Back to Shopping

BlueBird

1 Cart 2 Checkout 3 Payment 4 Review

Payment

Select payment method

 **Paypal**

 **Mastercard**

 **American Express**

 **Bank Giro**

 **Visa**  **Credit Card**

Card number:

Expiration date:

CVV:

Your order

 **T-shirt**
Size: S | Color: Blue | 1 x 1 = 30€

 **Hoodie**
Size: M | Color: Grey | 1 x 1 = 40€

 **Cap**
Size: One Size | Color: Black | 1 x 1 = 20€

Discount Code

Apply discount code **Apply**

Subtotal 100€

Discount -80

Shipping Costs -50

Grand Total 19€

[Continue to Review](#)

BlueBird

[All products](#) [Blog](#) [About](#) [Contact us](#)

[Facebook](#) [Instagram](#) [Twitter](#) [LinkedIn](#)

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Figure 102: Final designed Payment Method page of the online shop of 'BlueBird'.



Figure 104: Final designed Order Confirmation page of the online shop of 'BlueBird'.